

Questions:-

Q.1 Find the value of α and β such that:

$$\left. \begin{array}{l} x+y+z=5 \\ x+3y+3z=9 \\ \hookrightarrow x+2y+\alpha z=\beta \end{array} \right\} \begin{array}{l} \text{have infinite no. of solutions.} \\ \nearrow \text{N. Homogenous} \end{array}$$

$$\hookrightarrow \alpha - 2 = 0$$

$$\alpha = 2$$

$$\hookrightarrow \beta - 7 = 0$$

$$\beta = 7$$

$$\hookrightarrow \rho(A) = \rho(A:B) < 3 \quad \Rightarrow \quad \rho(A) = \rho(A:B) = 2$$

$$(A:B) = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 3 & 3 & 9 \\ 1 & 2 & \alpha & \beta \end{array} \right] \xrightarrow[\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 2 & 2 & 4 \\ 0 & 1 & \alpha - 1 & \beta - 5 \end{array} \right] \xrightarrow[\substack{R_3 \rightarrow R_3 - \frac{R_2}{2}}]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & \alpha - 2 & \beta - 7 \end{array} \right]$$

MSQ

Q. $\begin{cases} x+y+z=3 \\ x+2y+3z=4 \\ x+4y+kz=6 \end{cases}$ will not have unique solⁿ for $k =$
 (a) 6 (b) 5 (c) 7 (d) -2

↳

↳ $\rho(A) \neq \rho(A:B) \rightarrow$ NO solution

$\rho(A) = \rho(A:B) < 3 \rightarrow$ Infinite solⁿ

$$A:B = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 1 & 2 & 3 & : & 4 \\ 1 & 4 & k & : & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 3 & k-1 & : & 3 \end{bmatrix} \xrightarrow[\substack{R_3 \downarrow \\ R_3 - 3R_2}]{\substack{R_3 \downarrow \\ R_3 - 3R_2}} \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 0 & k-7 & : & 0 \end{bmatrix}$$

If $k=7$; $\rho(A) = \rho(A:B) = 2 \rightarrow$ Infinite solⁿ

If $k \neq 7$; $\rho(A) = \rho(A:B) = 3 \rightarrow$ Unique solⁿ

MSQ
Q.

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 2$$

$$x_1 + 4x_2 + ax_3 = 4$$

will have unique solⁿ for $a =$

~~(a) 1~~ ~~(b) -2~~ (c) 5 ~~(d) -5~~

$$\rho(A) = \rho(A:B) = 3$$

$$\begin{aligned} \hookrightarrow A:B &= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 4 & a & 4 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & a-2 & 3 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & 0 \end{array} \right] \end{aligned}$$

If $a = 5$; $\rho(A) = \rho(A:B) = 2 < 3 \rightarrow$ Infinite solⁿ

If $a \neq 5$; $\rho(A) = \rho(A:B) = 3 \rightarrow$ Unique solⁿ

MSQ
Q.

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 2$$

$$x_1 + 4x_2 + ax_3 = 5$$

will have NO solution for $a =$

(a) 1 (b) -2 (c) 5 (d) -5

$$\rho(A) \neq \rho(A:B)$$

$$\begin{aligned} \hookrightarrow A:B &= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 4 & a & 5 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & a-2 & 4 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & 1 \end{array} \right] \end{aligned}$$

If $a = 5$; $\rho(A) = 2$, $\rho(A:B) = 3 \Rightarrow \rho(A) \neq \rho(A:B)$

If $a \neq 5$; $\rho(A) = 3$, $\rho(A:B) = 3 \Rightarrow$ unique solⁿ

Q. $2x + 3y = 0$
 $6x + qy = 0$

Homogenous eqⁿ

The value of q for non-trivial solⁿ

(a) 7 (b) 6 (c) 9 (d) 2

→ Infinite

↳ $\rho(A) < 2$; A is a square matrix of 2×2 order $\Rightarrow |A| = 0$

$A = \begin{bmatrix} 2 & 3 \\ 6 & q \end{bmatrix}_{2 \times 2} \xrightarrow{\rho(A) < 2} \begin{bmatrix} 2 & 3 \\ 0 & q-9 \end{bmatrix}$ $q-9=0 \Rightarrow \boxed{q=9}$

$2q - 18 = 0$
 $\boxed{q=9}$

Q. $x_1 + 2x_2 + 3x_3 + 4x_4 = 7$
 $2x_1 + 4x_2 + 6x_3 + 8x_4 = 14$

Nature of solⁿ

(a) Consistent (b) Inconsistent
 (c) Trivial (d) Non-Trivial

Free variables = ?

↳ 4 variable, 2 eqⁿ

$A:B = \begin{bmatrix} 1 & 2 & 3 & 4 & : & 7 \\ 2 & 4 & 6 & 8 & : & 14 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 & 4 & : & 7 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$

$\rho(A) = \rho(A:B) = 1 < 4$
 free variable = $4 - 1 = 3$

Q. $x + 2y - 3z = a$

$2x + 3y + 3z = b$

$5x + 9y - 6z = c$

The system is consistent if

(a) $7a - b - c = 0$

(c) $3a - b + c = 0$

(b) $3a + b - c = 0$

(d) $7a - b + c = 0$

Unique
↗ ↘ Infinite } $\rho(A) = \rho(A:B)$

$$\hookrightarrow A:B = \begin{bmatrix} 1 & 2 & -3 & : & a \\ 2 & 3 & 3 & : & b \\ 5 & 9 & -6 & : & c \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & -1 & 9 & : & c-5a \end{bmatrix} \xrightarrow{A_3 \rightarrow A_3 - A_2} \begin{bmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & 0 & 0 & : & ? \end{bmatrix}$$

$c - 5a - b + 2a$

$c - b - 3a$

↗

$\rho(A) = 2 = \rho(A:B)$

$\hookrightarrow c - b - 3a = 0 \Rightarrow \boxed{3a + b - c = 0}$

Q.

Find the values of k for which the following system of equations has a non-trivial solution:

$$\left. \begin{aligned} (3k-8)x + 3y + 3z &= 0 \\ 3x + (3k-8)y + 3z &= 0 \\ 3x + 3y + (3k-8)z &= 0 \end{aligned} \right\} \begin{array}{l} \text{Infinite sol}^n \\ \rightarrow \text{Homogenous} \\ \rho(A) < 3 \end{array}$$

$$A = \begin{bmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{bmatrix}_{3 \times 3}$$

$$\underbrace{[A]_{3 \times 3}}_{\downarrow} \text{ and } \rho(A) < 3$$

$$|A| = 0$$

$$|A| = \begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2 + R_3} \begin{vmatrix} 3k-2 & 3k-2 & 3k-2 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 3k-2 \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix}$$

$$= 3k-2 \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} \xrightarrow[\substack{C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1}]{(3k-2)} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 3k-11 & 0 \\ 3 & 0 & 3k-11 \end{vmatrix}$$

$$\Rightarrow (3k-2)(3k-11) = 0$$

$$\boxed{k = \frac{2}{3} \text{ or } \frac{11}{3}} \quad \checkmark$$

Q.

If the system

$$Px + Qy + Rz = 0, \quad Qx + Ry + Pz = 0, \quad Rx + Py + Qz = 0$$

has a non-trivial solution, then which of the following are true?

Options:

Infinite

A. $P - Q + R = 0$ or $P = Q = -R$

B. $P + Q - R = 0$ or $P = -Q = -R$

✓ C. $P + Q + R = 0$ or $P = Q = R$

D. $P - Q + R = 0$ or $P = -Q = -R$

$\hookrightarrow \rho(A) < 3$

$\hookrightarrow [A]_{3 \times 3} = \begin{bmatrix} P & Q & R \\ Q & P & R \\ R & P & Q \end{bmatrix}$

$\Rightarrow |A| = 0$

$$\begin{vmatrix} P & Q & R \\ Q & P & R \\ R & P & Q \end{vmatrix} = 0$$

$$\Rightarrow P(RQ - P^2) - Q(Q^2 - PR) + R(QP - R^2) = 0$$

$$RPQ - P^3 - Q^3 + RPQ + RPQ - R^3 = 0$$

$$P^3 + Q^3 + R^3 - 3RPQ = 0$$

$$\boxed{a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)} \quad \checkmark$$

$$= (a+b+c) \times \frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= \frac{1}{2} (a+b+c) [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$p^3 + q^3 + r^3 - 3pqr = \frac{1}{2} (p+q+r) [(p-q)^2 + (q-r)^2 + (r-p)^2] = 0$$

$$\Downarrow$$

$$\boxed{p+q+r=0}$$

$$\text{OR } (p-q)^2 + (q-r)^2 + (r-p)^2 = 0$$

$$p-q=0, \quad q-r=0, \quad r-p=0$$

$$\hookrightarrow \boxed{p=q=r}$$

Q.

Consider the following system of equations:

3 variable and
4 eqⁿ
3 effective eqⁿ

$$x + y + z = 6$$

$$2x + y + z = 7$$

$$x + 2y + z = 8$$

$$x + y + 2z = 9$$

Which of the following statements are true?

Options:

- ☒ A. The system has a unique solution
- ☐ B. The system has one independent solution
- ☐ C. The system has two independent solutions
- ☐ D. The system has three independent solutions

↳

$$(A:B) = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 2 & 1 & 1 & : & 7 \\ 1 & 2 & 1 & : & 8 \\ 1 & 1 & 2 & : & 9 \end{bmatrix}$$

↳ Rank = 3 =

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & ? & ? & : & ? \\ 0 & 0 & ? & : & ? \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

↳ 1 free variable x

$\rho(A) = \rho(A:B) = 3 = \text{no. of variables}$

unique solⁿ



Q. ★★

Question:

A set of linear equations is represented by the matrix equation $AX = B$.
The necessary condition for the existence of a solution for this system is:

consistent

Options:

- A. A must be invertible
- ☒ B. B must be linearly dependent on the columns of A
- C. B must be linearly independent of the columns of A
- D. None of the above

$$\boxed{\rho(A) \leq \rho(A:B)}$$

Rank: no. of Independent
Rows and columns

$$\rho(A) = \rho(A:B)$$

$$C = (A:B) = \begin{bmatrix} \underbrace{a \quad b \quad c}_{c_1 \quad c_2 \quad c_3} : \underbrace{d}_{c_4} \\ e \quad f \quad g : h \\ i \quad j \quad k : l \end{bmatrix}$$

$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{2em}}_B$

$$\hookrightarrow \rho(A) = 3 = \rho(A:B) : \text{consistent [Unique]}$$

$$\rho(A:B) = 3 = \rho(C)$$

In matrix C , there are 3 Independent
columns and 1 dependent column

$\rho(A) = 2 = \rho(A:B)$: consistent [Infinite solⁿ]
 $\rho(A:B) = \rho(C) = 2$

$$\left[\begin{array}{cccc|c} c_1 & c_2 & c_3 & c_4 & \\ a & b & c & d & \\ 0 & \alpha & \beta & \gamma & \\ 0 & 0 & 0 & 0 & \end{array} \right]$$

$\underbrace{\hspace{10em}}_A \qquad \underbrace{\hspace{2em}}_B$

NOT Invertible

$\vec{D}$ depends on $\vec{A}, \vec{B}, \vec{C}$

$$\alpha \vec{A} + \beta \vec{B} + \gamma \vec{C} = \vec{D}$$

2 Ind. colⁿ, 2 dep. columns

$$c_4 = \alpha c_1 + \beta c_2$$

columns of A

$$\left\{ \begin{array}{l} 2x + 3y + 5z = 4 \\ 3x + 2y + 4z = 5 \\ 2x + 4y - 6z = 7 \end{array} \right.$$

$\vec{B}$

$$\boxed{B = f(A)}$$

Q.

Question:

Let A be a 4×4 matrix with $\text{rank}(A) = 2$.

It is given that the system of equations $AX = B$ is consistent.

Which of the following statements are true?

Options:

- ☒ A. $\text{rank}([A \mid B]) = 2$
- ☒ B. The system has a unique solution
- ☒ C. The system has infinite solutions
- ☒ D. The system has two independent solutions
- ☒ E. The system has three independent solutions

$\Rightarrow$ no. of columns = 4 = no. of variables

$$\rho(A) = \rho(A:B) = 2 < 4$$

$$\eta(A) = 4 - 2 = \underline{2}$$

Q.

Question:

Consider the following system of linear equations:

$$x_1 + 2x_2 = b_1, \quad 2x_1 + 4x_2 = b_2, \quad 3x_1 + 7x_2 = b_3, \quad 3x_1 + 9x_2 = b_4$$

Which of the following conditions ensure that a solution exists for the above system?

Options:

A. $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

B. $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

C. $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

D. None of the above

$$\rho(A) = \rho(A^r, B)$$

$$A:B = \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 2 & 4 & b_2 \\ 3 & 7 & b_3 \\ 3 & 9 & b_4 \end{array} \right] \xrightarrow[\substack{R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - 2R_1}]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 0 & b_2 - 2b_1 \\ 0 & 1 & b_3 - 3b_1 \\ 0 & 3 & b_4 - 3b_1 \end{array} \right]$$

$$\begin{bmatrix} \textcircled{1} & 2 & : & b_1 \\ 0 & 0 & : & b_2 - 2b_1 \\ 0 & \textcircled{1} & : & b_3 - 3b_1 \\ 0 & 3 & : & b_4 - 3b_1 \end{bmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - 3R_3 \\ R_3 \leftrightarrow R_2}]{} \begin{bmatrix} 1 & 2 & : & b_1 \\ 0 & 1 & : & b_3 - 2b_1 \\ 0 & 0 & : & b_2 - 2b_1 \\ 0 & 0 & : & b_4 - 3b_3 + 6b_1 \end{bmatrix} \begin{array}{l} \rightarrow \\ \rightarrow \end{array}$$

$$\rho(A) = 2 = \rho(A:B)$$

$$\begin{array}{l} b_2 - 2b_1 = 0 \\ b_4 - 3b_3 + 6b_1 = 0 \end{array}$$

Q.

Let

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

Let P be the matrix whose columns are

$$\mathbf{a}, \mathbf{b}, (3\mathbf{a} - \mathbf{b}), (\mathbf{a} - 3\mathbf{b})$$

in that order.

Free variable = ?

The number of linearly independent solutions of the homogeneous system $PX = 0$ is:

$$PX = 0$$

$$\eta(P) = 4 - \rho(P)$$

$$= 4 - 2$$

$$= \boxed{2} \rightarrow \text{Independent soln}$$

$$P = \begin{bmatrix} \mathbf{a} & \mathbf{b} & 3\mathbf{a}-\mathbf{b} & \mathbf{a}-3\mathbf{b} \\ 2 & 2 & 4 & -4 \\ 3 & 0 & 9 & 3 \\ 0 & 3 & -3 & -9 \end{bmatrix}_{4 \times 4}$$

$c_1 \quad c_2 \quad c_3 \quad c_4$

$$= 3c_1 - c_2 = c_1 - 3c_2 \rightarrow \text{no. of variables}$$

c_1 and c_2 are l.i.

c_3 is depending on c_1 and c_2

c_4 " " " c_1 " c_2

$\rightarrow$ 2 l.i. columns $\downarrow$

$$\text{Rank} = 2$$

Q.

Question:

Let

$$\eta(A) = 6 - \rho(A) = 6 - 1 = 5$$

$$A = \begin{bmatrix} 1 & b & b^2 & b^3 & b^4 & b^5 \\ 1 & b & b^2 & b^3 & b^4 & b^5 \\ 1 & b & b^2 & b^3 & b^4 & b^5 \\ 1 & b & b^2 & b^3 & b^4 & b^5 \\ 1 & b & b^2 & b^3 & b^4 & b^5 \\ 1 & b & b^2 & b^3 & b^4 & b^5 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ \hline R_3 \rightarrow R_3 - R_1 \\ \vdots \\ R_6 \rightarrow R_6 - R_1 \end{array}$$

 6×6

$$R_6 \rightarrow R_6 - R_1$$

$$A = \begin{bmatrix} \text{---} \\ 0 & 0 & 0 & 0 & \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \rightarrow \text{NON-ZERO}$$

be a 6×6 square matrix.The number of linearly independent solutions of the homogeneous system $AX = 0$ is:

5

Q.

NAT

If the system

$$Ax + y + z = 0, \quad x + By + z = 0, \quad x + y + Cz = 0$$

(where $A, B, C \neq 1$) has a non-trivial solution, then the value of

$$\frac{1}{1-A} + \frac{1}{1-B} + \frac{1}{1-C}$$

$$[A]_{3 \times 3}; \quad \rho(A) < 3$$

 $\Rightarrow$

is:

Numerical value $|A| = 0$

$$\rho(A) < 3$$

$$\begin{vmatrix} A & 1 & 1 \\ 1 & B & 1 \\ 1 & 1 & C \end{vmatrix} = 0$$

?

$$\Rightarrow A(BC - 1) - 1(C - 1) + 1(1 - B) = 0$$

$$ABC - A - C + 1 + 1 - B = 0$$

$$ABC - (A + B + C) + 2 = 0$$

$$ABC - (A+B+C) + 2 = 0$$

$$\boxed{A = -1, B = 2}, C = ?$$

$$-2C - (1+C) + 2 = 0$$

$$-2C - 1 - C + 2 = 0$$

$$\boxed{C = \frac{1}{3}}$$

$$\frac{1}{1-A} + \frac{1}{1-B} + \frac{1}{1-C}$$

Numerical
value

$$A, B, C \neq 1$$

$$\frac{1}{1+1} + \frac{1}{1-2} + \frac{1}{1-\frac{1}{3}}$$

$$= \frac{1}{2} - 1 + \frac{3}{2}$$

$$= \boxed{1} \text{ Ans.}$$

Q.

Let A be an $n \times \underline{n}$ real matrix such that $A^2 = I$, and let Y be an n -dimensional vector. The system $AX = Y$ has:

Options:

- A. No solution
- ☒ B. Unique solution
- C. Infinitely many solutions
- D. Cannot be determined

$$\hookrightarrow A^2 = I \Rightarrow A \cdot A = I \Rightarrow A = A^{-1} \leadsto A^{-1} \text{ exist} \Rightarrow |A| \neq 0$$

$$A \cdot B = I \Rightarrow B = A^{-1}$$

$$\boxed{\rho(A) = n}$$

$$\rho(A:Y) = n = \rho(A) = \text{no. of variables}$$

$$n \leq \rho(A:Y) \leq n \Leftarrow \begin{cases} \rho(A) \leq \rho(A:Y) \\ \rho(A:Y) \leq n \end{cases}$$

Q. If the vectors x_1, x_2, x_3, x_4 are linearly dependent and A is the coefficient matrix of the homogeneous system $\underline{A} \underline{X} = 0$ whose columns are $\underline{x_1}, \underline{x_2}, \underline{x_3}, \underline{x_4}$, then the system has:

- (A) Unique solution
- (B) Infinitely many solutions
- (C) Only zero solution
- (D) No solution

$$A = \begin{bmatrix} \downarrow x_1 & \downarrow x_2 & \downarrow x_3 & \downarrow x_4 \\ & & & \end{bmatrix}_{m \times 4}$$

$$\rho(A) < 4$$

< no. of variable

A: NOT full rank matrix

$$\underbrace{\rho(A) = n}_{\text{Unique}}$$

$$\underbrace{\rho(A) < n}_{\text{Infinite}}$$

Q.

If the vectors x_1, x_2, x_3, x_4 are linearly dependent and A is the coefficient matrix of the non-homogeneous system $AX = B$ whose columns are x_1, x_2, x_3, x_4 , then the system ~~is~~ may have:

- ~~(A)~~ Unique solution
~~(B)~~ Infinitely many solutions
 (C) Only zero solution
~~(D)~~ No solution

$\hookrightarrow [A] = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ \sim & \sim & \sim & \sim \\ \sim & \sim & \sim & \sim \\ \sim & \sim & \sim & \sim \\ \sim & \sim & \sim & \sim \end{bmatrix}_{4 \times 4}$

$$\rho(A) < 4$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} \sim & \sim & \sim & \sim & : & a \\ \sim & \sim & \sim & \sim & : & b \\ \sim & \sim & \sim & \sim & : & c \\ \sim & \sim & \sim & \sim & : & d \end{bmatrix}$$

$\hookrightarrow$ no. of variable

$$\rho(A) = 3, \rho(A:B) = 4 \Rightarrow \text{No sol}^n$$

$$\text{if } d=0, \rho(A) = 3 = \rho(A:B) = 3 \Rightarrow \text{Infinite sol}^n$$

Q.

$AX=B$: zero solⁿ ?
↳ Non-Homogenous =

$x_1=0, x_2=0, x_3=0$?

↳ $2x_1 + 3x_2 + 4x_3 = 7$ — (1)

$2x_1 + 4x_2 + 3x_3 = 0$ — (2)

$-4x_1 + 2x_2 + x_3 = 0$ — (3)

Non-Homogenous

↳ System of Non-Homogenous linear eqⁿ can never have zero solⁿ

↳ zero solⁿ $\neq$ No solⁿ
↳ $x=0$

↳ x : NOT POSSIBLE

* Representation of a vector:-

Column vector

$$\vec{A} = 3\hat{i} + 2\hat{j} + 5\hat{k} \rightarrow A = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}_{3 \times 1}; A = \begin{bmatrix} 3 & 2 & 5 \end{bmatrix}_{1 \times 3}$$

↓
Row vector

* length of a vector:-

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$|\vec{A}| = A = \sqrt{a^2 + b^2 + c^2}$$

$A^T A = ||A||^2$

$$A = \begin{bmatrix} a \\ b \\ c \end{bmatrix}_{3 \times 1} \quad [A^T]_{1 \times 3} [A]_{3 \times 1} = [a \quad b \quad c] \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$
$$||A|| = \sqrt{a^2 + b^2 + c^2} = a^2 + b^2 + c^2$$

Important NOTE:-

Let x be a column vector of dimension $n \times 1$.

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{x^T x}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$

What does Trace of xx^T represent = ?

$$\text{Tr}[xx^T] = ?$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}_{1 \times n} = \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_n \\ x_2 x_1 & x_2^2 & \dots & x_2 x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n x_1 & x_n x_2 & \dots & x_n^2 \end{bmatrix}_{n \times n}$$

$$\text{Tr}[xx^T] = x_1^2 + x_2^2 + \dots + x_n^2 = \|x\|^2 = x^T x$$

* Unit Vector / "Normalized" Vector:-

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$|\vec{A}| = \sqrt{a^2 + b^2 + c^2} = d$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{a\hat{i} + b\hat{j} + c\hat{k}}{\sqrt{a^2 + b^2 + c^2}}$$

$$= \frac{a}{d}\hat{i} + \frac{b}{d}\hat{j} + \frac{c}{d}\hat{k}$$

$$A = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$|A| = \sqrt{a^2 + b^2 + c^2} = d$$

$$\hat{A} = \frac{A}{|A|} = \begin{bmatrix} a/d \\ b/d \\ c/d \end{bmatrix}$$

* Orthogonal vectors:-

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{P} = p\hat{i} + q\hat{j} + r\hat{k}$$

$$\vec{A} \perp \vec{P} ; \vec{A} \cdot \vec{P} = 0 = ap + bq + cr$$

$$A = \begin{bmatrix} a \\ b \\ c \end{bmatrix}_{3 \times 1}$$

$$A^T P = 0$$

$$; P = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{3 \times 1}$$

$$ap + bq + cr = 0$$

* Orthonormal vectors:-

Same as orthogonal + length of vectors = 1

$$\vec{A} \cdot \vec{P} = 0 ; ap + bq + cr = 0$$

$$\sqrt{a^2 + b^2 + c^2} = 1 = \sqrt{p^2 + q^2 + r^2}$$

N.B. - let $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ are set of Mutually orthogonal vectors

then $\vec{A} \perp \vec{B} \perp \vec{C} \perp \vec{D}$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{C} = \vec{C} \cdot \vec{D} = \vec{D} \cdot \vec{A} = \vec{B} \cdot \vec{D} = \vec{A} \cdot \vec{C} = 0$$

↳ let $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ are set of Mutually orthonormal vectors

then

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{C} = \vec{C} \cdot \vec{D} = \vec{D} \cdot \vec{A} = \vec{B} \cdot \vec{D} = \vec{A} \cdot \vec{C} = 0$$

$$|\vec{A}| = |\vec{B}| = |\vec{C}| = |\vec{D}| = 1$$

Q.

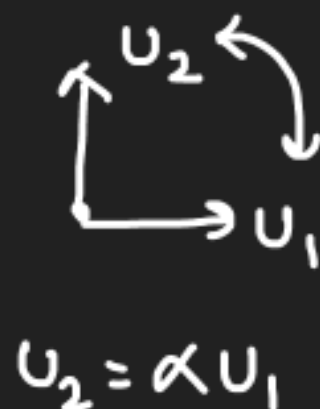
$$\|u_1\| = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1, \quad \|u_2\| = 1, \quad \|u_3\| = 1$$

Let

$$\vec{u}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \quad \vec{u}_2 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right), \quad \vec{u}_3 = (0, 0, 1)$$

Which of the following statements are true?

- ☒ (A) The vectors u_1, u_2, u_3 are linearly independent.
- ☒ (B) The vectors u_1, u_2, u_3 form an orthogonal set.
- ☒ (C) The vectors u_1, u_2, u_3 form an orthonormal set.
- ☒ (D) If a matrix A is formed with u_1, u_2, u_3 as its rows, then A is invertible.



$$\vec{u}_1 \cdot \vec{u}_2 = \frac{1}{2} - \frac{1}{2} + 0 = 0$$

$$\vec{u}_2 \cdot \vec{u}_3 = 0 + 0 + 0 = 0$$

$$\vec{u}_1 \cdot \vec{u}_3 = 0 + 0 + 0 = 0$$

↳

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \rightarrow u_1 \\ \rightarrow u_2 \\ \rightarrow u_3 \end{matrix}$$

3×3

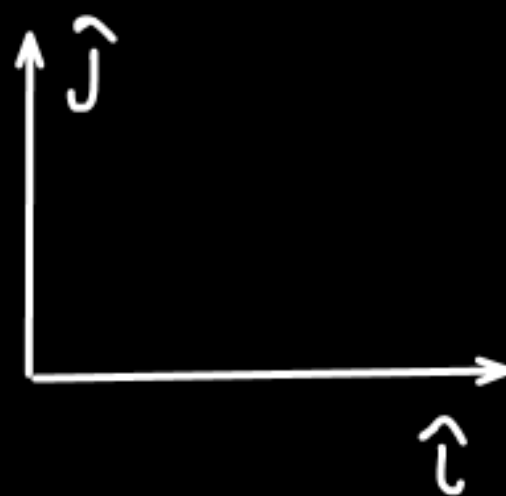
$$|A| \neq 0$$

$$\rho(A) = 3 \Rightarrow 3 \text{ Independent Rows}$$

$$B = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \hat{i} \quad B = \hat{j}$$

$$A \cdot B = 0$$



↳ Orthogonal vectors are always Independent of each other.

Let $\mathbf{x}$ be a 5×1 real column vector such that $\|\mathbf{x}\| = 5$. Define the matrix

$$A = \mathbf{x}\mathbf{x}^T$$

Then, the trace of matrix A is:

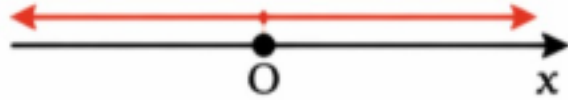
↳ $\|\mathbf{x}\| = 5 = \sqrt{\mathbf{x}^T \mathbf{x}}$

$$\text{Tr}[\mathbf{x}\mathbf{x}^T] = \|\mathbf{x}\|^2 = 5^2 = \boxed{25} \text{ Ans.}$$

Dimension and Geometric Interpretation

1 Dimension

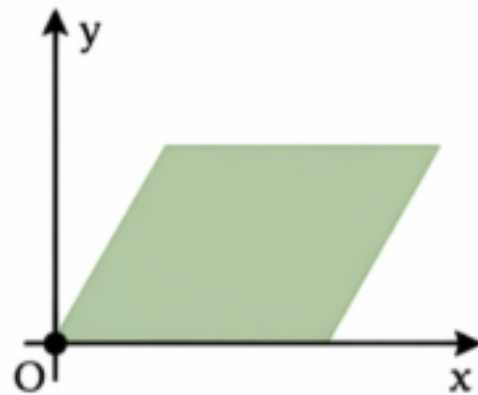
If we can get only **one dimension**, that means it will be a **LINE**.



Shape:
Line

2 Dimensions

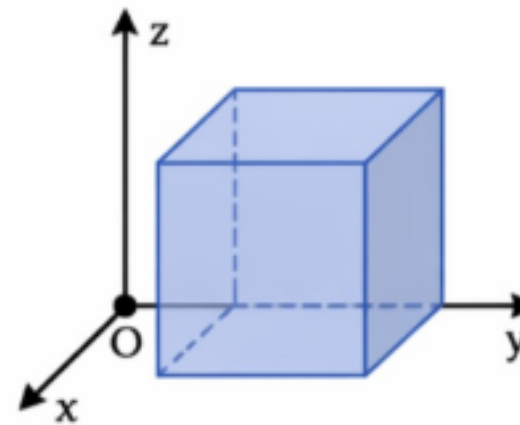
If we can get **two dimensions**, that means it will be a **PLANE**.



Shape:
Plane

3 Dimensions

If we can get **three dimensions**, that means it will be a **SPACE**.



Shape:
Space

n -dimension $\rightarrow$
 n -D space

* Dimension of a vector:-

Dimension $\neq$ span

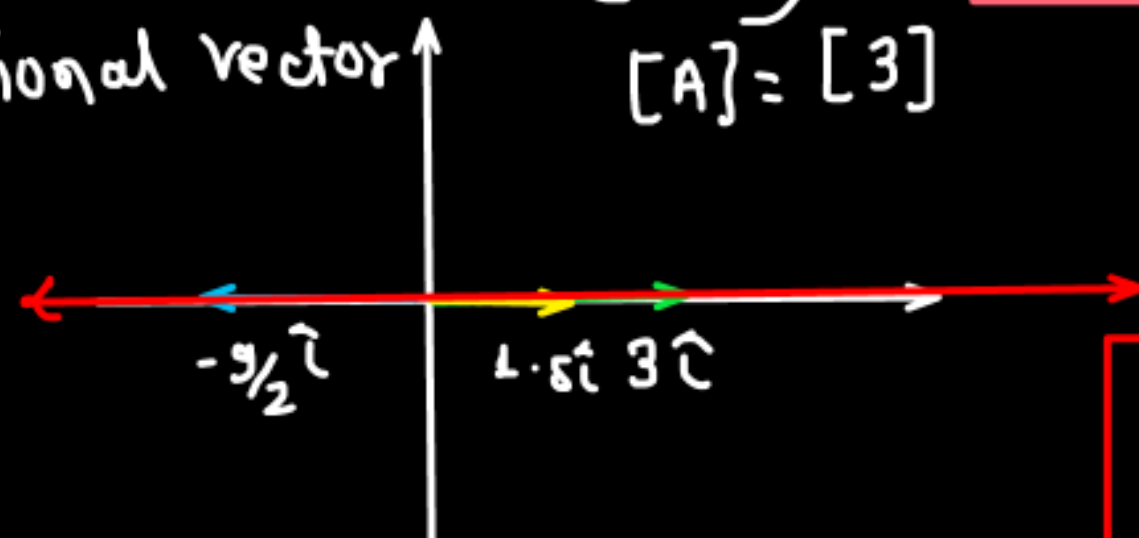
$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow 2-D ; B = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \rightarrow 3-D ; C = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 1 \end{bmatrix} \rightarrow 4-D ; D = [1] \rightarrow 1-D$$

* Span of a set of Vectors:-

Q. For the given vectors, Find their span. $\rho(A) = 1 \Rightarrow \text{span}$

(a) $\vec{A} = 3\hat{i}$ $\hookrightarrow$ 1-Dimensional vector

$$\alpha \vec{A} = \vec{p} ; \vec{p} = \frac{\vec{A}}{2} = \frac{3}{2}\hat{i} = 1.5\hat{i}$$



$$-3/2 \vec{A} = -9/2 \hat{i}$$

Span = 1-Dimensional
= line

(b) $\vec{B} = 3\hat{i} + 2\hat{j}$ $\hookrightarrow$ 2-Dimensional vector

$$\alpha \vec{B} = \vec{Q}$$

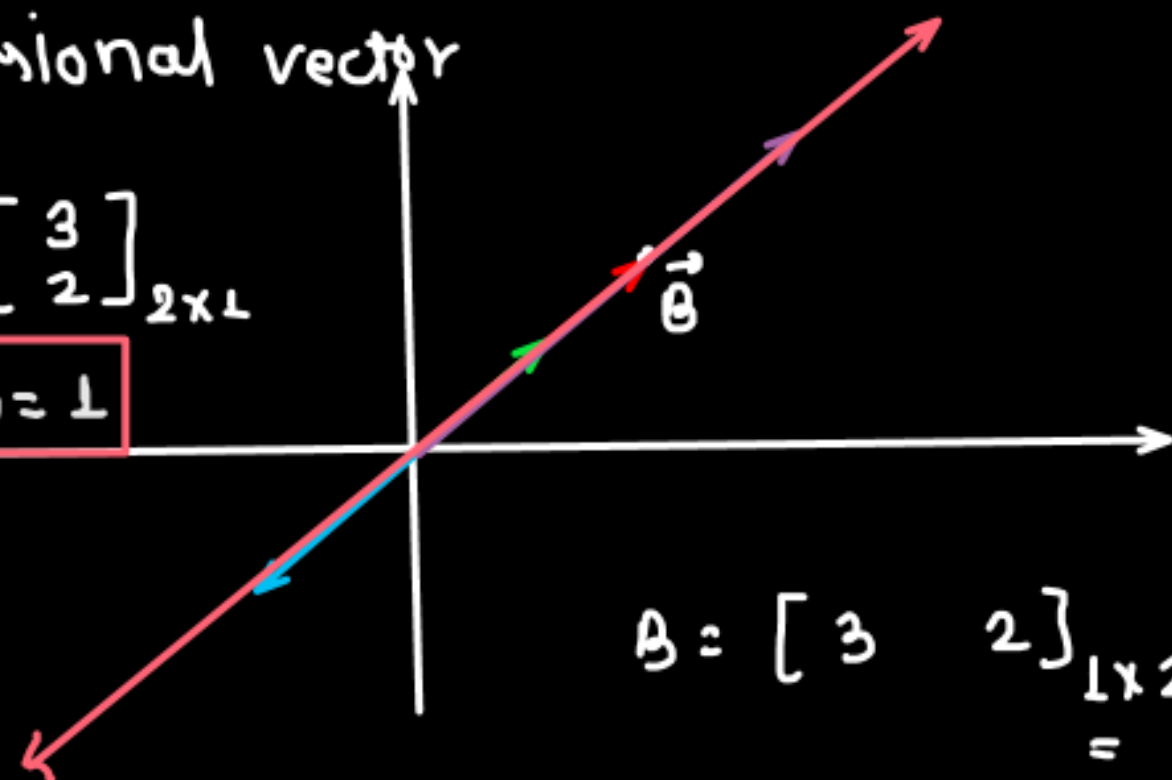
$$\vec{Q} = 1/2 \vec{B} = 1.5\hat{i} + \hat{j}$$

$$-3/2 \vec{B} = -9/2 \hat{i} - 3\hat{j}$$

$$4\vec{B} = 12\hat{i} + 8\hat{j}$$

$$B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{2 \times 1}$$

$$\rho(B) = 1$$



Span = 1-Dimensional
= Line

$$B = \begin{bmatrix} 3 & 2 \end{bmatrix}_{1 \times 2}$$

$$\rho(B) = 1 \Rightarrow \text{span}$$

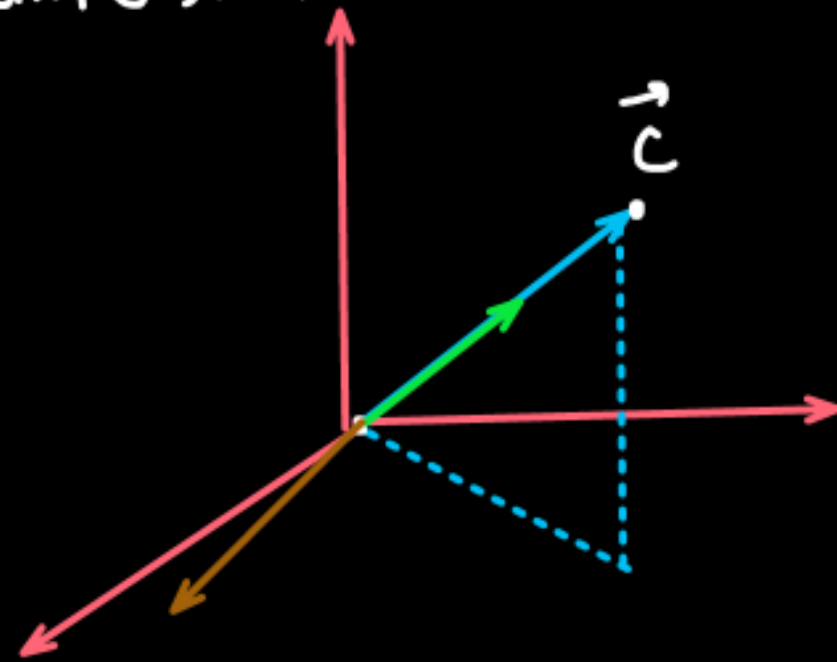
$$(c) \quad \vec{c} = 3\hat{i} + 2\hat{j} + 3\hat{k} \rightarrow \text{3-dimensional vector}$$

$$\frac{1}{2}\vec{c} = 1.5\hat{i} + \hat{j} + 1.5\hat{k}$$

$$-4\vec{c} = -(12\hat{i} - 8\hat{j} - 12\hat{k})$$

$$[\vec{c}] = [3 \quad 2 \quad 3]_{1 \times 3}$$

$$\rho(\vec{c}) = 1$$



$$\text{Span} = 1 - 0$$

$$(d) \quad \vec{D} = 3\hat{p} + 2\hat{q} + \hat{r} + 4\hat{s} \rightarrow \text{4-dimensional vector}$$

$$D = [3 \quad 2 \quad 1 \quad 4]$$

$$\rho(D) = 1$$

$$\text{Span} = 1 - 0$$

Q. For the given set of vectors, Find the span.

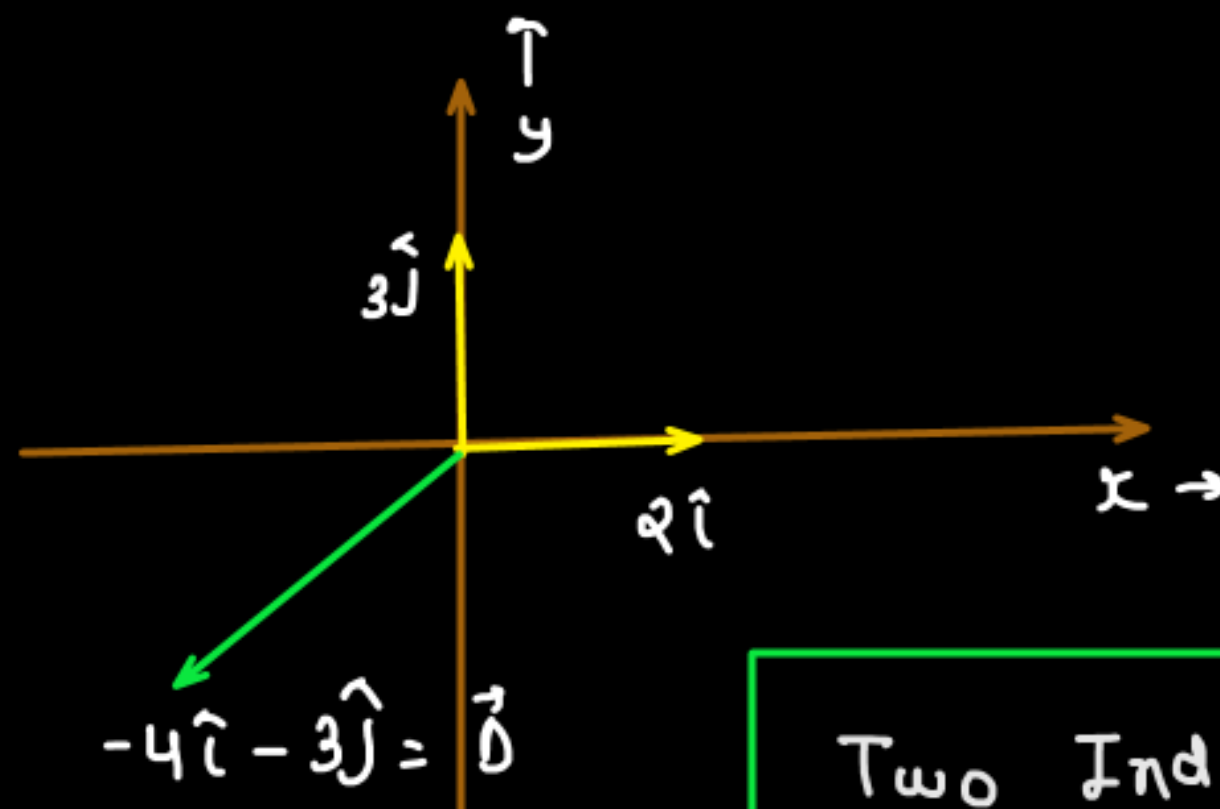
$$\left. \begin{array}{l} \text{(a) } \vec{A} = 2\hat{i} \\ \vec{B} = 3\hat{j} \end{array} \right\} \quad \left. \begin{array}{l} \vec{A} = 2\hat{i} + 0\hat{j} \\ \vec{B} = 0\hat{i} + 3\hat{j} \end{array} \right\} \rightarrow \text{2-dimensional, 2 independent vectors}$$

Set of
Indepen.
vectors

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} ; \rho(C) = 2$$

$$\text{Span} = 2\text{-D span}$$

$$\vec{D} = -2\vec{A} - \vec{B}$$



$$\alpha \vec{A} + \beta \vec{B} = -4\hat{i} - 3\hat{j}$$

$$2\alpha\hat{i} + 3\beta\hat{j} = -4\hat{i} - 3\hat{j}$$

$$\alpha = -2, \beta = -1$$

Two Independent 2-D vector
created a 2-D span.

$$(b) \begin{cases} \vec{A} = 2\hat{i} + 3\hat{j} \\ \vec{B} = 3\hat{i} - 4\hat{j} \end{cases} \quad \underbrace{\text{2-dimensional, 2-Independent vectors}}$$

$$C = \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix} ; \rho(C) = 2$$

$$\text{Span} = 2D$$

Two Independent 2-D vector
created a 2-D span.

$$(c) \begin{cases} \vec{A} = 2\hat{i} \\ \vec{B} = 3\hat{i} \end{cases} \rightarrow \begin{matrix} 1\text{-Dimensional} \\ 1\text{-Independent vector} \end{matrix} \quad \vec{B} = \frac{3}{2}\vec{A}$$

$$C = \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{2 \times 1} \xrightarrow{R_2 \rightarrow 2R_2 - 3R_1} \begin{bmatrix} 2 \\ 0 \end{bmatrix}_{2 \times 1} \quad \rho(C) = \underline{1}$$

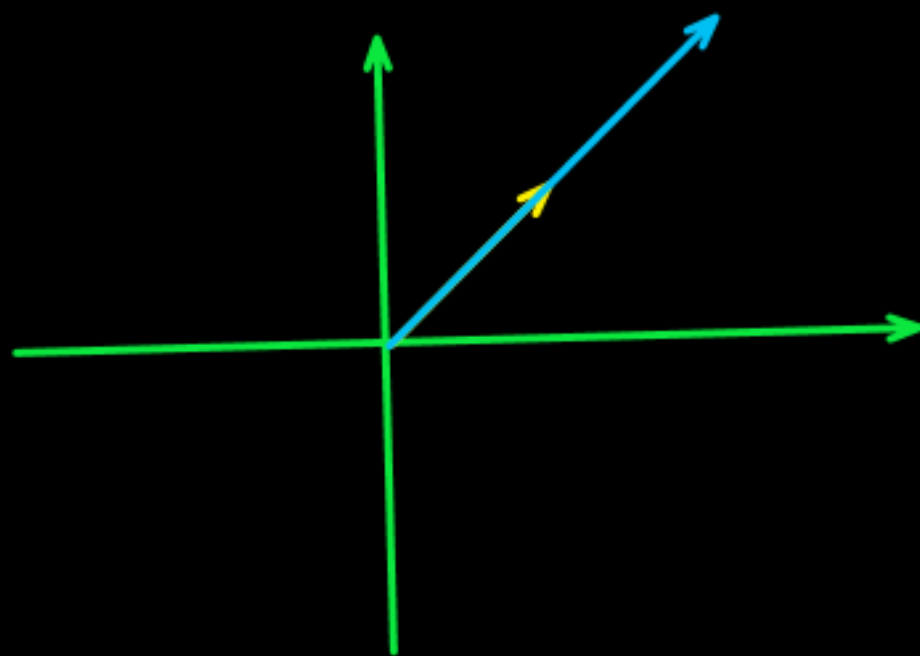
1-D span

$$\left. \begin{aligned} \vec{A} &= 2\hat{i} + 3\hat{j} \\ \vec{B} &= 6\hat{i} + 9\hat{j} \end{aligned} \right\} \begin{array}{l} \text{2-Dimensional} \\ \text{1-Independent vector} \end{array}$$

$$C = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix} \quad \text{rank}(C) = 1$$

$$\text{Span} = 1-D$$

$$\vec{B} = 3\vec{A}$$



$$2\alpha + 6\beta = 3$$

$$3\alpha + 9\beta = 0$$

$$\left\{ \begin{array}{l} \alpha + 3\beta = 1.5 \\ \alpha + 3\beta = 0 \end{array} \right\} \alpha, \beta: \text{No solution}$$

$$\alpha\vec{A} + \beta\vec{B} = 3\hat{i} \quad ?$$

$$2\alpha\hat{i} + 3\alpha\hat{j} + 6\beta\hat{i} + 9\beta\hat{j} = 3\hat{i}$$

$$(\underbrace{2\alpha + 6\beta})\hat{i} + (3\alpha + 9\beta)\hat{j} = \underbrace{3}\hat{i} + 0\hat{j}$$

$$(e) \left. \begin{aligned} \vec{A} &= 2\hat{i} + 3\hat{j} + 5\hat{k} \\ \vec{B} &= 3\hat{i} + 4\hat{j} + 6\hat{k} \end{aligned} \right\} \text{3-dimensional, } \underbrace{2 \text{ independent vectors}}$$

$$C = \begin{bmatrix} 2 & 3 & 5 \\ 3 & 4 & 6 \end{bmatrix} \xrightarrow{R_2 \rightarrow 2R_2 - 3R_1} \begin{bmatrix} 2 & 3 & 5 \\ 0 & -1 & -3 \end{bmatrix}$$

$$\rho(C) = 2$$

$$; \quad \boxed{\text{span} = 2D}$$

$$(f) \quad \left. \begin{aligned} \vec{A} &= 2\hat{i} + 3\hat{j} + 4\hat{k} \\ \vec{B} &= 4\hat{i} + 6\hat{j} + 8\hat{k} \end{aligned} \right\} \quad \vec{A} = \vec{B}/2$$

3 dimensional, 1 - independent vector
↳ 0 span

$$[C] = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \end{bmatrix} \quad \rho(C) = 1$$

$$\begin{aligned}
 (g) \quad \vec{A} &= 2\hat{i} = 2\hat{i} + 0\hat{j} + 0\hat{k} \\
 \vec{B} &= 3\hat{j} = 0\hat{i} + 3\hat{j} + 0\hat{k} \\
 \vec{C} &= 5\hat{k} = 0\hat{i} + 0\hat{j} + 5\hat{k}
 \end{aligned}$$

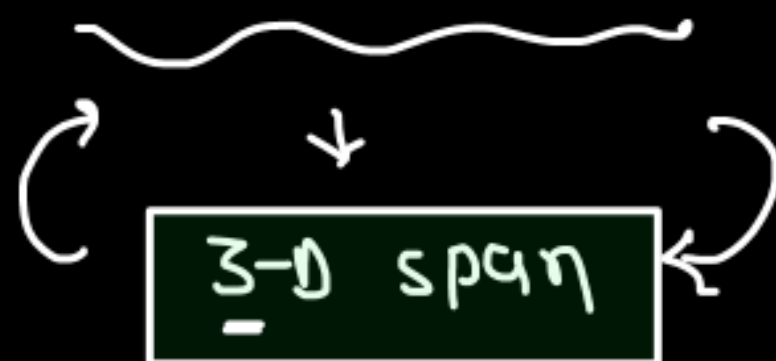
3 dimensional, 3 independent
vector

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}_{3 \times 3} \rightarrow J(0) = 3$$

$$\text{Span} = 3D$$

$$\begin{aligned}
 (h) \quad \vec{A} &= 2\hat{i} + 3\hat{j} \\
 \vec{B} &= 3\hat{j} + 4\hat{k} \\
 \vec{C} &= 2\hat{i} - 3\hat{j} + 2\hat{k}
 \end{aligned}
 \left. \vphantom{\begin{aligned} \vec{A} \\ \vec{B} \\ \vec{C} \end{aligned}} \right\}$$

3 dimensional, 3-independent vectors



$$D = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 3 & 4 \\ 2 & -3 & 2 \end{bmatrix}$$

$$\rho(D) = 3$$

$$|D| = 2[6 + 12] + 2(12) \neq 0$$

n-independent vectors
 $\swarrow$ n-D span $\searrow$

$$\left. \begin{aligned} \vec{A} &= 2\hat{i} + 3\hat{j} + 4\hat{k} \\ \vec{B} &= 3\hat{i} + 4\hat{j} + 3\hat{k} \\ \vec{C} &= 6\hat{i} + 8\hat{j} + 6\hat{k} \end{aligned} \right\}$$

3-dimensional, 2 independent vectors
2-D span ↙

$$D = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 3 \\ 6 & 8 & 6 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow 2R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \rightarrow$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & -1 & -6 \\ 0 & -1 & -6 \end{bmatrix}$$

$$p(0) = 2$$

Q. using a set of 1-dimensional vectors Try creating a 2-D span. You can use any number of 1-dimensional vector.

$$L \quad S = \{ \underbrace{[1]}, \underbrace{[-1]}, \underbrace{[2/3]}, \underbrace{[7/5]}, \underbrace{[-3]} \dots \} = \mathbb{R}^1$$

$$(a) \left. \begin{array}{l} \vec{A} = 2\hat{i} \\ \vec{B} = 3\hat{i} \\ \vec{C} = -2/3\hat{i} \end{array} \right\} \text{set of 1-Dimensional vectors}$$

$$(b) \left. \begin{array}{l} \vec{A} = 2\hat{i} \\ \vec{B} = 3\hat{i} \\ \vec{C} = -2\hat{i} \\ \vec{D} = 3/2\hat{i} \\ \vec{E} = 9\hat{i} \end{array} \right\}$$

$$D = \begin{bmatrix} 2 \\ 3 \\ -2/3 \end{bmatrix}_{3 \times 1}$$

$$\rho(D) = 1$$



$$1\text{-D span}$$

$$F = [2 \quad 3 \quad -2 \quad \dots]_{1 \times 1000}$$

$$\rho(F) = 1$$

Q. Using a set of 2-dimensional vector, Try creating a 3-D span. You can use any number of 2-dimensional vector.

$$\hookrightarrow S = \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \dots \right\}$$

$$\left. \begin{array}{l} (a) \vec{A} = 2\hat{i} + 3\hat{j} \\ \vec{B} = 3\hat{i} + 2\hat{j} \\ \vec{C} = -2\hat{i} + 4\hat{j} \end{array} \right\} \rightarrow \underbrace{\text{2-dimensional vectors}} \Rightarrow \underbrace{\text{2-independent vectors}}$$

$\Downarrow$

$D = \begin{bmatrix} 2 & 3 & -2 \\ 3 & 2 & 4 \end{bmatrix}_{2 \times 3}$

$P(D) = 2$

2-D span

 $\begin{bmatrix} a & c & e & \dots \\ b & d & f & \dots \end{bmatrix}_{2 \times n}$

$\nearrow$
 $\text{max}^n \text{Rank} = 2$

* Meaning of $\mathbb{R}^1, \mathbb{R}^2, \mathbb{R}^3 \dots \mathbb{R}^n$:- Set of vectors : vector space

$\mathbb{R}^1$: set of ^{all} 1-Dimensional vectors whose co-efficients are real

$\mathbb{R}^2$: set of 2 - " " " " " " " "

$\mathbb{R}^3$: set of 3 - " " " " " " " "

$\mathbb{R}^n$: set of all n - Dimensional vector " " " " real.

$\mathbb{R}^1 = \{ [1], [3/2], [-5], [3], \dots \}$ $\rightarrow$ 1 Independent vectors

$\mathbb{R}^2 = \{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -3/2 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \end{bmatrix} \dots \}$ $\rightarrow$ 2-Independent vectors

$\mathbb{R}^3 = \{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} \dots \}$ $\rightarrow$ 3-Independent vectors =

vector space
Q. A set of vector is defined as

$$V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R} \right\}$$

3-D span

(a) write some elements of this set.

(b) find the number of independent vectors from this set.

all 3-D vectors

$$\hookrightarrow V = \mathbb{R}^3 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \dots \right\} \xrightarrow{\text{all possible 3-D vectors}}$$

$$D = \begin{bmatrix} 1 & 2 & -1 & 2 & \alpha & a \\ 2 & 3 & -2 & 4 & \beta & b \\ 3 & 1 & 3 & 1 & \gamma & c \end{bmatrix} \dots \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} 3 \times n$$

$$\rho(D) = 3$$

Q. $\hookrightarrow$ 3-dimensional vector space $\rightarrow$ 2 Ind. vectors

Let $V = \{(x, y, z) \in \mathbb{R}^3 \mid \underline{y = 2x}\}$. Find the span of V .

$$V = \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 7 \end{bmatrix} \right\}$$

$\hookrightarrow$ Few 3-D vectors
✓ but not ALL

$$D = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ -1 & -2 & 3 \\ 3 & 6 & 7 \end{bmatrix}_{4 \times 3} \xrightarrow[\substack{R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - 3R_1}]{\substack{R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - 10R_2}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 10 \end{bmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - 10R_2}]{\substack{R_3 \rightarrow R_3 - 2R_2}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\rho(\omega) = 2$ $\hookrightarrow$ 2 Independent vectors^{**}

$$\vec{A} = x\hat{i} + y\hat{j} + z\hat{k} \quad \leadsto \quad \boxed{\hat{i} \perp \hat{j} \perp \hat{k}}$$

$$y = 2x$$

$$\vec{A} = x\hat{i} + 2x\hat{j} + z\hat{k}$$

$$\vec{A} = \frac{\sqrt{5}x}{\sqrt{5}} [\hat{i} + 2\hat{j}] + z(\hat{k})$$

$$\boxed{\frac{\hat{i} + 2\hat{j}}{\sqrt{5}} \perp \hat{k}}$$

$$\boxed{\vec{A} = \sqrt{5}x\hat{i} + z\hat{k}}$$

2-Dimensional vector

Q. $V = \{(x \ y \ z) \in \mathbb{R}^3 \mid y = 2x, x+y+z=0\}$ span of V ? $\hookrightarrow$ 1-D span

$$\begin{aligned} \vec{A} &= x\hat{i} + y\hat{j} + z\hat{k} \\ &= x\hat{i} + 2x\hat{j} - 3x\hat{k} \\ &= \sqrt{14}x \frac{[\hat{i} + 2\hat{j} - 3\hat{k}]}{\sqrt{14}} \end{aligned}$$

$$\vec{A} = \sqrt{14}x \hat{j}$$

$$p(0) = 1 \quad \downarrow$$

1-D span

$$y = 2x$$

$$\begin{aligned} x + y + z &= 0 \\ 3x + z &= 0 \end{aligned}$$

$$z = -3x$$

NOT a SET OF ALL possible 3-D vectors, Few 3-D vectors ✓

$$V = \left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix} \dots \right\}$$

$$D = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ -1 & -2 & 3 \\ 2 & 4 & -6 \end{bmatrix} \xrightarrow[R_4 \rightarrow R_4 - 2R_1]{R_3 \rightarrow R_3 + R_1} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ n -independent vectors $\Rightarrow n$ -D span

↳ n -dimensional vector space $\Rightarrow \max^m$ n -independent vectors $\Rightarrow n$ -D span

↳ Vector space of ALL possible n -dimensional vectors: $\mathbb{R}^n$

$\Downarrow$
 n independent vector $\Rightarrow n$ -D span

↳ Vector space of few n -dimensional vectors: $\mathbb{R}^n$ | condition

$\Downarrow$
 k independent vector $\Rightarrow k$ -D span

$$k < n$$

→ 4-dimensional vector space $\Rightarrow$ 2- indep. vect.

Q. $V = \{ (a \ b \ c \ d) \in \mathbb{R}^4 \mid \underline{a+b+c=0}, \underline{d=3a} \}$ span of V ?

↳
$$\begin{aligned} c &= -(b+a) \\ d &= 3a \end{aligned}$$

$$\vec{A} = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 3 \end{bmatrix}$$

$$\vec{B} = \begin{bmatrix} 2 \\ -6 \\ 4 \\ 6 \end{bmatrix}$$

$\left. \begin{array}{l} a \text{ free} \\ b \text{ free} \\ c \text{ dependent} \\ d \text{ dependent} \end{array} \right\}$; $\left. \begin{array}{l} a \text{ free} \\ b \text{ dependent} \\ c \text{ free} \\ d \text{ dependent} \end{array} \right\}$

2 independent variables

Span = 2-D

↓
2-independent vectors
=

* Linear span:-

Q. check if vector $[2 \ 4]$ belongs to the linear span of vectors $[1 \ 3]$ and $[-1 \ 2]$? $\rightarrow$ YES

OR

check if vector $[2 \ 4]$ is spanned by vectors $[1 \ 3]$ & $[-1 \ 2]$?

OR

check if vectors $[1 \ 3]$ & $[-1 \ 2]$ spans vector $[2 \ 4]$?

- ↳ $\vec{C} = \alpha \vec{A} + \beta \vec{B}$ $\rightarrow$ using the linear combⁿ of $\vec{A}$ and $\vec{B}$, you can create vector $\vec{C}$
- ↳ α, β simultaneously, $\vec{C}$ belongs to the linear span of $\vec{A}$ and $\vec{B}$.
- ↳ can be zero $\rightarrow \vec{C}$ is spanned by $\vec{A}$ and $\vec{B}$.
- ↳ $\vec{A}$ and $\vec{B}$ spans $\vec{C}$.

$$\alpha \underbrace{\begin{bmatrix} 1 & 3 \end{bmatrix}}_{\hat{i} + 3\hat{j}} + \beta \underbrace{\begin{bmatrix} -1 & 2 \end{bmatrix}}_{-\hat{i} + 2\hat{j}} = \underbrace{\begin{bmatrix} 2 & 4 \end{bmatrix}}_{2\hat{i} + 4\hat{j}}$$

$$\begin{bmatrix} \alpha & 3\alpha \end{bmatrix} + \begin{bmatrix} -\beta & 2\beta \end{bmatrix} = \begin{bmatrix} 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} \alpha - \beta & 3\alpha + 2\beta \end{bmatrix} = \begin{bmatrix} 2 & 4 \end{bmatrix}$$

$$\begin{aligned} \beta &= -0.4 \\ \alpha &= 1.6 \end{aligned}$$

$$\begin{aligned} \alpha - \beta &= 2 \\ 3\alpha + 2\beta &= 4 \end{aligned}$$

↪ If there exists any α and β which satisfy this linear system?

⇓

$$A:B = \begin{bmatrix} 1 & -1 & : & 2 \\ 3 & 2 & : & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & : & 2 \\ 0 & 5 & : & -2 \end{bmatrix}$$

consistent solⁿ
[Unique, Infinite solⁿ]

$$\rho(A) = \rho(A:B) \rightarrow \alpha, \beta \text{ exist}$$

N.B. -

$$\underline{[1 \ 3] \text{ and } [-1 \ 2]}$$

$$\begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} \Rightarrow \text{Rank} = 2$$

$\Downarrow$
2-independent vectors

$\Downarrow$
2-D span

$\Downarrow$
Any 2-Dimensional vector

$$\{ \dots [0 \ 0], [2 \ 4], [1 \ 3], [2, 4] \dots \}$$

set of all possible 2-D dimensional vectors = $\mathbb{R}^2$

$\hookrightarrow [1 \ 3] \text{ and } [-1 \ 2] \text{ spans } \underline{\mathbb{R}^2}$

Q. check if vectors $[1 \ 3]$ & $[-1 \ 2]$ spans $\mathbb{R}^2$?

↳

↕
2-independent vectors

↳ 2-D span

↕
2-D span

YES, $[1 \ 3]; [-1 \ 2]$
span $\mathbb{R}^2$

Q. check if vectors $[1 \ 3]$ and $[2 \ 6]$ spans $\mathbb{R}^2$?

↳

$[1 \ 3] \ [2 \ 6]$
doesn't span $\mathbb{R}^2$.

$$B = 2A$$

only 1-independent vector

↳ 1-D span

↳ 2-D span

NO

Q. which of the following set of vectors span $\mathbb{R}^2$? 2-dimⁿ

✓ (a) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

2-Ind.

✗ (b) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ 1 ind. x

$$\begin{bmatrix} 1 & 0.5 \\ 2 & 1 \end{bmatrix}$$

2-dimⁿ, 2 Independent vectors

✓ (c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ 2-Ind.

$$\begin{bmatrix} 1 & 0.5 & 3 \\ 2 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.5 & 3 \\ 0 & 0 & -7 \end{bmatrix}$$

Rank = 2

✗ (d) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 8 \\ 12 \end{bmatrix}, \begin{bmatrix} 16 \\ 24 \end{bmatrix}$ 1 ind. x

$$\begin{bmatrix} 2 & 4 & 8 & 16 \\ 3 & 6 & 12 & 24 \end{bmatrix} \xrightarrow{R_2 \rightarrow 2R_2 - 3R_1} \begin{bmatrix} 2 & 4 & 8 & 16 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank = 1 =

Language:-

↳ $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ Forms the basis of vector space $\mathbb{R}^2$
spans

↳ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ form the basis of vector space $\mathbb{R}^2$
spans

2-dimⁿ, 2-ind. $\Rightarrow$ Span $\mathbb{R}^2$

↳ $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \Rightarrow$ doesn't $\mathbb{R}^2$

dep.

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ doesn't form basis of vector space $\mathbb{R}^2$.

$\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ forms the basis of $\mathbb{R}^2$.

Q. which of the following set of vectors spans the space

$$V = \{ (x \ y \ z) \in \mathbb{R}^3 \mid y = 2x \}$$

$$\mathbb{R}^3 \mid \underline{y = 2x}$$

3-dimensional vectors

2-Independent vectors

(a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$

2-ind. vector

~~(b)~~ $\begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}$ $y \neq 2x$

2-ind. vectors $\neq$ condition

* Requirement: 3-dimensional, at least 2-ind. vectors satisfying $y = 2x$

~~(c)~~ $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \times 2 \rightarrow \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} \times 3 \rightarrow \begin{bmatrix} 6 \\ 12 \\ 24 \end{bmatrix}$

1-ind. vector

(d) $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \times 2 \rightarrow \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 7 \end{bmatrix}$

2. indep. vectors

(e) $\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}^x, \begin{bmatrix} 4 \\ 8 \\ -3 \end{bmatrix}$

2-Independent, 3-D
 $y = 2x$

2-ind. vector, 3-Dim.
 $y = 2x$

~~(f)~~ $\begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \end{bmatrix}$

2-dim.

$$R^3|_{y=2x} = \alpha \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} + \underline{0} \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix} + \gamma \begin{bmatrix} 4 \\ 8 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} \end{bmatrix} + 0 \begin{bmatrix} \end{bmatrix} + 0 \begin{bmatrix} \end{bmatrix}$$

Q. which of the following set of vectors spans the space

$$V = \{(x \ y \ z) \in \mathbb{R}^3 \mid y=2x, x+y+z=0\}$$

$$\mathbb{R}^3 \mid y=2x$$

$$x+y+z=0$$

3-dimensional vector

$$z = -x - y = -3x$$

(a) $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ -12 \end{bmatrix}, \begin{bmatrix} 6 \\ 12 \\ -18 \end{bmatrix}$

(c) $\begin{bmatrix} 3 \\ 6 \\ -9 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$

$$x+y+z \neq 0$$

3-dimensional & Independent vector: $y=2x$ and $x+y+z=0$

~~(d)~~ $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 7 \end{bmatrix}$

$$x+y+z=0$$

1-independent vector

Which of the following statements are true?

- ☒ (A) A set of n linearly independent vectors in $\mathbb{R}^n$ spans $\mathbb{R}^n$.
- ☒ (B) Any set containing fewer than n linearly independent vectors in $\mathbb{R}^n$ cannot span $\mathbb{R}^n$.
- ☒ (C) Any set containing more than n vectors in $\mathbb{R}^n$ is necessarily linearly dependent.
- ☒ (D) A set of vectors in $\mathbb{R}^n$ spans $\mathbb{R}^n$ if and only if it contains at least n linearly independent vectors.

↳ (a) To span $\mathbb{R}^n$

n -D span $\Rightarrow$ n -linearly independ. vectors

(b) ↗

(c) ✓

Q.

Let

$$\mathbf{v}_1 = (2, -1, 3), \quad \mathbf{v}_2 = (1, 4, -2), \quad \mathbf{v}_3 = (\lambda, 5, 1)$$

be vectors in $\mathbb{R}^3$. Which of the following statements are true?

- ☒ (A) The vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ form a basis of $\mathbb{R}^3$ for all $\lambda \neq \frac{22}{5}$.
☒ (B) The vectors are linearly dependent when $\lambda = \frac{22}{5}$.
☒ (C) The dimension of the span of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is 3 for all values of λ .
☒ (D) When $\lambda = \frac{22}{5}$, the vectors span a plane in $\mathbb{R}^3$.

→ 3-D span for $\lambda \neq \frac{22}{5}$

4 $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ form a basis of $\mathbb{R}^3$
spans

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ spans $\mathbb{R}^3$
↓
Independent of each other
↳ 3 independent vectors.

plane in $\mathbb{R}^3$

Span = 2D

3-dimensional vectors giving

2-D span

2 L.I.D. vectors

$$D = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ \lambda & 5 & 1 \end{bmatrix}_{3 \times 3}$$

3 L.I.D. vectors $\Rightarrow$ 3 L.I.D. Rows

$\Downarrow$

$$\rho(D) = 3$$

$$|D| \neq 0$$

$$2(14) + 1(1+2\lambda) + 3(5-4\lambda) \neq 0$$

$$28 + 2\lambda + 1 + 15 - 12\lambda \neq 0$$

$$10\lambda \neq 44$$

$$\boxed{\lambda \neq 22/5}$$

$\rightarrow \lambda = 22/5$; $\rho(D) = 2 \hookrightarrow$ 2 L.I.D. $\Rightarrow$ span = 2-D [plane]

↳ set $S_1 = \{1, 2, 3\}$ ✓

set $S_2 = \{2, 3\}$ ✓

S_2 is a subset of S_1

(a) $S_2 = \{2, 3\} \rightarrow$ set

(b) $S_1 = \{1, 2, 3\} \rightarrow$ set

$S_2 = \{2, 3\} \rightarrow$ subset

↳ (a) $V: \mathbb{R}^3$
↓
vector space

(b) $V: \mathbb{R}^3 | x=y=z$
↓
vector space

(c) $V_1: \mathbb{R}^3 \hookrightarrow$ vector space

$V_2: \mathbb{R}^3 | x=y=z \hookrightarrow$ vector subspace

$$\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \dots \quad \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \dots$$

Q.

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ be m non-zero vectors in a vector space. Consider the set**

$$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m, -\mathbf{x}_1, -\mathbf{x}_2, \dots, -\mathbf{x}_m\}$$

Let d denote the dimension of the subspace spanned by this set. Which of the following statements are true?

- (A) If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are mutually orthogonal, then $d = m$.
 (B) If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are linearly independent, then $d = m$.
 (C) If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are linearly dependent, then $d < m$.
 (D) The dimension d is equal to the dimension of $\text{Span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$.

$$\text{span}[v_2] = \text{span}[v]$$

$$v_2 = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$$

$$v = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m, -\mathbf{x}_1, -\mathbf{x}_2, \dots, -\mathbf{x}_m\}$$

→ independent

↳ $v = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m, -\mathbf{x}_1, -\mathbf{x}_2, \dots, -\mathbf{x}_m\}$ Total $2m$ vectors =

$d = \text{span of vector space } v$

$d \leq m$

$\rho(D) \leq m$

h.o. of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$

max m
I.O. v

Q.

Let A be an $n \times n$ matrix and $\mathbf{x}$ be a non-zero column vector in $\mathbb{R}^n$. Suppose that

$$\underline{A\mathbf{x} = \mathbf{0}}.$$

Which of the following statements are true?

(A) $\text{rank}(A) < n$.

(B) The matrix A is not invertible.

(C) The homogeneous system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution.

(D) The nullity of A is at least 1.

↳

$$A\mathbf{x} = \mathbf{0}$$

$$\mathbf{x} = \text{solution vector} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{n \times 1}$$

Unique solⁿ

$$\underline{x_1 = x_2 = x_3 = 0}$$

$$\rho(A) = n$$

Infinite solⁿ [Non-Trivial]

$$\underline{\underline{\rho(A) < n}}$$

$$|A| = 0$$

$$n(A) = 3 - 2 = 1$$

$$ax_1 + bx_2 + cx_3 = 0$$

$$dx_1 + ex_2 + fx_3 = 0$$

$$gx_1 + hx_2 + ix_3 = 0$$

$$\underline{AX=0}$$

Unique

Infinite

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

2 solⁿ are possible.

$\Downarrow$
Infinite solⁿ are possible.

* Basic Question:-

Q. Are $\overset{\vec{A}}{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}$ and $\overset{\vec{B}}{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$ independent of each other? $\rightarrow$ NO

$\vec{B} = 0\vec{A}$

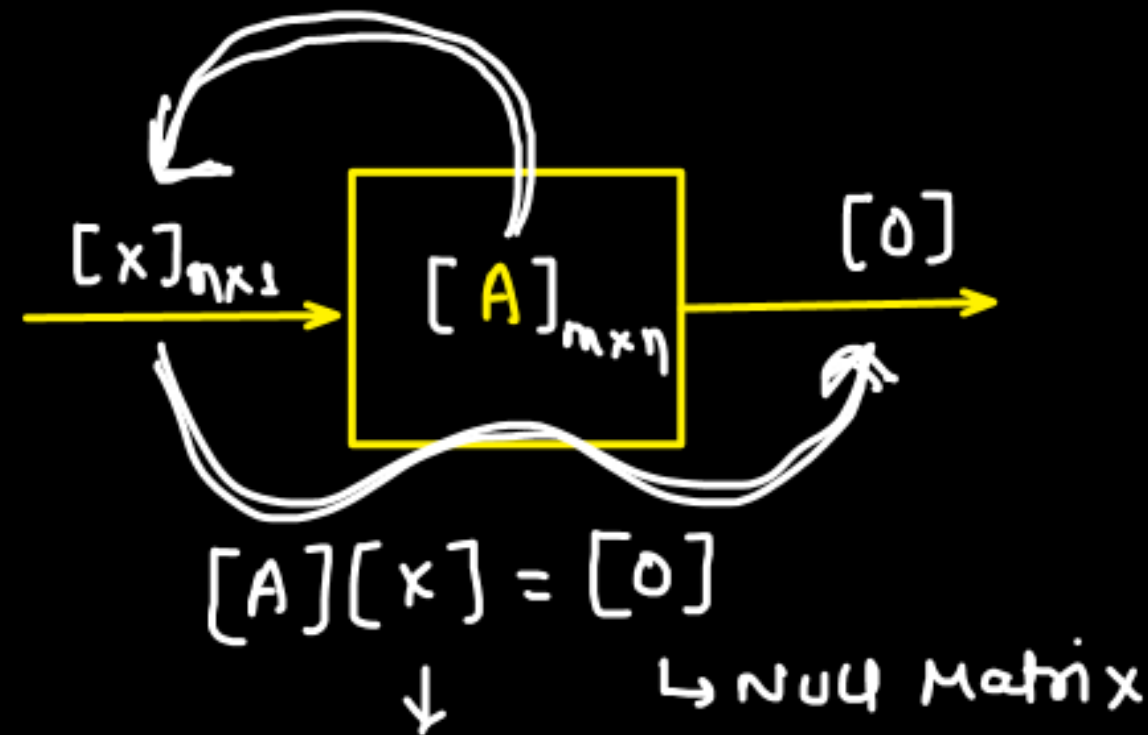
$\hookrightarrow C = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\rho(C) = 1 \rightsquigarrow$ 1 L.I.D. vector
1 L.D. vector

Q. Are $\begin{bmatrix} 1 & 2 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \end{bmatrix}$ independent of each other? $\rightarrow$ NO

$\hookrightarrow C = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ $\rho(C) = 1 \rightsquigarrow$ 1 L.I.D.
1 L.D. $\vec{B} = 0\vec{A}$

* A Null vector can never be independent.

* Understanding the Null space of a Matrix A:-



We take a matrix A and multiply it by a vector x . If the result is the zero vector, then x belongs to the null space of A . Hence, the null space is the set of all vectors that are mapped to zero by the matrix A .

Q. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$; show that $\underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{v_1}$ and $\underbrace{\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}}_{v_2}$ belongs to the null space of A.

→ $AX = ?$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AV_1 = [0]$$

$$AV_2 = [0]$$

$$AV_\alpha = [0] ?$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} -6 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Downarrow$

$$3 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AB = 0$$

$$A[3B] = 0$$

$$3AB = 0$$

$$\hookrightarrow AB = 0$$

If v_1 belongs to the Null space then αv_1 also belongs to Null space.
 If v_2 " " " " " then βv_2 " " " " "

$$Av_1 = 0 \Rightarrow A[\alpha v_1] = 0 \Rightarrow \alpha Av_1 = 0$$

$$Av_2 = 0 \Rightarrow A[\beta v_2] = 0 \Rightarrow \beta Av_2 = 0$$

$$A[\alpha v_1] = 0 \text{ and } A[\beta v_2] = 0$$

$$A[\alpha v_1] + A[\beta v_2] = 0$$

$$A[\alpha v_1 + \beta v_2] = 0$$

$$A[\alpha v_1 + \beta v_2] = 0$$

$\Downarrow$

$\alpha v_1 + \beta v_2$ also belong to the Null space.

Null space = $\left\{ \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}}_{v_2}, \overset{\nearrow 2v_2}{\begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix}}, \overset{\nearrow 3v_1}{\begin{bmatrix} -6 \\ 2 \\ 0 \end{bmatrix}}, \overset{\nearrow 3v_1 + 4v_2}{\begin{bmatrix} -18 \\ 3 \\ 4 \end{bmatrix}}, \dots, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

Set of vector $\downarrow$

∞ vectors v_1 and v_2 are h.i.d.

$$\text{Null space of } A = \left\{ \underbrace{(v_1, v_2)}_{2 \text{ h.i.d. vectors}}, 2v_2, 3v_1, 3v_1 + 4v_2, 2v_2 + 3v_1, \dots, [0] \right\}$$

2 h.i.d. vectors

Null space $\in \mathbb{R}^3$ | condition $\Rightarrow$ 2-D span

$$AX=0$$

$x \in$ Null space of A

Unique solⁿ

$$\rho(A) = n$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\underline{\eta[A] = 0}$$

$$\text{Null space} = \left\{ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$$

Infinite solⁿ

$$\rho(A) < n$$

↳ No. of linearly independent solⁿ of $AX=0$
 $\Downarrow$
 No. of Free variable
 $\Downarrow$
 nullity of $[A]$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3} ; \text{ Find out Null space} = ?$$

↳ Find out the span of Null Space

↓

How many L.I.D. you will have in the Null space of $[A]$

↓

$$\rho(A) = 1, \quad \eta(A) = 3 - 1 = 2 =$$

$$\hookrightarrow x: \text{Null space} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$Ax = 0$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 + 2x_2 + 3x_3 &= 0 & \text{--- (1)} \\ 2x_1 + 4x_2 + 6x_3 &= 0 \Rightarrow x_1 + 2x_2 + 3x_3 = 0 & \text{--- (2)} \\ 3x_1 + 6x_2 + 9x_3 &= 0 \Rightarrow x_1 + 2x_2 + 3x_3 = 0 & \text{--- (3)} \end{aligned}$$

$$x_1 + 2x_2 + 3x_3 = 0$$

3 variable
1 eqn

$$x_3 = c_1 \quad \checkmark$$

$$x_2 = c_2 \quad \checkmark$$

$$x_1 = -(2x_2 + 3x_3)$$

$$= -2c_2 - 3c_1 \quad \checkmark$$

2 free variables

2-independent vectors

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3c_1 - 2c_2 \\ c_2 \\ c_1 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}}_{v_2} + c_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{v_1}$$

$$\text{vector space } X = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 2x_2 + 3x_3 = 0 \right\} \Rightarrow \text{span} = 2-0$$

$$AX=0$$



$\eta(A)$ = nullity of A = no. of free variables

OR

no. of Independent solⁿ of $AX=0$

OR

no. of Independent vector in the
null space x of matrix $[A]$

OR

Dimension of span of null space of
matrix $[A]$

OR

Dimension of null space = $\eta(A)$

Q.

$$A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}_{2 \times 2}$$

; Find Null space of A.

$$\hookrightarrow \rho(A) = 1, \eta(A) = 2 - 1 = 1 \hookrightarrow \text{Span of Null-space} = 1 - 0$$

Dimension of Null space = 1

$$\hookrightarrow AX = 0$$

$$\begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$2x_1 + 4x_2 = 0 \quad \text{--- (1)}$$

$$-x_1 - 2x_2 = 0 \quad \text{--- (2)}$$

$$\rightarrow x_1 + 2x_2 = 0 \quad \text{--- (1)}$$

$$\rightarrow x_1 + 2x_2 = 0 \quad \text{--- (2)}$$

$$x_2 = c_1, x_1 = -2c_1$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2c_1 \\ c_1 \end{bmatrix} = c_1 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\underbrace{\text{Null space}}_{\substack{\downarrow \text{span} \\ 1-0}} = \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0.5 \end{bmatrix}, \begin{bmatrix} -3 \\ 1.5 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

1 L.I.D. vector

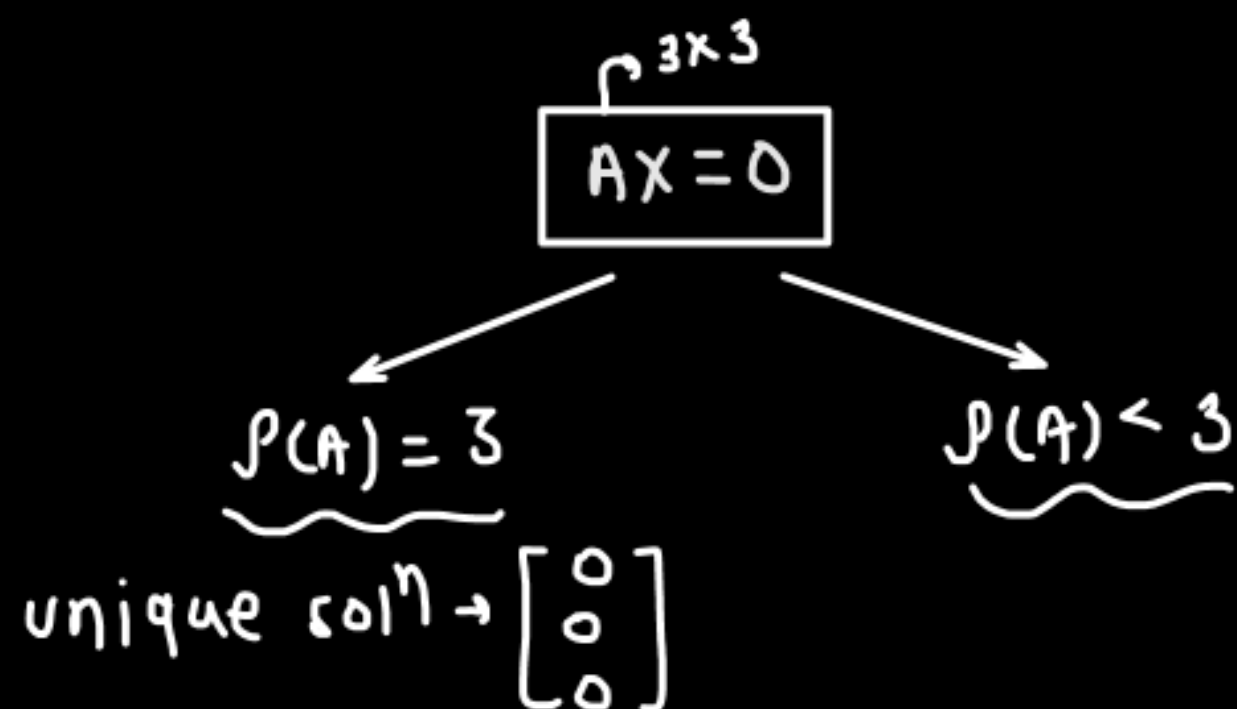
Q. $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & 1 \\ 2 & 4 & 1 \end{bmatrix}$; Null space of A?

$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & 1 \\ 2 & 4 & 1 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - 3R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & -1 \end{bmatrix}$

$$\rho(A) = 3$$

$$\eta(A) = 0$$

↳ R.E. ↙



$$\text{Null space} = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

↑

Q. $A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 4 & 6 & 2 & 1 \\ 6 & 9 & 12 & 1 \end{bmatrix}_{3 \times 4}$ Find the dimension of
Null space of $A = \underline{1}$

↳ $\begin{bmatrix} 2 & 3 & 4 & 1 \\ 4 & 6 & 2 & 1 \\ 6 & 9 & 12 & 1 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 0 & -6 & -1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

$\rho(A) = 3$
 $\eta(A) = 4 - 3 = 1$

In the Null space of $[A]$, there will be 1 L.I.D. vector
 Dimension of span of Null space of $[A]$ will be 1-0.

$$AX=0$$

$$\begin{bmatrix} 2 & 3 & 4 & 1 \\ 4 & 6 & 2 & 1 \\ 6 & 9 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \equiv \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 0 & -6 & -1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 3x_2 + 4x_3 + x_4 = 0 \quad \text{--- (1)}$$

$$-6x_3 - x_4 = 0 \rightarrow \boxed{x_3 = 0}$$

$$-2x_4 = 0 \rightarrow \boxed{x_4 = 0}$$

$$2x_1 + 3x_2 = 0$$

$$x_1 = c_1 \quad x_2 = -\frac{2c_1}{3}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} c_1 \\ -2c_1/3 \\ 0 \\ 0 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} 1 \\ -2/3 \\ 0 \\ 0 \end{bmatrix}}_{\text{L.I.O.}}$$

* Revision Question:-

Q. Multiply A and B ; $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$\hookrightarrow |A| = 0$ $|B| = 0$

$\hookrightarrow AB = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$
 $\neq [0]$

Q. Multiply A and B ; $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$

$\hookrightarrow |A| = 0$ $\hookrightarrow |B| = 0$

$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = [0]$

$[A] \neq 0$	$[B] \neq 0$
$[A][B] = [0]$	

* Discussion on $[A][B]=0$

↙ Considering $[A]$ and $[B]$ are square Matrices.

$$\text{Need } [A][B]=0$$

✓ Case-1:- $[A]$ and $[B]$ both are Null Matrix.

$$[A]=[0], [B]=[0] \quad ; \quad [A][B]=[0][0]=[0]$$

✓ Case-2:- $[A]=0$, $[B] \neq 0$

$$[A][B]=[0][B]=[0]$$

✓ Case-3:- $[A] \neq 0$, $[B]=0$

$$[A][B]=[A][0]=[0]$$

Case-4:- $[A]$ and $[B]$ both are non-Null Matrices.

$$\hookrightarrow [A] \neq 0, [B] \neq 0 \quad \downarrow \Rightarrow \text{Given}$$

$$[A][B] = AB = 0$$

$$\hookrightarrow AB = 0$$

* pre multiplication with A^{-1} ← This step shouldn't be true

$$A^{-1}AB = A^{-1}[0]$$

$$IB = [0]$$

$$\times \times \boxed{B = [0]} \leftarrow$$

$\hookrightarrow B$ is a null Matrix $\downarrow$

NOT Desired.

$$[B] \neq [0]$$

Assumption: A^{-1} exists

$$\downarrow \\ \underline{\underline{|A| \neq 0}}$$

Shouldn't happen ←

A^{-1} shouldn't exist

$$|A| = 0$$

$$\underline{[A] \neq [0]}, [B] \neq 0 \text{ But still } [A][B] = [0]$$

↳ post multiplication with B^{-1}

$$[A]B B^{-1} = [0] \cdot B^{-1}$$

$$\boxed{[A] = [0]}$$

↳ NOT desired

shouldn't be possible

$$\Downarrow$$

$$\boxed{|B| = 0}$$

↳ Given ; $[A] \neq 0, [B] \neq 0$ and $[A][B] = [0]$

$$\Downarrow$$

$$\underline{|A| = 0 \text{ and } |B| = 0}$$

A and B both should be non-invertible.

MSQ
Q.

Let A and B be non-zero square matrices of the same order. Which of the following statements are true?

☒ (A) If $AB = 0$, then $\det(A) = 0$ and $\det(B) = 0$.

☐ (B) If $\det(A) = 0$ and $\det(B) = 0$, then $AB = 0$.

☐ (C) If $AB = 0$, then $\det(A) = 0$ or $\det(B) = 0$.

☐ (D) If $AB = 0$, then exactly one of $\det(A)$ or $\det(B)$ is zero.

↳ $[A] \neq 0, [B] \neq 0, [A][B] = [0] \Rightarrow |A| \neq 0 \text{ and } |B| \neq 0$

Q.

Let A be a non-zero matrix of order $m \times n$ and B be a non-zero column vector of order $n \times 1$, where $m > n$. Given that

$$AB = 0,$$

$$[B] \neq [0] \rightarrow \rho(A) \neq n$$

which of the following statements about $\text{rank}(A)$ are true?

~~(A)~~ $\text{rank}(A) > m$

(B) $\text{rank}(A) \leq n$

~~(C)~~ $\text{rank}(A) > n$ and $\text{rank}(A) < m$

(D) $\text{rank}(A) < n$

$$B = X$$

↳

$$[A]_{m \times n}, m > n$$

$$\rho(A)_{\max} = \min[m, n] \\ = n$$

$$\rho(A) \leq n \quad \rho(A) = n ?$$

$$AB = 0, \quad \underline{AX = 0}$$

$$\hookrightarrow \underline{\rho(A) = n} \quad \text{unique sol}^n \\ X = B = \begin{bmatrix} 0 \\ 0 \\ \vdots \end{bmatrix}$$

$$\rho(A) < n \\ \text{u}$$

Q.

Let $c_1, c_2, \dots, c_n$ be scalars, not all zero, such that

$$\sum_{i=1}^n c_i \mathbf{a}_i = \mathbf{0},$$

where $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$ are column vectors. Let

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]_{n \times n}, \quad \mathbf{B} = \sum_{i=1}^n \mathbf{a}_i.$$

Consider the system of linear equations

$$A\mathbf{x} = \mathbf{B}.$$

↗ Constant Number

$c_1, c_2, \dots, c_n$: scalar

$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \\ \vdots \end{bmatrix}_{n \times 1} \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 3 \\ \vdots \end{bmatrix}_{n \times 1}$$

Then the system has:

~~(A)~~ A unique solution

(B) No solution

✓ (C) Infinitely many solutions

(D) None of the above

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + c_3 \mathbf{a}_3 + \dots + c_n \mathbf{a}_n = \mathbf{0}$$

$$\lambda_1 \vec{A}_1 + \lambda_2 \vec{A}_2 + \lambda_3 \vec{A}_3 + \dots = \mathbf{0} \rightarrow \vec{A}_1, \vec{A}_2, \dots, \vec{A}_n \text{ are l.i.d.}$$

$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are forming a set of l.i.d. vectors.

$$[A] = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \dots \ \mathbf{a}_n]_{n \times n} \quad \leadsto \boxed{\rho(A) < n}$$

$a_1, a_2, \dots, a_n$ are set of L.D. vectors.

At least one vector is L.D.

$$\boxed{\rho(A) < n}$$

$$\begin{array}{c} AX = B \\ \swarrow \quad \downarrow \quad \searrow \\ \underbrace{\rho(A) \neq \rho(A:B)} \quad \rho(A) < \rho(A:B) \quad \rho(A) = \rho(A:B) < n \\ \quad \quad \quad = n \end{array}$$

$$\rho(A) < n$$

$$B = a_1 + a_2 + a_3 + \dots + a_n \leftarrow$$

$$= \begin{bmatrix} \quad \end{bmatrix}_{n \times 1} + \begin{bmatrix} \quad \end{bmatrix}_{n \times 1} + \begin{bmatrix} \quad \end{bmatrix}_{n \times 1} + \dots \begin{bmatrix} \quad \end{bmatrix}_{n \times 1}$$

$$\underbrace{[B]}_{\downarrow} = \begin{bmatrix} \quad \end{bmatrix}_{n \times 1}$$

L.D. on

$a_1, a_2, \dots, a_n$

$$\begin{array}{c}
 \text{h.d.} \rightarrow a_1 \quad a_2 \quad \dots \quad a_n : \underbrace{a_1 + a_2 + \dots + a_n}_{\text{Dependent}} \rightarrow \text{h.d.} \\
 \left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] \\
 \underbrace{\hspace{10em}}_A \\
 \underbrace{\hspace{15em}}_{A:B}
 \end{array}$$

$(A:B) =$

Let a_1 is dependent

$$P(A) = \underline{n-1}$$

$$P(A:B) = (n+1) - 2 = \underline{n-1}$$

$$\underline{P(A) = P(A:B) = n-1 < n}$$

$\Downarrow$
Infinite solⁿ
=

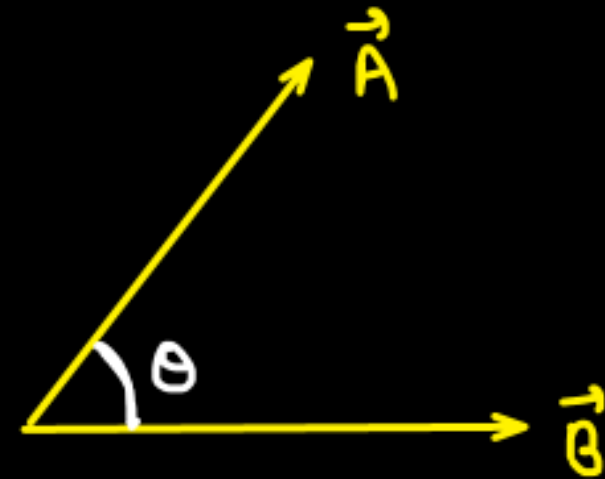
↳ Some More points on vectors:-

Dot product:- $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
 $\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\vec{A} \cdot \vec{B} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$= |\vec{A}| |\vec{B}| \cos \theta$$

$$\vec{A} \cdot \vec{B} = \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2} \cos \theta$$



Q.

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in $\mathbb{R}^2$ such that $\|\mathbf{u}\| = 2\|\mathbf{v}\|$. If the vector $\underline{\mathbf{w}} = \mathbf{u} + \alpha\mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$, then find the value of α .

↳

u and v are 2-D real vector

$$\|u\| = 2\|v\|$$

$$\boxed{v = \hat{i} + \hat{j}}$$

$$\|v\| = \sqrt{2}$$

$$u = -2(\hat{i} + \hat{j})$$

$$\|u\| = 2\sqrt{2}$$

$$\boxed{u = -2\hat{i} - 2\hat{j}}$$

$$w = u + \alpha v = -2\hat{i} - 2\hat{j} + \alpha\hat{i} + \alpha\hat{j}$$

$$\boxed{w = (\alpha - 2)\hat{i} + (\alpha - 2)\hat{j}}$$

w bisects the angle b/w u and v

Angle b/w $\vec{w}$ and $\vec{v}$ is 90°

$$\vec{w} \cdot \vec{v} = (w||v|\cos 90^\circ$$

$$\boxed{\vec{w} \cdot \vec{v} = 0}$$

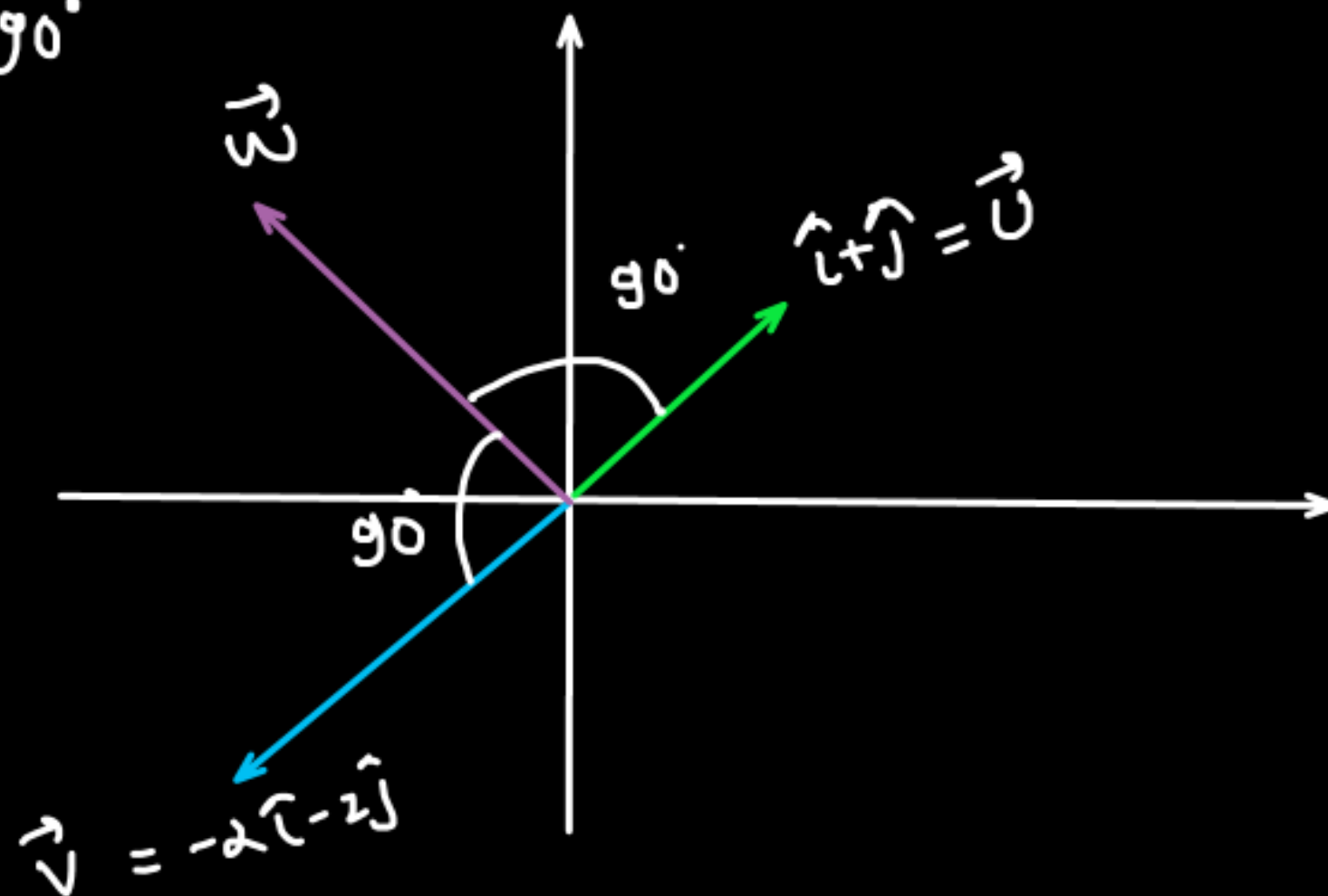
$$\vec{w} = (\alpha - 2)\hat{i} + (\alpha - 2)\hat{j}$$

$$\vec{v} = \hat{i} + \hat{j}$$

$$\vec{w} \cdot \vec{v} = 0$$

$$\vec{w} \cdot \vec{v} = (\alpha - 2) + (\alpha - 2) = 0$$

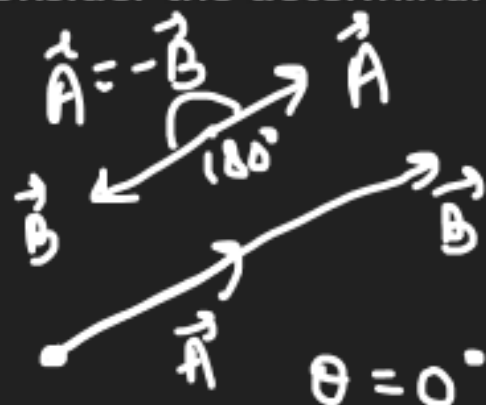
$$\hookrightarrow \boxed{\alpha = 2} \text{ Ans.}$$



Q.

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. Let $\mathbf{A} \cdot \mathbf{B}$ denote their dot product. Consider the determinant

$$\eta = \begin{vmatrix} \vec{\mathbf{A}} \cdot \vec{\mathbf{A}} & \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} \\ \vec{\mathbf{B}} \cdot \vec{\mathbf{A}} & \vec{\mathbf{B}} \cdot \vec{\mathbf{B}} \end{vmatrix}$$



$$\vec{\mathbf{A}} = \gamma \vec{\mathbf{B}}$$

$$\theta = 0^\circ$$

$$\vec{\mathbf{A}} \neq \gamma \vec{\mathbf{B}}; \theta \neq 0^\circ$$

$$|\mathbf{A}| = \alpha$$

$$|\mathbf{B}| = \beta$$

Which of the following statements are true?

- ☒ (A) It is zero if $\mathbf{A}$ and $\mathbf{B}$ are linearly dependent.
- ☒ (B) It is positive if $\mathbf{A}$ and $\mathbf{B}$ are linearly independent.
- ☒ (C) Both (A) and (B) are true.
- (D) None of the above.

$$\begin{aligned} \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} &= |\mathbf{A}| |\mathbf{B}| \cos \theta \\ &= \alpha \beta \cos \theta \end{aligned}$$

$$\vec{\mathbf{A}} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \quad ; \quad \vec{\mathbf{B}} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{A}} = a_1^2 + a_2^2 + a_3^2 = |\mathbf{A}|^2 \quad ; \quad \vec{\mathbf{B}} \cdot \vec{\mathbf{B}} = b_1^2 + b_2^2 + b_3^2 = |\mathbf{B}|^2$$

$$\begin{vmatrix} \alpha^2 & \alpha \beta \cos \theta \\ \alpha \beta \cos \theta & \beta^2 \end{vmatrix} = \alpha^2 \beta^2 - \alpha^2 \beta^2 \cos^2 \theta = \alpha^2 \beta^2 (1 - \cos^2 \theta)$$

$$\eta = \alpha^2 \beta^2 (1 - \cos^2 \theta) > 0$$

$$\rightarrow A^T A = I$$

Q.

Let A be a 2×2 real orthogonal matrix and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be a real column vector. Find the relation between $\|Ax\|$ and $\|x\|$. $\rightarrow \|Ax\| = \|x\|$

$$\rightarrow A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2}$$

orthogonal

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$$

$x = x_1 \hat{i} + x_2 \hat{j}$

$$Ax = \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix}_{2 \times 1}$$

$$Ax = (ax_1 + bx_2)\hat{i} + (cx_1 + dx_2)\hat{j}$$

$$a^2 + b^2 = 1$$

$$a^2 + c^2 = 1$$

$$c^2 + d^2 = 1$$

$$b^2 + d^2 = 1$$

$$ac + bd = 0$$

$$ab + cd = 0$$

$$\|Ax\| = \sqrt{x_1^2 + x_2^2}$$

$$\|x\| = \sqrt{x_1^2 + x_2^2}$$

$$\|Ax\| = \sqrt{(ax_1 + bx_2)^2 + (cx_1 + dx_2)^2}$$

$$= \sqrt{a^2 x_1^2 + b^2 x_2^2 + 2abx_1 x_2 + c^2 x_1^2 + d^2 x_2^2 + 2cdx_1 x_2}$$

$$= \sqrt{\underbrace{(a^2 + c^2)}_1 x_1^2 + \underbrace{(b^2 + d^2)}_1 x_2^2 + 2 \underbrace{(ab + cd)}_0 x_1 x_2}$$

$$\|B\|^2 = B^T B \quad \text{Tr}[B B^T] = \|B\|^2$$

Method 2:-

$$\|Ax\|^2 = (Ax)^T (Ax) = x^T \underbrace{A^T A}_{I} x$$

$$= x^T I x$$

$$= x^T x = \|x\|^2$$

$$\boxed{\|Ax\| = \|x\|}$$

$$(AB)^T = B^T \cdot A^T$$

$$\|Ax\|^2 = \|x\|^2$$

$$\hookrightarrow \boxed{\|Ax\| = \|x\|}$$

Let $P = [\mathbf{a}_1 \ \mathbf{a}_2]$ be a 2×2 matrix, where $\mathbf{a}_1, \mathbf{a}_2$ are column vectors. Suppose

$$P^{-1} = \begin{bmatrix} \mathbf{b}_1^T \\ \mathbf{b}_2^T \end{bmatrix},$$

where $\mathbf{b}_1^T$ and $\mathbf{b}_2^T$ are row vectors. Which of the following statements are true?

$$(A) \ \mathbf{b}_1^T \mathbf{a}_1 = 1$$

$$(B) \ \mathbf{b}_2^T \mathbf{a}_2 = 1$$

$$(C) \ \mathbf{b}_1^T \mathbf{a}_2 = 0$$

$$(D) \ \mathbf{b}_2^T \mathbf{a}_1 = 0$$

} all four correct

$$\mathbf{a}_1 = \begin{bmatrix} \quad \\ \quad \end{bmatrix}_{2 \times 1} \quad \mathbf{a}_2 = \begin{bmatrix} \quad \\ \quad \end{bmatrix}_{2 \times 1}$$

$$P = \begin{bmatrix} \sim & \sim \end{bmatrix}_{2 \times 2}$$

$$P^{-1} = \begin{bmatrix} \mathbf{b}_1^T \\ \mathbf{b}_2^T \end{bmatrix}_{2 \times 2} \quad \mathbf{b}_1^T = [\sim \ \sim]_{1 \times 2} \\ \mathbf{b}_2^T = [\sim \ \sim]_{1 \times 2}$$

$$\rho \rho^{-1} = \rho^{-1} \rho = \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$\rho = \begin{bmatrix} \downarrow & \downarrow \\ a_1 & a_2 \end{bmatrix}_{1 \times 2} \quad \rho^{-1} = \begin{bmatrix} b_1^T \\ b_2^T \end{bmatrix}_{2 \times 1}$$

$$\rho^{-1} = \begin{bmatrix} b_1^T \\ b_2^T \end{bmatrix}_{2 \times 1} ; \quad \rho = [a_1 \quad a_2]_{1 \times 2}$$

$$\rho^{-1} \rho = \begin{bmatrix} b_1^T a_1 & b_1^T a_2 \\ b_2^T a_1 & b_2^T a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b_1^T a_1 = b_2^T a_2 = 1, \quad b_1^T a_2 = b_2^T a_1 = 0$$

Method - 2

$$P = \begin{matrix} & \nearrow a_1 & \nearrow a_2 \\ \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \end{matrix}$$

$$|P| = 4 - 3$$

$$P^{-1} = \frac{1}{1} \times \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{matrix} \nearrow b_1^T \\ \nearrow b_2^T \end{matrix}$$

$$a_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$b_1^T = [2 \quad -1], \quad b_2^T = [-3 \quad 2]$$

$$b_1^T a_1 = 1, \quad b_2^T a_2 = -3 + 4 = 1$$

$$b_1^T a_2 = 2 - 2 = 0, \quad b_2^T a_1 = -6 + 6 = 0$$

Q.

Let $A = [a_{ij}]$, where $1 \leq i, j \leq n$, $n \geq 3$, and $a_{ij} = ij$. Find rank(A).

- (A) 0 (B) 1 (C) $n - 1$ (D) n

$$i \geq 1, j \leq n; n \geq 3$$

$$\underline{n=3}$$

if $n=3$

- (A) 0 (B) 1 (C) 2 (D) 3

$$[A] = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{bmatrix}$$

$$\rightarrow \underline{P(A) = 1}$$

$$[A] = [a_{ij}] \quad a_{ij} = ij$$

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

$$\downarrow \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

Q.

Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ be linearly independent vectors. Form the matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2]$ (i.e., $\mathbf{v}_1, \mathbf{v}_2$ as columns). Define $B = AA^T$. Find $\text{rank}(B)$.

↳

$$\text{Rank}[AA^T] = \text{Rank}[A] = 2$$

$$\hookrightarrow \rho(A) = \rho(A^{-1}) = \rho(A^0) = \rho(A^T) = \rho(AA^T) = \rho(AA^0) \downarrow$$

Indep.

$$A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3 \times 2}$$

$$\rho(A) = 2$$

Q.

Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ be linearly independent vectors, and $\mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^3$ be linearly independent vectors. Form the matrices

$A = [\mathbf{v}_1 \ \mathbf{v}_2]$ (columns) and

$B = \begin{bmatrix} \mathbf{v}_3^T \\ \mathbf{v}_4^T \end{bmatrix}$ (rows).

Let $C = AB$. Find $\text{rank}(C)$. = 2 ✓

$$2 \leq \rho(AB) \leq 2$$

$$\hookrightarrow A = [\mathbf{v}_1 \ \mathbf{v}_2]_{3 \times 2} \quad \rho(A) = 2$$

$$B = \begin{bmatrix} \mathbf{v}_3^T \\ \mathbf{v}_4^T \end{bmatrix}_{2 \times 3} \quad \rho(B) = 2$$

$$C = AB$$

$$\rho(C) = \rho(AB)$$

$$\hookrightarrow \rho(AB) \leq 2$$

$$\hookrightarrow \rho(AB) \leq \min[\rho(A), \rho(B)]$$

$$\rho(AB) \geq \rho(A) + \rho(B) - n$$

$$[A]_{m \times n} \quad [B]_{n \times p}$$

$$\rho(AB) \geq 2 + 2 - 2$$

$$\hookrightarrow \rho(AB) \geq 2$$

Q.

Let $M^4 = I$, where I is the identity matrix, and suppose $M \neq I$, $M^2 \neq I$, $M^3 \neq I$. For any natural number k , find M^{-1} :

- (A) M^{4k+1} (B) M^{4k+2} (C) M^{4k+3} (D) M^{4k}

↳

$$M^4 = I$$

$$M \cdot M^3 = I$$

$$\underbrace{M^{-1} M}_{I} M^3 = M^{-1} I$$

$$M^3 = M^{-1}$$

$$M^{-1} = M^3$$

$$= M^3 \cdot I$$

$$= M^3 \cdot (M^4)^k$$

$$M^{-1} = M^{4k+3}$$

$$M^4 = I$$

$$(M^4)^k = (I)^k$$

$$M^{4k} = I$$

$$I^2 = I^3 = \dots$$

$$= I^k = I$$

Q.

Let

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}, \quad B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}.$$

If $\text{rank}(A) = N$, find $\text{rank}(B)$.

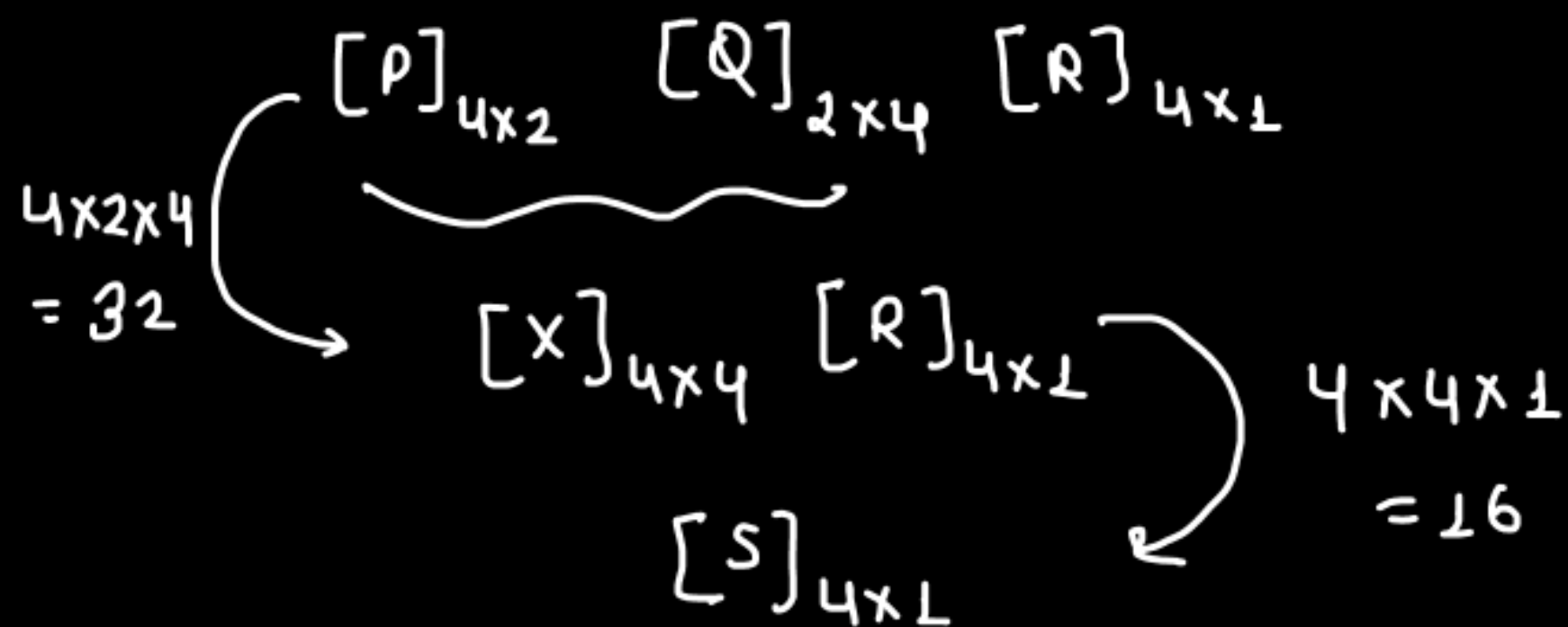
- (A) $N/2$ (B) $N - 1$ (C) N (D) $2N$

$$B = A A^T$$

$$\rho(A) = \rho(A A^T) = N$$

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

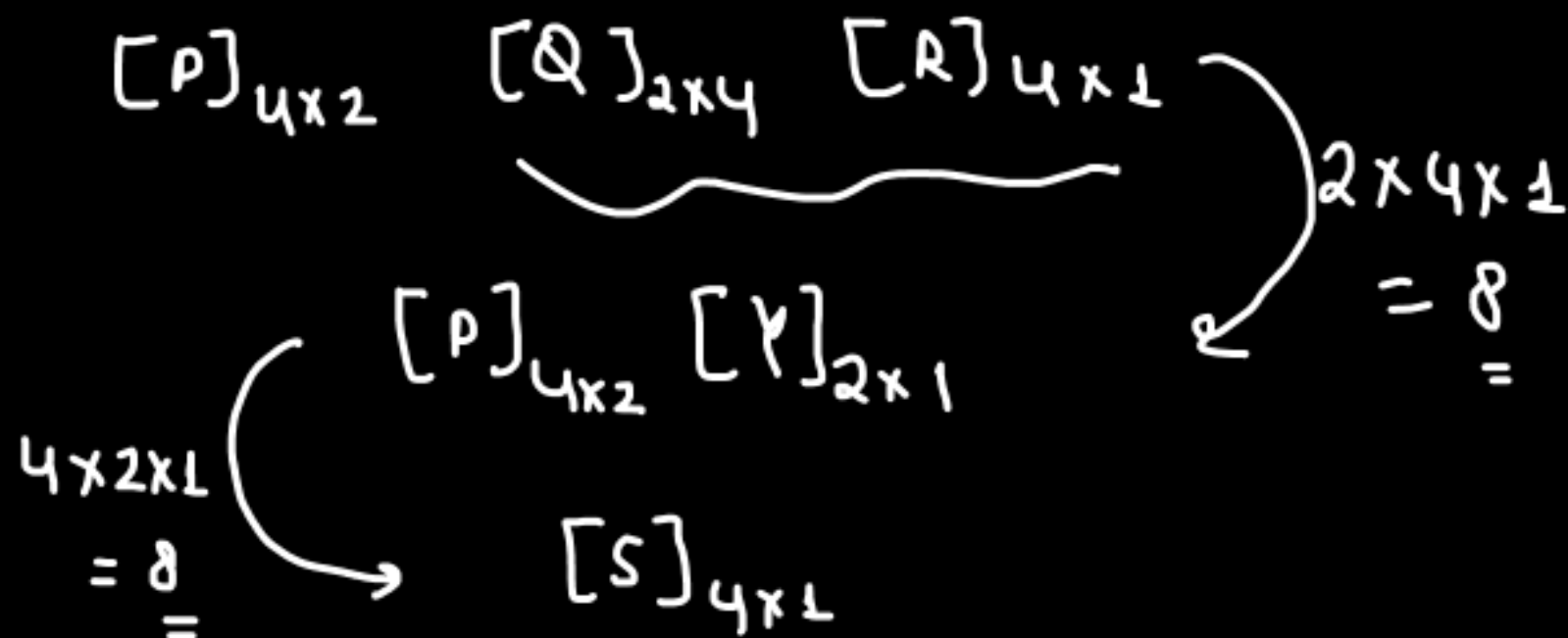
$$B = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \quad \begin{array}{l} \rho(A) = 2 = N \\ \rho(B) = 2 = N \end{array}$$



Multiplication = $32 + 16 = \boxed{48}$ $\times \times \times \times$

$PQR = (PQ)R = P(QR)$

Multiplication = 16



Q. Given $ABCD = I$; Find B^{-1} .

↳

$$ABCD = I$$

$$A^{-1}ABCD = A^{-1}I$$

$$BCD = A^{-1}$$

$$BCD \cdot D^{-1} = A^{-1}D^{-1}$$

$$BC = A^{-1}D^{-1}$$

$$BC C^{-1} = A^{-1}D^{-1}C^{-1}$$

$$\boxed{B = A^{-1}D^{-1}C^{-1}}$$

$$[PQR]^{-1} = R^{-1}Q^{-1}P^{-1}$$

$$B = A^{-1}D^{-1}C^{-1}$$

$$B^{-1} = [A^{-1}D^{-1}C^{-1}]^{-1}$$

$$= (C^{-1})^{-1}(D^{-1})^{-1}(A^{-1})^{-1}$$

$$\boxed{B^{-1} = CDA}$$

↳ Concept of Eigen value and Eigen vector:-



$$AX = \lambda X$$

$$\begin{array}{lcl} x_1 & \rightarrow [A] & \rightarrow \lambda_1 x_1 \\ x_2 & \rightarrow [A] & \rightarrow \lambda_2 x_2 \\ & \vdots & \vdots \\ x_3 & \rightarrow [A] & \rightarrow \lambda_3 x_3 \end{array}$$

Let A be a square matrix. If there exists a non-zero vector X and a scalar λ such that

$$AX = \lambda X,$$

then:

- X is called an eigenvector of the matrix A
- λ is called the eigenvalue of A corresponding to the eigenvector X

↳ X is the eigen vector corresponding to eigen value λ

Q.

Question:

Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

$$Ax = ?$$

$= \lambda x$ $\rightarrow$ Eigen vector

Which of the following vectors are eigenvectors of A ? For each valid eigenvector, also determine the corresponding eigenvalue.

(A) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^{x_1}$ (B) $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}^{x_2}$ (C) $\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}^{x_5}$

(E) $\begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$ (F) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^{x_6}$ (G) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}^{x_7}$ (H) $\begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$ (I) $\begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix}$

$x_1 \perp x_5$

$Ax_1 = \lambda_1 x_1$
 $Ax_2 = \lambda_1 x_2$
 $Ax_6 = \lambda_2 x_6$

$\rightarrow$ same E.V.

$$\textcircled{1} + \textcircled{2}$$

$$A[x_1 + x_2] = \lambda_1 [x_1 + x_2]$$

$$AY = \lambda_1 Y$$

$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$A[x_1 + x_2 + x_6] = \lambda_1 [x_1 + x_2] + \lambda_2 x_6$$

$$AY = \sim Y \rightarrow \text{NOT E. vector}$$

$$\hookrightarrow \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{x_1} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \underbrace{2}_{\lambda_1} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{x_1}$$

Eigen value

$$Ax_1 = \lambda_1 x_1$$

Eigen vector

$$\lambda_1 = 2$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{x_2} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \underbrace{2}_{\lambda_1} \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{x_2} \checkmark$$

$$= 2x_2 = \lambda_1 x_2$$

$$Ax_1 = \lambda_1 x_1$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \underbrace{\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}}_{x_3 = 5x_1} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} = 2 \underbrace{\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}}_{x_3 = 5x_1} \checkmark$$

$$= 2x_3 = \lambda_1 x_3$$

$$\begin{aligned} A[5x_1] &= 5Ax_1 \\ &= 5\lambda_1 x_1 \\ &= \lambda_1 [5x_1] \end{aligned}$$

$$A[5x_1] = \lambda_1 [5x_1]$$

$$\hookrightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_4 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = \lambda_1 x_4$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 15 \end{bmatrix} = \underbrace{3}_{\lambda_2} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}}_{x_5} = \lambda_2 x_5$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \underbrace{3}_{\lambda_2} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{x_6} = \lambda_2 x_6$$

$$\hookrightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \overset{x_7 \rightarrow \text{NOT E-vector}}{=} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \overset{x_7 \rightarrow \text{NO}}{=} 2 \begin{bmatrix} 1 \\ 1 \\ 3/2 \end{bmatrix}$$

$$Ax_1 \neq \lambda x_7$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} \overset{\text{NOT E-vector}}{=} \begin{bmatrix} 10 \\ 10 \\ 15 \end{bmatrix} = 2 \begin{bmatrix} 5 \\ 5 \\ 15/2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 12 \end{bmatrix} = 2 \begin{bmatrix} 5 \\ 5 \\ 6 \end{bmatrix}$$

NOT E.V.

↳ Eigen vectors corresponding to Eigen value $\lambda = 2$

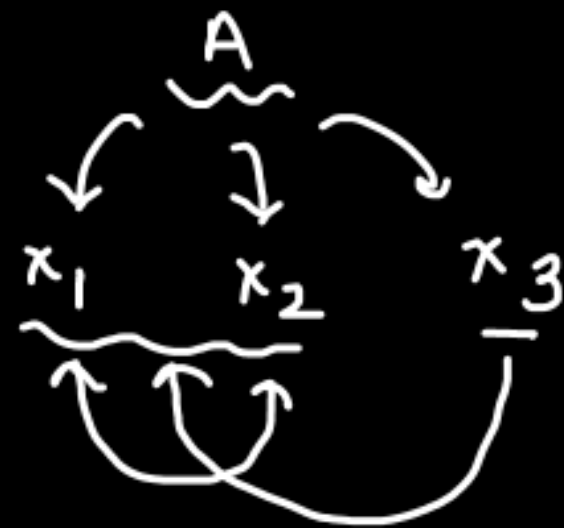
$$= \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \\ 0 \end{bmatrix} \dots \right\}$$

↳ Eigen vectors corresponding to Eigen value $\lambda = 3$

$$= \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix} \dots \right\}$$

$\lambda = 2$: 2 Independent Eigen vectors

$\lambda = 3$: 1 Independent Eigen vector



MCQ
Q.

Let

$$A = \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}.$$

Which of the following vectors is an eigenvector of A ?

~~(A)~~ $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ ~~(B)~~ $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ ☒ (C) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ ~~(D)~~ $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$\hookrightarrow \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -6 \\ 18 \end{bmatrix} \neq \sim \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -10 \\ 25 \end{bmatrix} \neq \sim \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -10 \\ 5 \end{bmatrix} = \underbrace{-5}_{\sim} \underbrace{\begin{bmatrix} 2 \\ -1 \end{bmatrix}}_{\sim}$$

$$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 8 \\ 2 \end{bmatrix}}_{\sim}$$

* How to Find Eigen value of a Matrix A :-

Let λ be a eigen value corresponding to Eigen vector x of $[A]_{n \times n}$

$$Ax = \lambda x$$

$$Ax - \lambda x = 0$$

λ : Scalar / Constant

A, x : Matrix ; $A - \lambda I$: Matrix

$$[A - \lambda I]x = 0$$

$$\Rightarrow Cx = 0$$

$$[C]_{n \times n}$$

NOT
Desired

$$\left\{ \begin{array}{l} \text{Unique sol}^n \\ \rho[A - \lambda I] = n \\ x = \begin{bmatrix} 0 \\ 0 \\ \vdots \end{bmatrix} \end{array} \right.$$

Infinite solⁿ

$$\left\{ \begin{array}{l} \rho[A - \lambda I] < n \\ x = \begin{bmatrix} \\ \\ \vdots \end{bmatrix} \end{array} \right. \text{Desired}$$

$\in \mathbb{R}, \mathbb{C}$

$$|A - \lambda I| = 0$$

Q.

Question:

Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

3×3 matrix

- (A) Find the characteristic polynomial and hence the characteristic equation of A .
- (B) Determine the eigenvalues of A .
- (C) Find the eigenvectors corresponding to each eigenvalue of A .
- (D) For each eigenvalue, determine its ~~algebraic~~ (arithmetic) multiplicity and geometric multiplicity.
- (E) Comment on the nullity of $A - \lambda I$ for each eigenvalue and explain what it represents.

characteristic poly.

$$p(\lambda) = |A - \lambda I|$$

↳ (B) Eigen values = ?

$$\boxed{|A - \lambda I| = 0} \quad \text{characteristic eqn}$$

$$C = A - \lambda I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 5-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 5-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(2-\lambda)(5-\lambda) = 0$$

$$- (\lambda-2)(\lambda-2)(\lambda-5) = 0 \quad \leadsto \text{c. eq.}^\uparrow$$

↳ $p(\lambda) = -(\lambda-2)(\lambda-2)(\lambda-5)$: characteristic polynomial

↳ $p(\lambda) = 0$: characteristic equation

$$(\lambda-2)(\lambda-2)(\lambda-5) = 0$$

$$\hookrightarrow \boxed{\lambda = 2, 2, 5}$$

$\lambda = 2$: Repeated Twice , $\lambda = 5$: Came once
A.M. A.M.

Arithmetic Multiplicity of Eigen value $\lambda = 2$:-

$$\text{A.M.} [\lambda = 2] = 2$$

$$\text{A.M.} [\lambda = 5] = 1$$

$$\text{Eigen values} = 2, 2, 5 \quad [\lambda_1, \lambda_2, \lambda_3] \quad \left\{ \begin{array}{l} \lambda_1 = 2 \\ \lambda_2 = 2 \\ \lambda_3 = 5 \end{array} \right.$$

$$\text{characteristic eq}^n \rightarrow (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = 0$$

$\downarrow$

max^m power of $\lambda = 3 = n$

$$[A]_{n \times n} : \text{max}^m \text{ number of different E. value} = n$$

Eigen vectors:

$$Ax = \lambda x \rightarrow \lambda \in \{2, 5\}$$

Eigen vector corresponding to $\lambda = 2$

$$Ax = 2x$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$2x_1 = 2x_1 \rightarrow \boxed{x_1 = x_1} \rightarrow \text{Always: free variable}$$

$$2x_2 = 2x_2 \rightarrow \boxed{x_2 = x_2} \rightarrow \text{Always: free variable}$$

$$5x_3 = 2x_3$$

$$5x_3 - 2x_3 = 0$$

$$3x_3 = 0 \rightarrow \boxed{x_3 = 0}$$

$$\boxed{\begin{matrix} x_1 = x_1 = c_1 \\ x_2 = x_2 = c_2 \end{matrix}}$$

} Two free variables

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ 0 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{span } v_1} + c_2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\text{span } v_2}$$

w.r.t. $\lambda = 2$; Eigen vectors = $\left\{ \underbrace{v_1, v_2}_{\text{2 Independent Eigen vectors}}, \alpha v_1, \beta v_2, \dots, \alpha v_1 + \beta v_2 \right\}$

Eigen vector corresponding to $\lambda = 5$:-

$$Ax = 5x$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 5 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$2x_1 = 5x_1, \quad 2x_2 = 5x_2,$$

$$\underline{5x_3 = 5x_3}$$

$$-3x_1 = 0; \quad -3x_2 = 0;$$

Always: free var.

$$x_1 = 0; \quad x_2 = 0;$$

$$\boxed{x_3 = c_3}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ c_3 \end{bmatrix} = c_3 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\text{eigen vector}} \overset{v_3}{\curvearrowright}$$

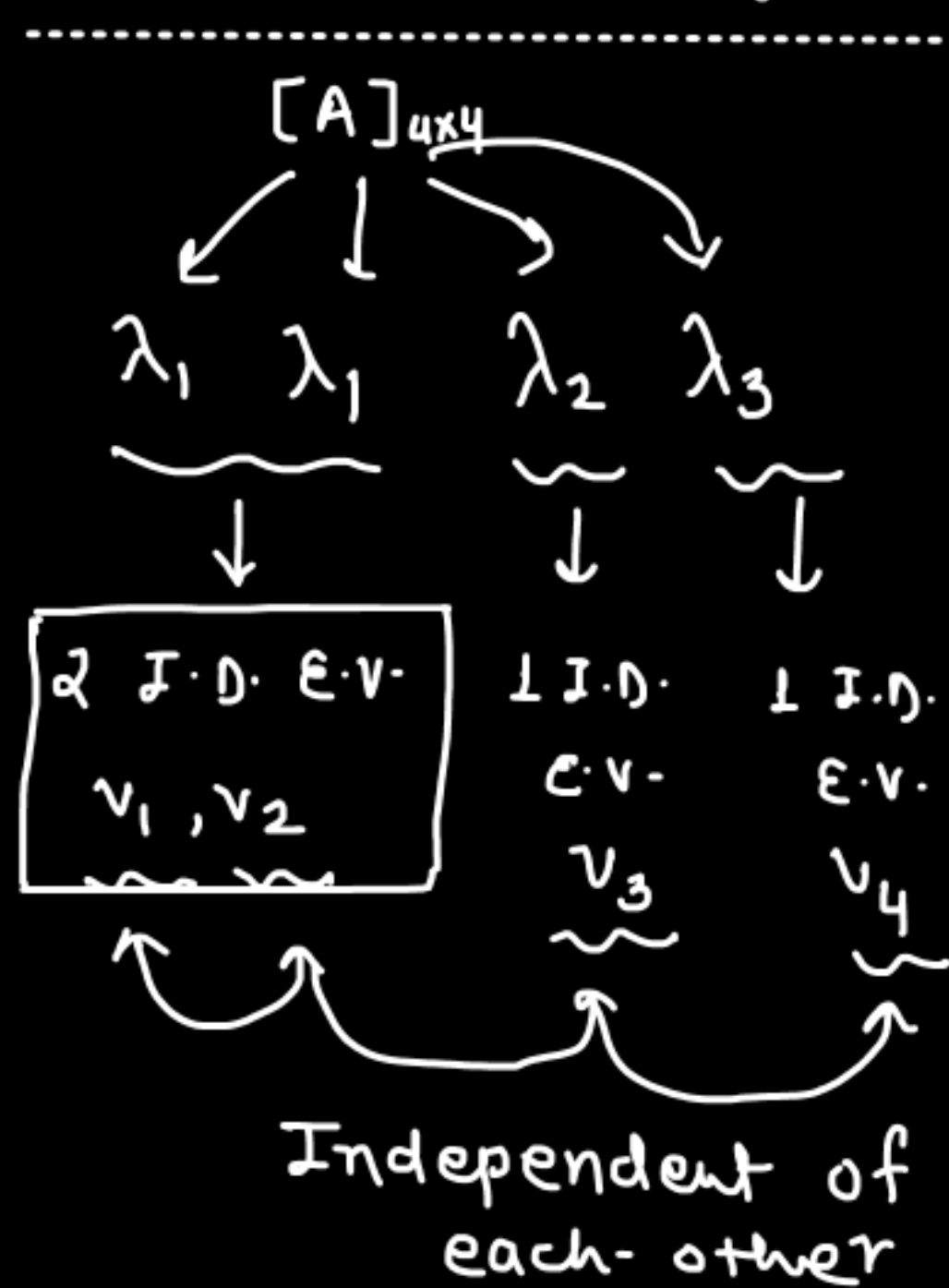
$$\underline{\text{eigen vector corresponding to } \lambda = 5} = \left\{ \underbrace{v_3}, 2v_3, \dots, \gamma v_3 \right\}$$

1 Independent eigen vector

Geometric Multiplicity (GM): - No. of Independent eigen vector corresponding to each eigen value.

$\text{A.M.} [\lambda=2] = 2 \rightarrow$ Two independent E. vectors $\underline{v_1}, \underline{v_2}$

$\text{A.M.} [\lambda=5] = 1 \rightarrow$ One independent E. vector $\underline{v_3}$



$\lambda = 2, 2, 5 \rightarrow v_1, v_2, v_3$

$\text{A.M.} [\lambda=2] = 2 \quad ; \quad \text{G.M.} [\lambda=2] = 2$

$\text{A.M.} [\lambda=5] = 1 \quad ; \quad \text{G.M.} [\lambda=5] = 1$

$$Ax = \lambda x \rightarrow [A - \lambda I]x = 0$$

$$\downarrow$$
$$\rho[A - \lambda I] < n \Rightarrow \underbrace{n[A - \lambda I] > 0}$$

No. of Independent
E. vectors w.r.t. E.V. λ

$\downarrow$
No. of free variables

Let $\lambda = 2$; $\eta[A - 2I] = ?$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \leadsto \eta[A - 2I] = 2$$

↓
2 Independent E.v.

Let $\lambda = 5$; $\eta[A - 5I] = ?$

$$\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \leadsto \eta[A - 5I] = 1$$

↓
1 Independent E.v.

$$\eta[A - \lambda_1 I] = G.M. [\lambda_1]$$

↳ $\eta[A - \lambda_1 I]$: The no. of independent eigen vectors w.r.t. eigen value $\lambda = \lambda_1$

Q.

$$A = \begin{bmatrix} 4 & -2 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 1 & -1 & 0 & 5 \end{bmatrix}_{4 \times 4}$$

(A) Find the eigenvalues of the matrix A .

(B) For each eigenvalue, determine its algebraic multiplicity and geometric multiplicity.

(C) Comment on the relation between algebraic multiplicity and geometric multiplicity for each eigenvalue.

(D) Find the total number of linearly independent eigenvectors of the matrix A .

(E) Find the trace and determinant of matrix A . Hence, comment on their relation with the eigenvalues of A .

$$\text{Tr}(A) = 12$$

$$|A| = 60$$

$$\lambda = \lambda_1, \lambda_2, \lambda_3, \lambda_4$$
$$\lambda = 2, 2, 3, 5$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = \text{Tr}(A)$$

$$\lambda_1 \lambda_2 \lambda_3 \lambda_4 = |A|$$

$$[A]_{n \times n} \rightarrow \downarrow |A - \lambda I| = 0$$

$$(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4) = 0$$

$$\lambda = \lambda_1, \lambda_2, \lambda_3, \lambda_4$$

characteristic equation: $(\lambda-2)^2(\lambda-3)(\lambda-5)=0$

$$\lambda = 2, 2, 3, 5$$

$$\hookrightarrow \text{A.M.}[\lambda=2] = 2,$$

$$\hookrightarrow \text{A.M.}[\lambda=3] = 1; \text{G.M.}[\lambda=3] = 1$$

$$\hookrightarrow \text{A.M.}[\lambda=5] = 1; \text{G.M.}[\lambda=5] = 1$$

$$\hookrightarrow \text{G.M.}[\lambda=3] = \eta[A-3I] = 1$$

$$A-3I = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 2 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & 2 \end{bmatrix}$$

$$\xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_4 \rightarrow R_4 - R_1}]{\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{bmatrix}} \xrightarrow{R_4 \rightarrow R_4 - R_2}$$

$$\rho(A-3I) = 3$$

$$\eta(A-3I) = 4-3 = 1$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xleftarrow{R_4 \rightarrow R_4 - 2A_3} \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$G.M. [\lambda=2] = \eta [A-2I] = 1$$

$$\begin{bmatrix} 2 & -2 & 1 & 0 \\ 2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 3 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 - R_1]{R_4 \rightarrow R_4 - \frac{1}{2}R_1} \begin{bmatrix} 2 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\rho(A-2I) = 3$
 $\eta(A-2I) = 4 - 3 = 1$

$G.M. [\lambda=2] = 1$

 $\rightarrow$ only 1 Independent Eigen vector

$$(A-2I)(x) = 0$$

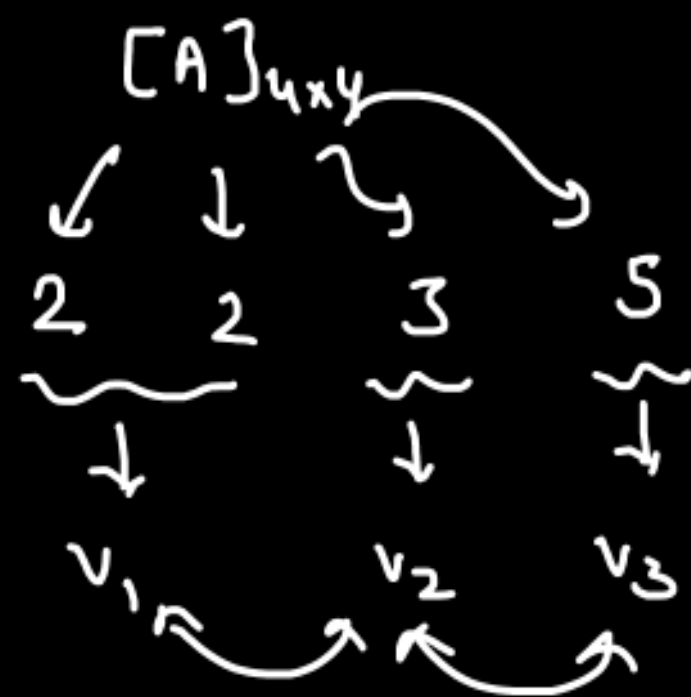
$$\begin{bmatrix} 2 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$\begin{aligned} 3x_4 &= 0 \rightarrow x_4 = 0 \\ x_3 + x_4 &= 0 \rightarrow x_3 = 0 \\ 2x_1 - x_2 + x_3 &= 0 \\ \hookrightarrow 2x_1 &= x_2 \end{aligned}$$

let $x_1 = c$
 $x_2 = 2c$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} c \\ 2c \\ 0 \\ 0 \end{bmatrix} = c \underbrace{\begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}}$$

$$\left. \begin{aligned} \hookrightarrow A.M. [\lambda=2] &= 2, G.M. [\lambda=2] = 1 \\ \hookrightarrow A.M. [\lambda=3] &= 1; G.M. [\lambda=3] = 1 \\ \hookrightarrow A.M. [\lambda=5] &= 1; G.M. [\lambda=5] = 1 \end{aligned} \right\}$$



Total 3 I.D. Eigen
vectors

$$A.M. > G.M.$$

$$[A]_{7 \times 7}$$



$$\underbrace{\lambda_1}_{\downarrow}, \underbrace{\lambda_2}_{\downarrow}, \underbrace{\lambda_3, \lambda_3}_{\downarrow}, \underbrace{\lambda_4, \lambda_4, \lambda_4}_{\downarrow}$$

$$1 \text{ E.V.}$$

$$\hookrightarrow v_1$$

$$1 \text{ E.V.}$$

$$v_2$$

$$v_3$$

$$v_4$$

$$\left\{ \begin{aligned} A.M. [\lambda_1] &= 1, G.M. [\lambda_1] = 1 \\ A.M. [\lambda_2] &= 1, G.M. [\lambda_2] = 1 \\ A.M. [\lambda_3] &= 2, G.M. [\lambda_3] = 1 \text{ or } 2 \\ A.M. [\lambda_4] &= 3, G.M. [\lambda_4] = 1, 2 \text{ or } 3 \end{aligned} \right.$$

$$= 1$$

$$= 1$$

Question:

For a square matrix A of order $n \times n$, which of the following statements are true?

☒ (A) If $\text{nullity}(A) \geq 1$, then 0 is an eigenvalue of A . $\rightarrow |A| = 0$

☒ (B) $\text{nullity}(A)$ represents the number of linearly independent eigenvectors corresponding to the eigenvalue 0.

☒ (C) $\text{nullity}(A)$ represents the algebraic (arithmetic) multiplicity of the eigenvalue 0, i.e., the number of times 0 appears as an eigenvalue of A .

☒ (D) If 0 is an eigenvalue of A , then $\det(A) = 0$.

$$\text{A.M.}[\lambda=0] \geq \text{G.M.}[\lambda=0]$$

$$\text{A.M.}[\lambda=0] \geq 1$$

$\hookrightarrow [A]_{n \times n} \rightarrow \boxed{\eta[A - \lambda_1 I] = \text{G.M.}[\lambda = \lambda_1]}$

Let $\lambda_1 = 0$ is one of the eigen values of A .

$\downarrow \boxed{\eta[A] = \text{G.M.}[\lambda = 0]}$ $\underbrace{\text{G.M.}[\lambda = 0] \geq 1}$

The no. of Independent Eigen vectors corresponding to $\lambda = 0$.

$$\eta[A] = 2$$

$$G.M. [\lambda=0] = 2$$

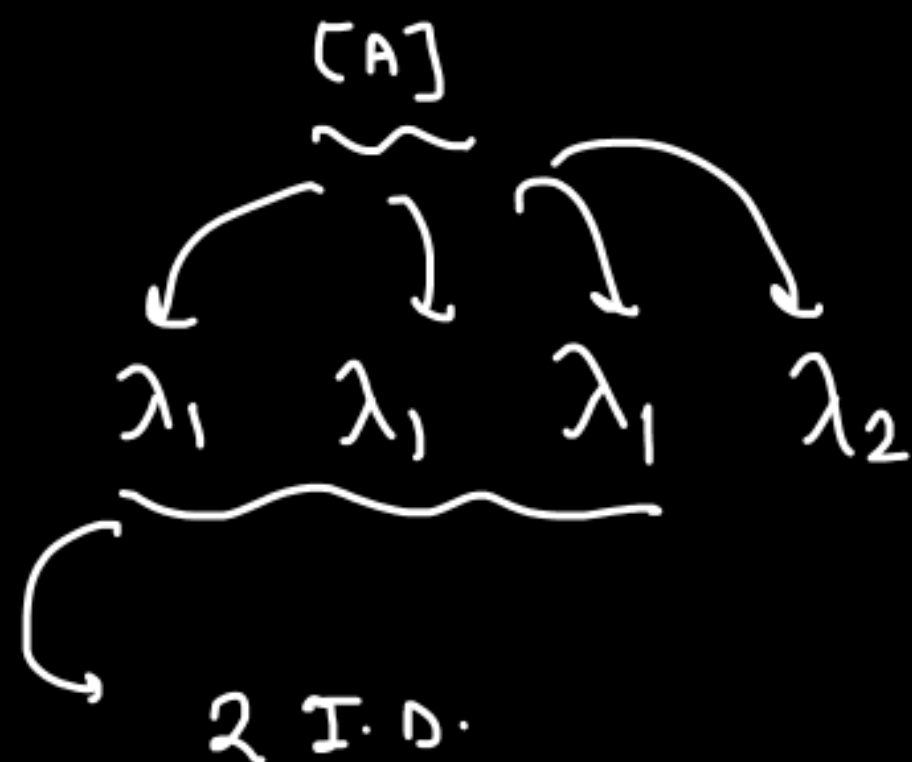
$$\Rightarrow \eta[A - 0I] = 2$$

$$\eta[A] = 2$$

$$A.M. > G.M.$$

$$A.M. [\lambda=0] > 2$$

The number of times, Eigen value 0 has appeared is at least 2.



$$[A] \longrightarrow \lambda_1, \lambda_2, \lambda_3, 0$$

$$|A| = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot 0$$

$$|A| = 0$$

↳ Diagonal and Triangular Matrices:-

↳ Diag.

$$\begin{bmatrix} \textcircled{a} & 0 & 0 \\ 0 & \textcircled{b} & 0 \\ 0 & 0 & \textcircled{c} \end{bmatrix}$$

$$\text{Trace} = a + b + c$$

$$\text{Det} = abc$$

$$\lambda \rightarrow a, b, c$$

↳ Upper Tri

$$\begin{bmatrix} \textcircled{a} & d & e \\ 0 & \textcircled{b} & f \\ 0 & 0 & \textcircled{c} \end{bmatrix}$$

$$\text{Trace} = a + b + c$$

$$\text{Det} = abc$$

$$\lambda \rightarrow a, b, c$$

↳ Lower Tri

$$\begin{bmatrix} \textcircled{a} & 0 & 0 \\ g & \textcircled{b} & 0 \\ h & i & \textcircled{c} \end{bmatrix}$$

$$\text{Trace} = a + b + c$$

$$\text{Det} = abc$$

$$\lambda \rightarrow a, b, c$$

In a triangular or diagonal matrix, the eigenvalues are simply the elements on the main diagonal.

Properties of Eigenvalues & Eigenvectors

1. Basic Definition ✓

- If $AX = \lambda X$, then
 X is an eigenvector of A and λ is the corresponding eigenvalue.
- $X \neq 0$

2. Diagonal / Triangular Matrices ✓

- For any diagonal, upper triangular, or lower triangular matrix:

Eigenvalues = diagonal elements

3. Characteristic Polynomial & Equation ✓

- Characteristic polynomial:

$$|A - \lambda I| = p(\lambda) = \det(A - \lambda I)$$

- Characteristic equation:

$$\hookrightarrow |A - \lambda I| = \det(A - \lambda I) = 0$$

- Roots of this equation = eigenvalues

$$[A]_{n \times n} \rightarrow \lambda_1, \lambda_2, \dots, \lambda_n$$

4. Trace and Determinant

- Sum of eigenvalues:

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = \boxed{\text{tr}(A) = \sum \lambda_i}$$

- Product of eigenvalues:

$$\lambda_1 \lambda_2 \lambda_3 \dots \lambda_n = \boxed{\det(A) = \prod \lambda_i}$$

5. Nullity and Eigenvalue 0 ✓

- If:

$$\text{nullity}(A) \geq 1$$

then:

0 is an eigenvalue

$$A \cdot M \geq C \cdot M$$

- Also:

$$\hookrightarrow C \cdot M [\lambda = 0]$$

$\text{nullity}(A)$ = number of independent eigenvectors for $\lambda = 0$

6. Geometric Multiplicity (GM) ✓

- For eigenvalue λ :

$$\text{GM} = \text{nullity}(A - \lambda I)$$

- It represents:
Number of linearly independent eigenvectors corresponding to λ

7. Algebraic Multiplicity (AM) ✓

- Number of times an eigenvalue appears as a root of the characteristic equation

8. Relation between AM and GM ✓

$$\text{AM} \geq \text{GM} \geq 1$$

$$1 \leq \text{GM} \leq \text{AM}$$

9. Total Independent Eigenvectors ✓

$$\text{Total} = \sum \text{GM of each eigenvalue}$$

- ✓ Maximum possible = n (for $n \times n$ matrix)

$[A]_{n \times n}$
max^m n different ϵ -values
max^m n different ϵ -vectors =

10. Independence of Eigenvectors ✓

- Eigenvectors corresponding to distinct eigenvalues are always linearly independent

$[A]$
 $\lambda_1 \quad \lambda_1 \quad \lambda_2 \quad \lambda_3$

$$\begin{aligned} \text{G.M.} [A - \lambda_1 I] &= 1 \rightarrow v_1 \\ \text{G.M.} [A - \lambda_2 I] &= 1 \rightarrow v_2 \\ \text{G.M.} [A - \lambda_3 I] &= 1 \rightarrow v_3 \end{aligned}$$

Question:

Let A be an invertible square matrix and suppose λ is an eigenvalue of A with corresponding eigenvector $X \neq 0$. Determine the eigenvalues of the following matrices (in terms of λ):

- (A) $A^2 \longrightarrow \lambda^2; X$
 (B) A^n , where n is a integer $\longrightarrow \lambda^n; X$
 (C) $A^{-1} \longrightarrow \lambda^{-1}; X$
 (D) $\text{adj}(A) = |A| A^{-1} \longrightarrow |A| \lambda^{-1}; X$
 (E) kA , where k is a scalar $\longrightarrow k\lambda; X$
 (F) $A + cI$, where c is a scalar $\longrightarrow \lambda + c; X$

$$A \rightarrow \lambda; X$$

$$A \cdot \text{adj}(A) = |A| I$$

$$\text{adj}(A) = |A| A^{-1}$$

$$A^{-1} \rightarrow \frac{1}{\lambda}; X$$

$$|A| A^{-1} \rightarrow \frac{|A|}{\lambda}; X$$

↳

$$\text{Given } AX = \lambda X$$

pre multiplication by A

$$A \cdot AX = \lambda \underline{AX}$$

$$\begin{aligned} A^2 X &= \lambda \cdot \lambda X \\ &= \lambda^2 X \end{aligned}$$

$$\{ AX = \lambda X \}$$

↳ e-value and e-vector of $B = A^2$

$$BY = \alpha Y$$

$$A \rightarrow \lambda; X$$

$$A^T \rightarrow \lambda; Y = ?$$

$$\hookrightarrow A^2[x] = \underbrace{\lambda^2}_{\substack{\downarrow \\ \text{Eigen value}}} \underbrace{[x]}_{\text{Eigen vector}}$$

$$B[x] = \alpha[x]$$

$$\begin{array}{l} A \rightarrow \lambda; x \\ A^2 \rightarrow \lambda^2; x \end{array}$$

$$\hookrightarrow \text{For } A^3 = ? \quad ; \quad Ax = \lambda x$$

$$A^2 Ax = \lambda A^2 x$$

$$A^3 x = \lambda A(Ax)$$

$$A^3 x = \lambda A[\lambda x] = \lambda^2 \underbrace{Ax}_{\lambda x} = \lambda^2 [\lambda x] = \lambda^3 x$$

$$A^3 x = \lambda^3 x$$

$$[c][x] = \underbrace{\lambda^3}_{\substack{\downarrow \\ \text{Eigen value}}} \underbrace{[x]}_{\text{Eigen vector}}$$

$$\begin{array}{l} A \rightarrow \lambda; x \\ A^2 \rightarrow \lambda^2; x \\ \vdots \\ A^n \rightarrow \lambda^n; x \end{array}$$

$$\begin{array}{l} A \rightarrow \lambda; x \\ A^{-1} \rightarrow \lambda^{-1}; x \\ \quad = \frac{1}{\lambda} \end{array}$$

$$\hookrightarrow Ax = \lambda x$$

$$A^{-1}Ax = \lambda A^{-1}x$$

$$x = \lambda [A^{-1}x]$$

$$A^{-1}x = \frac{1}{\lambda}x \rightarrow [A^{-1}][x] = \frac{1}{\lambda}[x]$$

$$A \rightarrow \lambda; x$$

$$A^{-1} \rightarrow \frac{1}{\lambda}; x$$

$$\hookrightarrow \text{Given } Ax = \lambda x; \quad kA = ? \quad \text{where } k = \text{scalar}$$

$$kAx = k\lambda x$$

$$[\underline{kA}][\underline{x}] = [\underline{k\lambda}][\underline{x}]$$

$$A \rightarrow \lambda; x$$

$$kA \rightarrow k\lambda; x$$

$$\hookrightarrow Ax = \lambda x; \quad A + cI = ?; \quad \text{where } c = \text{scalar}$$

$$Ax + cx = \lambda x + cx$$

$$[A + cI]x = [\lambda + c]x$$

$$[A + cI][x] = [\lambda + c][x]$$

$$A \rightarrow \lambda; x$$

$$A + cI \rightarrow \lambda + c; x$$

Question:

Let A be a 3×3 matrix with eigenvalues $\underbrace{2, 2, 3}_{v_1, v_2}$. A new matrix B is defined as

$$B = 3A^2 + 12A - 4I,$$

where I is the identity matrix. Find $\det(B)$.

Ans. $\rightarrow 60416$

$$B = 3A^2 + 12A - 4I$$

$$\lambda_B = 3\lambda_A^2 + 12\lambda_A - 4(1)$$

$$\lambda_A = 2 ; \quad \lambda_B = 12 + 24 - 4 = 32$$

$$\lambda_A = 3 ; \quad \lambda_B = 27 + 36 - 4 = 59$$

$$[B]_{3 \times 3}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $\lambda_1, \lambda_2, \lambda_3$

$$|B| = \lambda_1 \lambda_2 \lambda_3$$

$$[B]$$

$\downarrow$
 $32, 32, 59$

$$|B| = 32 \times 32 \times 59 = 60416$$

↳ Basic Quadratic equation:-

$$ax^2 + bx + c = 0 \rightarrow -x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \left. \begin{array}{l} \text{Roots} \\ \alpha \\ \beta \end{array} \right\}$$

$$\begin{aligned} \alpha + \beta &= -b/a \\ \alpha\beta &= c/a \end{aligned}$$

↳ Basic Cubic Equation:-

$$ax^3 + bx^2 + cx + d = 0 \rightarrow -x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0 \quad \left. \begin{array}{l} \alpha \\ \beta \\ \gamma \end{array} \right\}$$

$$\begin{aligned} \alpha + \beta + \gamma &= -b/a \\ \alpha\beta\gamma &= -d/a \end{aligned}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$$

$$\hookrightarrow ax^n + bx^{n-1} + cx^{n-2} + \dots + d = 0$$

$$x^n + \frac{b}{a}x^{n-1} + \frac{c}{a}x^{n-2} + \dots + \frac{d}{a} = 0 \quad \hookrightarrow \begin{matrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{matrix}$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_n = -\frac{b}{a}$$

$$\alpha_1 \alpha_2 \dots \alpha_n = \begin{cases} +\frac{d}{a} & ; \quad n = \text{Even} \\ -\frac{d}{a} & ; \quad n = \text{odd} \end{cases}$$

* characteristic eqⁿ for 2x2 Matrix:-

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}_{2 \times 2}$$

$$\text{Trace}(A) = 5 = \text{Tr}(A)$$

$$\text{Det}(A) = 4 = |A|$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(3-\lambda) - 2 = 0$$

$$(\lambda-2)(\lambda-3) - 2 = 0$$

$$\lambda^2 - 5\lambda + 6 - 2 = 0$$

$$\boxed{\lambda^2 - 5\lambda + 4 = 0}$$

$$\lambda^2 - \text{Tr}(A)\lambda + |A| = 0$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$\left. \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix} \right\}$ Eigen value

$$\lambda_1 + \lambda_2 = 5 = \text{Tr}(A)$$

$$\lambda_1 \lambda_2 = 4 = |A|$$

↳ Cayley Hamilton Theorem:-

Every square matrix satisfies its own characteristic equation.

$$[A]^2 = A \cdot A$$

$$A^2 = 5A - 4I$$

→

$$\hookrightarrow A^2 - 5A + 4I = 0 \text{ --- (1)}$$

$$A^3 = ?$$

pre M. of A

$$\hookrightarrow A^3 - 5A^2 + 4A = 0$$

$$A^3 - 5[5A - 4I] + 4A = 0$$

$$A^3 - 25A + 20I + 4A = 0$$

$$A^3 = 21A - 20I$$

$$A^4 = 85A - 84I$$

Eg:- $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}_{2 \times 2}$

$$\hookrightarrow \lambda^2 - 5\lambda + 4 = 0$$

$$[A]^2 - 5[A] + 4[I] = [0]$$

$$A^4 = 21A^2 - 20A$$

$$= 21[5A - 4I] - 20A$$

$$A^4 = 105A - 84I - 20A$$

Eg.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

↙

$$\lambda^3 - 3\lambda^2 + 3\lambda - 3 = 0$$

$$\hookrightarrow A^3 - 3A^2 + 3A - 3I = 0$$

$$A^3 = 3A^2 - 3A + 3I$$

$$A^4 = 3A^3 - 3A^2 + 3A$$

$$= 3[3A^2 - 3A + 3I] - 3A^2 + 3A$$

$$= 9A^2 - 9A + 3I - 3A^2 + 3A$$

$$A^4 = 6A^2 - 6A + 3I$$

$$A^5 = 6A^3 - 6A^2 + 3A$$

$$= 6[3A^2 - 3A + 3I] - 6A^2 + 3A$$

$$= 18A^2 - 18A + 18I - 6A^2 + 3A$$

$$A^5 = 12A^2 - 15A + 18I$$

↳ For a 2×2 Matrix:- $k \geq 2$

$$\hookrightarrow A^k = c_1 A + c_2 I$$

↳ For a 3×3 Matrix:- $k \geq 3$

$$\hookrightarrow A^k = c_1 A^2 + c_2 A + c_3 I$$

↳ For $n \times n$ Matrix:- $k \geq n$

$$\hookrightarrow A^k = c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_n I$$

↳ For 5×5 Matrix:-

$$\hookrightarrow A^{99} = c_1 A^4 + c_2 A^3 + c_3 A^2 + c_4 A + c_5 I$$

$$A^{99} \neq c_1 A + c_2 I$$

Q. $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}_{2 \times 2}$ Find A^{10} in terms of A and I ?

$\hookrightarrow A^{10} = c_1 A + c_2 I$

$\lambda^{10} = c_1 \lambda + c_2$

$$\lambda^{10} = c_1 \lambda + c_2$$

$$1 = c_1 + c_2 \quad \text{--- (1)}$$

$$c_2 = 1 - c_1$$

$$4^{10} = 4c_1 + c_2 \quad \text{--- (2)}$$

$$= 1 - \left[\frac{4^{10} - 1}{3} \right]$$

$$3c_1 = 4^{10} - 1$$

$$= \frac{3 - 4^{10} + 1}{3}$$

$$c_1 = \frac{4^{10} - 1}{3}$$

$$c_2 = \frac{4 - 4^{10}}{3}$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda = 1, 4$$

$$A^{10} = \left(\frac{4^{10} - 1}{3} \right) A + 4 \left(\frac{1 - 4^9}{3} \right) I$$

Ans.

Q.

Consider the matrix

$$A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}_{2 \times 2}$$

~~MSQ~~

Q1. A satisfies:

$$\checkmark (A) A + 3I + 2A^{-1} = 0, \quad \times (B) A^2 + 2A + 2I = 0$$

$$\checkmark (C) (A + I)(A + 2I) = 0, \quad \times (D) e^A = 0$$

Q2. $A^9 =$

$$\checkmark (A) 511A + 510I, \quad \times (B) 309A + 104I$$

$$\times (C) 154A + 155I, \quad \times (D) e^{9A}$$

$$\hookrightarrow (\lambda + 1)(\lambda + 2) = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$A^2 + 3A + 2I = 0$$

$$A^2 + 2A + A + 2I = 0$$

$$A[A + 2I] + I[A + 2I] = 0$$

$$(A + I)(A + 2I) = 0$$

$$A^2 + 3A + 2I = 0$$

$$A^{-1}A \cdot A + 3A^{-1}A + 2A^{-1}I = 0$$

$$\boxed{A + 3I + 2A^{-1} = 0} \quad \hookrightarrow \quad \boxed{A^{-1} = -\left(\frac{A + 3I}{2}\right)}$$

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

$$A^9 = c_1 A + c_2 I$$

$$\hookrightarrow \lambda^9 = c_1 \lambda + c_2$$

$$\lambda = -1 \text{ and } -2$$

$$(-1)^9 = c_1(-1) + c_2$$

$$-1 = -c_1 + c_2 \quad \text{--- (1)}$$

$$(-2)^9 = c_1(-2) + c_2$$

$$-512 = -2c_1 + c_2 \quad \text{--- (2)}$$

$$\text{(1)} - \text{(2)}$$

$$511 = c_1$$

$$-1 = -511 + c_2$$

$$c_2 = 510$$

$$A^9 = 511A + 510I$$

Ans:

$$(9) \quad 511A + 510I \rightarrow 511\lambda + 510;$$

$$\lambda = -1$$

$\Downarrow$

$$\hookrightarrow -1$$

$$\lambda = -2$$

$\Downarrow$

$$\hookrightarrow -512$$

$$A \rightarrow -1, -2$$

$$A^9 \rightarrow (-1)^9, (-2)^9$$

$$\hookrightarrow A^9 \rightarrow -1, -512$$

Q.

Question:

Consider a non-singular 2×2 matrix A . If $\text{trace}(A) = 4$ and $\text{trace}(A^2) = 5$, then the determinant of matrix A is:

$$A \rightarrow \lambda_1, \lambda_2$$

$$A^2 \rightarrow \lambda_1^2, \lambda_2^2$$

$$|A| = \lambda_1 \lambda_2$$

↳

$$\lambda_1 + \lambda_2 = 4 \rightarrow \lambda_1^2 + \lambda_2^2 + 2\lambda_1\lambda_2 = 16$$

$$\lambda_1^2 + \lambda_2^2 = 5$$

$$5 + 2\lambda_1\lambda_2 = 16$$

↪

$$\lambda_1 \lambda_2 = 5 \cdot 5$$

$$|A| = 5 \cdot 5$$

Ans.

Q.

Question:

Let

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}_{3 \times 3}$$

and

$$B = A^3 - A^2 - 4A + 5I,$$

where I is the 3×3 identity matrix.

The determinant of B is: $\rightarrow \prod \lambda_B$

$$A \rightarrow \lambda_A = ?$$

$$B \rightarrow \lambda_B$$

$$B = A^3 - A^2 - 4A + 5I$$

$$\lambda_B = \lambda_A^3 - \lambda_A^2 - 4\lambda_A + 5$$

Eigen value of A = ?

↓
 $\lambda_A = -2, 2, 1$

$$|A - \lambda I| = 0 = \begin{vmatrix} 1-\lambda & 0 & -1 \\ -1 & 2-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix}$$

$$= (-2-\lambda)(2-\lambda)(1-\lambda)$$

$$\lambda_B = (-2)^3 - (-2)^2 - 4(-2) + 5 = -8 - 4 + 8 + 5 = 1$$

$$= (2)^3 - (2)^2 - 4(2) + 5 = 8 - 4 - 8 + 5 = 1$$

$$= (1)^3 - (1)^2 - 4(1) + 5 = 1 - 1 - 4 + 5 = 1$$

$$\lambda_B = 1, 1, 1$$



$$|A| = 1 \times 1 \times 1 = 1$$

✱ Question:

Let A be a 7×7 matrix with characteristic equation

$$2\lambda^7 - 3\lambda^6 + 4\lambda^4 + 7\lambda^2 + 16 = 0.$$

Which of the following statements are true?

(A) $\text{trace}(A) = 3$, $\det(A) = -16$

(B) $\text{trace}(A) = -3$, $\det(A) = 16$

☒ (C) $\text{trace}(A) = 1.5$, $\det(A) = -8$

(D) $\text{trace}(A) = -1.5$, $\det(A) = 8$

↳

$$2\lambda^7 - 3\lambda^6 + 4\lambda^4 + 7\lambda^2 + 16 = 0$$

$$\boxed{\lambda^7 - \frac{3}{2}\lambda^6 + 2\lambda^4 + \frac{7}{2}\lambda^2 + 8 = 0}$$

$$\left\{ \begin{array}{c} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_7 \end{array} \right.$$

$$\text{Tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_7 = -(-\frac{3}{2}) = 1.5$$

$$|A| = \lambda_1 \lambda_2 \dots \lambda_7 = -8$$

Q. Let $[A]_{4 \times 4}$ Matrix has 4 Eigen value $\lambda_1, \lambda_2, \lambda_3, \lambda_4$

find its characteristic polynomial and characteristic eqⁿ

$\hookrightarrow [A]_{4 \times 4} \rightarrow \lambda_1, \lambda_2, \lambda_3, \lambda_4$

eqⁿ :- $\underbrace{|A - \lambda I| = 0}_{\hookrightarrow \lambda = \lambda_1, \lambda_2, \lambda_3, \lambda_4}$

$\boxed{(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4) = 0}$ is characteristic eqⁿ
 $\hookrightarrow \lambda = \lambda_1, \lambda_2, \lambda_3, \lambda_4$

characteristic polynomial = $P(\lambda) = |A - \lambda I| = k(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4)$

Let $\boxed{p(\lambda) = 3(\lambda+3)(\lambda+2)}$: ch. poly. $\leftarrow **$

$\downarrow$

$$\text{ch. Eq. } p(\lambda) = 0 \Rightarrow 3(\lambda+3)(\lambda+2) = 0$$

$$\hookrightarrow \boxed{(\lambda+3)(\lambda+2) = 0} \text{ is ch. eq.}$$

$\downarrow$

$$p(\lambda) = k(\lambda+3)(\lambda+2)$$

★
Q.

polynomial

Matrix A is a 4×4 matrix whose characteristic is is

$$2\lambda^4 + \lambda^2 + 1 = P(\lambda)$$

Define

$$B = 2A^8 + A^6 + 2A^5 + A^4 + A^3 + 4A + 3I,$$

where I is the identity matrix. 4×4

Which of the following statements are true?

- ☒ (A) $\text{trace}(A) = 0$
- ☒ (B) $\det(A) = 0.5$
- ☒ (C) $A^{-1} = -2A^3 - A$
- ☒ (D) $\det(B) = 324$

$$\lambda_1, \lambda_2, \lambda_3, \lambda_4$$

∴

$$2\lambda^4 + \lambda^2 + 1 = 0$$

$$2A^4 + A^2 + I = 0$$

$$2A^3 + A + A^{-1} = 0$$

$$\hookrightarrow \boxed{A^{-1} = -2A^3 - A}$$

$$\hookrightarrow \text{Ch. eq.} \Rightarrow P(\lambda) = 0 ; 2\lambda^4 + \lambda^2 + 1 = 0$$

$$\lambda^4 + \frac{1}{2}\lambda^2 + \frac{1}{2} = 0$$

$$\text{Tr}(A) = -0 = 0$$

$$|A| = +0.5$$

$$\boxed{\lambda^4 + 0 \cdot \lambda^3 + 0.5\lambda^2 + 0\lambda + 0.5 = 0}$$

$$B = \underbrace{2A^8} + \underbrace{A^6} + \underbrace{2A^5} + \underbrace{A^4} + A^3 + 4A + 3I$$

$$B = A^4 \underbrace{[2A^4 + A^2 + I]}_{=0} + A \underbrace{[2A^4 + A^2 + I + 3I]}_{=0} + 3I$$

$$\underbrace{2A^4 + A^2 + I = 0}$$

$$B = A^4[0] + A[0 + 3I] + 3I$$

$$\boxed{B = 3A + 3I} \leftarrow$$

$$|B| = |3A + 3I|$$

$$\boxed{|kC| = k^n |C|}$$

$$\boxed{|B| = (3)^4 |A + I|}$$

Target: - $|A + I| = ?$

$$\det[A + I] = ?$$

$$A \rightarrow \lambda_1, \lambda_2, \lambda_3, \lambda_4$$

$$\text{Let } C = A + I ; |C| = |A + I|$$

$$C \rightarrow \lambda_1 + 1, \lambda_2 + 1, \lambda_3 + 1, \lambda_4 + 1$$

$$|C| = (\lambda_1 + 1)(\lambda_2 + 1)(\lambda_3 + 1)(\lambda_4 + 1) = ?$$

$$(\lambda_1 + 1)(\lambda_2 + 1)(\lambda_3 + 1)(\lambda_4 + 1) = ? = |A + I|$$

char. polynomial of Matrix $A = \boxed{p(\lambda) = 2\lambda^4 + \lambda^2 + 1}$

char. eqⁿ of matrix $A \Rightarrow (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4) = 0 \rightarrow$

$$p(\lambda) = k(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4)$$

$$= k[\lambda^4 + \sim \lambda^3 + \dots]$$

$$= \underline{k}\lambda^4 + \sim \lambda^3 + \sim \lambda^2 + \dots$$

$$\boxed{k=2}$$

$$p(\lambda) = 2(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)(\lambda - \lambda_4)$$

if $\lambda = -1$; $p(-1) = 2(-1 - \lambda_1)(-1 - \lambda_2)(-1 - \lambda_3)(-1 - \lambda_4)$

$$\underline{p(-1)} = 2(\lambda_1 + 1)(\lambda_2 + 1)(\lambda_3 + 1)(\lambda_4 + 1) = 4$$

$$\hookrightarrow (\lambda_1 + 1)(\lambda_2 + 1)(\lambda_3 + 1)(\lambda_4 + 1) = 2 = |A + I|$$

$$|B| = (3)^4 (2) = 162 \quad \text{Ans.}$$

$$p(\lambda) = 2\lambda^4 + \lambda^2 + 1$$

$$\underline{p(-1) = 2 + 1 + 1 = 4}$$

4 A real Matrix with complex Eigen values:-

$\Downarrow$
All elements
are real.

$$[A] = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 = 1+j \\ \lambda_2 = 1-j \end{bmatrix}$$

complex conjugate } $\rightarrow$ Real
 $\lambda_2 = \lambda_1^*$

$$\text{Characteristic eqn} \Rightarrow \lambda^2 - 2\lambda + 2 = 0$$

$$\lambda = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2j}{2} = 1 \pm j$$

one of the element should be complex
↳ A complex Matrix with complex Eigen Values:-

$$A = \begin{bmatrix} 3+i & -i & 0 \\ 1+i & 2-i & 0 \\ 1+i & -i & 4 \end{bmatrix}$$

$$\lambda = 2+i, 3-i, 4$$

no conjugate

Eigenvalues: $2+i, 3-i, 4$

Matrices

REAL



Eigen values = Real / Complex
↓
conjugate

complex



Eigen values = Real / complex
↓
not necessary conjugate

1. For a real matrix:

If a complex number $\underline{a + ib}$ is an eigenvalue, then its complex conjugate $\underline{a - ib}$ will also be an eigenvalue.

2. For a complex matrix:

There is no such restriction; eigenvalues need not appear in conjugate pairs.

Soln:- $\rightarrow$ REAL
 $[A]_{3 \times 3} \rightarrow \underline{\lambda_1, \lambda_2, \lambda_3}$

$$\begin{cases} \lambda_1 = 4 + 3i \\ \lambda_2 = 4 - 3i \\ \lambda_3 = \text{REAL } \cancel{4} \end{cases}$$

$$4 + 3i \neq 4 - 3i \\ \lambda_1 \neq \lambda_2 \neq \lambda_3$$

Question:

Let A be a real 3×3 matrix. One of its eigenvalues is $4 + 3i$. Which of the following statements are true?

☒ (A) A must have one real eigenvalue and two complex eigenvalues.

☒ (B) A will surely have three linearly independent eigenvectors. $|A| = 21$

☒ (C) A may have repeating eigenvalues.

☒ (D) If $\det(A) = 21$, then the real eigenvalue is 3.

$$(4 + 3i)(4 - 3i)\lambda_3 = 21$$

$$25\lambda_3 = 21$$

$$\boxed{\lambda_3 = \frac{21}{25}} \leftarrow$$

Q.

Let A be a real 5×5 matrix. Two of its eigenvalues are $4 + 3i$ and $2 - 4i$. It is given that $\text{rank}(A) = 4$. Find $\text{tr}(A)$.

$$\hookrightarrow \rho(A) = 4 < 5 \rightarrow |A| = 0$$

$$\eta(A) = 1 \rightsquigarrow \lambda = 0 \checkmark$$

$$[A] \rightarrow 4+3i, 4-3i, 2-4i, 2+4i, 0$$

$$\text{Tr}(A) = 4+3i + 4-3i + 2-4i + 2+4i$$

$$= \boxed{12} \text{ Ans.}$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

Q.

Let A be a 7×7 matrix whose eigenvalues are 1, 1, 1, 2, 2, 3, 4.

(A) What is the minimum possible number of linearly independent eigenvectors of A ? $\rightsquigarrow 4$

(B) What is the maximum possible number of linearly independent eigenvectors of A ? $\rightsquigarrow 7$

$$\hookrightarrow \begin{array}{ccccccc} 1 & 1 & 1 & 2 & 2 & 3 & 4 \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \text{At least 1} & \text{At least 1} & 1 & 1 & 1 & 1 & 1 \\ \text{Max}^m 3 & \text{Max}^m 2 & & & & & \end{array}$$

$$\begin{array}{l} \text{A.M.}[1] = 3 \quad ; \quad \text{G.M.}[1] \leq 3 \\ \text{A.M.}[2] = 2 \quad ; \quad \text{G.M.}[2] \leq 2 \end{array}$$

Consider a 4×4 matrix

$$A = \begin{matrix} & \begin{matrix} 15 & 15 & 15 & 15 \end{matrix} \\ \begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \end{matrix} & \begin{bmatrix} 1 & 2 & 5 & 7 \\ 2 & 5 & 7 & 1 \\ 5 & 7 & 1 & 2 \\ 7 & 1 & 2 & 5 \end{bmatrix} \end{matrix} \begin{matrix} \rightarrow 15 \\ \rightarrow 15 \\ \rightarrow 15 \\ \rightarrow 15 \end{matrix}$$

One of the possible eigenvalues of matrix A is:

- ✓ (A) 15
- (B) 2
- (C) 3
- (D) 12

$$(15 - \lambda) \text{ ~~~~~ } = 0$$

$$\boxed{\lambda = 15}$$

$$\Leftarrow (15 - \lambda)$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 5 & 7 \\ 2 & 5-\lambda & 7 & 1 \\ 5 & 7 & 1-\lambda & 2 \\ 7 & 1 & 2 & 5-\lambda \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 + R_2 + R_3 + R_4$$

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 2 & 5-\lambda & 7 & 1 \\ 5 & 7 & 1-\lambda & 2 \\ 7 & 1 & 2 & 5-\lambda \end{vmatrix} = 0$$

Q.

Consider a 2×2 matrix

$$A = \begin{bmatrix} \sigma & X \\ \omega & \sigma \end{bmatrix}$$

where X is unknown.

If the eigenvalues of matrix A are $\underbrace{\sigma + j\omega}$ and $\underbrace{\sigma - j\omega}$, then find the value of X .

↳

$$\lambda^2 - (\underbrace{2\sigma})\lambda + (\sigma^2 - \omega^2) = 0$$

$$\sigma^2 + \omega^2 = \sigma^2 - \omega^2 X$$

$$\boxed{X = -1}$$

$$\lambda_1 = \sigma + j\omega$$

$$\lambda_2 = \sigma - j\omega$$

$$\lambda_1 + \lambda_2 = 2\sigma$$

$$\begin{aligned} \lambda_1 \lambda_2 &= \sigma^2 + \omega^2 \\ &= \sigma^2 - \omega^2 X \end{aligned}$$

Q.

Question:

For the matrix

$$A = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

one eigenvalue is -2 . Which of the following is an eigenvector?

~~$\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$~~ ,
 ~~$\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$~~ ,
 ~~$\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$~~ ,
 $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

$$\hookrightarrow \begin{bmatrix} 3 & -2 & 2 \\ \sim & \sim & \sim \\ \sim & \sim & \sim \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ \sim \\ \sim \end{bmatrix} = 5 \begin{bmatrix} 3 \\ \sim \\ \sim \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 3 & -2 & 2 \\ \sim & \sim & \sim \\ \sim & \sim & \sim \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ \sim \\ \sim \end{bmatrix} = -2 \begin{bmatrix} 2 \\ \sim \\ \sim \end{bmatrix}$$

Q.

For the matrix

$$A = \begin{bmatrix} 2 & -2 & 3 \\ -2 & -1 & 6 \\ 1 & 2 & 0 \end{bmatrix}$$

Given that $|A - 3I| = 0$

Find all eigen values.

$$|A - 3I| = 0$$

$$|A - \lambda I| = 0$$



$\lambda \rightarrow$ Eigen values

$\lambda = 3$: eigen value

$$\lambda_1 + \lambda_2 + 3 = 2 - 1 + 0$$

$$\lambda_1 + \lambda_2 = -2$$

$$\lambda_1 \lambda_2 (3) = -45$$

$$\lambda_1 \lambda_2 = -15$$

$$\lambda_1 = -5$$

$$\lambda_2 = 3$$

$$\lambda_3 = 3$$

Q.

Question:

For a 2×2 matrix A , the eigenvalues and corresponding eigenvectors are given as:

$$\underline{\lambda_1 = 8}, v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \underline{\lambda_2 = 4}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$\lambda_1 + \lambda_2 = 12 = \text{Tr}(A)$$

$$\lambda_1 \lambda_2 = 32$$

Then the matrix A is:

$$\checkmark (A) \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}, \quad \cancel{(B)} \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}, \quad \cancel{(C)} \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}, \quad \cancel{(D)} \begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$$

$$\hookrightarrow A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$Ax = \lambda x$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$$

$$a + b = d \quad \text{--- (1)}$$

$$a = 6$$

$$\underline{c + d = 8} \quad \text{--- (2)}$$

$$b = 2$$

$$c = 2$$

$$d = 6$$

$$a - b = 4 \quad \text{--- (3)}$$

$$c - d = -4 \quad \text{--- (4)}$$

Q.

Question:

The eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

are of the form $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$.

Find the value of $a + b$.

$$[A - \lambda I][x] = 0$$

$$\lambda = 1, 2$$

$$\text{E.v. w.r.t. } \lambda = 1$$

$$\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x_2 = 0$$

$$x_2 = 0$$

$$\Rightarrow \underline{x_2 = 0}$$

$$\begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \checkmark$$

$$\text{E.v. w.r.t. } \lambda = 2$$

$$\begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_1 + 2x_2 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ \frac{c_1}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

Ans

$$a = 0$$

$$b = 0.5$$

$$a + b = 0.5$$

Q.

The eigenvalues of a 2×2 matrix A are -2 and -3 .

Find the eigenvalues of the matrix

$$B = (A + I)(A + 3I)(A + 5I)^{-1}.$$

↳

$$\lambda_B = (\lambda_A + 1)(\lambda_A + 3)(\lambda_A + 5)^{-1}$$

$$\lambda_{B_1} = (-2 + 1)(-2 + 3)(-2 + 5)^{-1} = (-1)(1)(3)^{-1} = -1/3$$

$$\lambda_{B_2} = (-3 + 1)(-3 + 3)(-3 + 5)^{-1} = (-2)(0)(2)^{-1} = 0$$

$$\boxed{-1/3, 0}$$

Q.

The eigenvalues of the matrix

$$A = \begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}_{3 \times 3}$$

are:

- ✓ (A) $3, -1 + 3j, -1 - 3j$
- ✗ (B) $3 + j, 3 - j, -5$
- ✗ (C) $-6 + 5j, 3 + j, 3 - j$
- ✗ (D) $3 + j, 3 - j, 3$

$$|A - \lambda I| = 0$$

→ REAL → complex conjugate

$$\text{Tr}(A) = 3 - 1 - 1 = 1$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$|A| = 3[1 + 9] = 30$$

$$\lambda_1 \lambda_2 \lambda_3 = 30$$

$$\boxed{3, -1 + 3j, -1 - 3j}$$

$$\hookrightarrow (3 + j)(3 - j)(-5)$$

$$\hookrightarrow 0(-5) = -50 \times \times$$

Q.

The matrix

$$M = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

has eigenvalues $-3, -3, 5$, and an eigenvector corresponding to eigenvalue 5 is $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

One of the eigenvectors of M^3 is:

(A) $\begin{bmatrix} 1 \\ 8 \\ -1 \end{bmatrix}$, (B) $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, (C) $\begin{bmatrix} 1 \\ \sqrt[3]{2} \\ -1 \end{bmatrix}$, (D) $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$[A]$ and $[A]^3$
 $\downarrow$ $\downarrow$
 5 $(5)^3$

$$Ax = \lambda x$$

$$A^3 x = \lambda^3 x$$

Eigen vectors
remain same

$$[A] \rightarrow \lambda_1, \lambda_2, \lambda_3$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$v_1 \quad v_2 \quad v_3$$

$$\uparrow \quad \uparrow \quad \uparrow$$

$$[A]^3 \rightarrow \lambda_1^3, \lambda_2^3, \lambda_3^3$$

Q.

Let

$$P = \begin{bmatrix} 0 & -2 & -3 \\ -1 & 1 & -1 \\ a & 2 & b \end{bmatrix}, \quad a, b \in \mathbb{R}$$

$\hookrightarrow$ 3×3

Suppose 1 and 2 are eigenvalues of P , and

$$P \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$\nearrow v_3$

Then

$$P^4 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = ?$$

$\nwarrow$ $\searrow$

E-vector = ?

$$P \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$a - b = -3 \quad \text{--- (1)}$$

$$\text{Tr}(P) = 1 + b = 1 + 2 + 3$$

$$b = 5$$

$$a = 2$$

$$[P]^4 \rightarrow 1, 16, 81 \quad \nearrow v_3$$

$$[P] \rightarrow 1, 2, 3$$

$\swarrow$
 v_3

$$P x = 3x$$

$$P^4 x = 1 \cdot x = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$P^4 x = 16x = \begin{bmatrix} 16 \\ -16 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -3 \\ -1 & 1 & -1 \\ 2 & 2 & 5 \end{bmatrix} \underbrace{\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}_{v_2 \text{ corresponding to } \lambda=2} = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

For $[P]^4$; Eigen vector corresponding to $(2)^4 = v_2$

$$[P]^4 [v_2] = 16 v_2$$

$$P^4 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = 16 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 16 \\ -16 \\ 0 \end{bmatrix}}_{\underline{\underline{\text{Ans.}}}}$$

Q.

Let A be a matrix with eigenvalue λ and corresponding eigenvector x .

For the matrix e^A , which of the following statements are true?

- ☒ (A) One of the eigenvalues is e^λ
- ☒ (B) The corresponding eigenvector is x
- ☒ (C) Both (A) and (B) are correct
- ☐ (D) None of the above are correct

$$A \rightarrow \lambda, x$$

$$A^2 \rightarrow \lambda^2, x$$

$\vdots$

$$A^n \rightarrow \lambda^n, x$$

↳

$$B = e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \dots$$

$$= I + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots = \underbrace{e^\lambda}$$

$A \rightarrow \lambda$
$e^A \rightarrow e^\lambda$

Q.

Let A be a matrix with a given set of eigenvectors (eigenspace).

Which of the following statements are true?

☒ (A) The eigenspaces of e^A , $\sin A$, $\cos A$, A^2 , A^3 , A^{-1} (whenever defined) are the same as those of A

☐ (B) The eigenspaces of A^T are the same as those of A

☐ (C) Both (A) and (B) are correct

☐ (D) None of the above are correct

$$A \rightarrow \lambda, x$$

$$A^T \rightarrow \lambda, y$$

↳

$$e^A = I + A + \frac{A^2}{2!} + \dots = 1 + \lambda + \frac{\lambda^2}{2!} + \dots = e^\lambda, x$$

$$\sin A = A - \frac{A^3}{3!} + \frac{A^5}{5!} - \dots = \lambda - \frac{\lambda^3}{3!} + \frac{\lambda^5}{5!} - \dots = \sin \lambda, x$$

$$\cos A = 1 - \frac{A^2}{2!} + \frac{A^4}{4!} - \dots = 1 - \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!} - \dots = \cos \lambda, x$$

Q.

Question:

If the eigenvalues of a matrix A are 1, 4, 3, then find the determinant of A^T .

$$\hookrightarrow A^T \rightarrow 1, 4, 3$$

$$|A| = |A^T| = 1 \times 4 \times 3 = 12$$

$$|A^T| = 1 \times 4 \times 3 = 12$$

Q.

If a matrix A has eigenvalues 1, 2, -1 , find the value of
trace $(A^3 + 4A^2)$.

$$B = A^3 + 4A^2$$

$$\begin{array}{ccc} \swarrow & \downarrow & \searrow \\ 1+4 & 8+16 & -1+4 \\ = 5 & = 24 & = 3 \end{array}$$

$$\begin{aligned} \text{Tr}[B] &= 24 + 3 + 5 \\ &= \boxed{32} \text{ Ans.} \end{aligned}$$

msc
Q.

Let A be a 5×5 matrix with $\text{nullity}(A) = 3$ and $\text{trace}(A) = \underline{5} = \sum \lambda$

Which of the following sets can **possibly** represent the eigenvalues of A ?

- ~~(A)~~ $0, 0, 0, 1, 4$
- ~~(B)~~ $0, 0, 0, 0, 5$
- ~~(C)~~ $0, 0, 1, 1, 3$
- ~~(D)~~ $0, 1, 1, 1, 2$

$\hookrightarrow \quad \text{nullity}(A) = 3 \quad ; \quad \text{nullity}(A) = \dim(\text{Nul}(A)) = 3$

$\dim(\text{Nul}(A)) \geq \dim(\text{Col}(A))$

$\dim(\text{Nul}(A)) \geq \underline{3}$

$\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 0, \lambda_4 = 0?$

↳ Diagonalization of a Matrix:-

Let A be a square matrix of order $n \times n$.

Then, A is diagonalizable if and only if it has n linearly independent eigenvectors.

Q. Which of the matrices are diagonalizable?

✓ (a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}_{3 \times 3}$

$$\lambda \in 1, 2, 3$$

↓
3 I.D. E-vector

✗ (b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$$\lambda = 0, 0, 0$$

$$G.M. [\lambda \in 0] = \eta(A) = 3 - 2 = 1$$

↓
1 I.D. E-vector

✓ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{3 \times 3}$

$$\lambda \in \underbrace{1, 2, 2}_{\substack{\downarrow \\ 2 \text{ I.D.} \\ \text{E-vector}}}$$

$$\eta(A - 2I) = 2$$

3 I.D. Eigen vectors

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\eta = 3 - 1 = 2$$

Q. Find Diagonal Matrix of Matrix A.

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$B = ?$$

↓
Diagonal Matrix of $[A] = ?$

↳ Eigen values = 2, 2, 4

Eigen vector corresponding to $\lambda = 2$

$$(A - 2I)X = 0 \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= -c_1 \\ x_2 &= c_1 \end{aligned} \quad x_3 = c_2$$

$$x_1 + x_2 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -c_1 \\ c_1 \\ c_2 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{V_1} + c_2 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{V_2}$$

Eigen vectors corresponding to $\lambda=4$

$$(A-4I)x=0 \Rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_2 = 0$$

$$x_1 - x_2 = 0 \quad \hookrightarrow x_1 = x_2 = c_1$$

$$-2x_3 = 0 \quad \hookrightarrow x_3 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \hookrightarrow v_3$$

$$P = [v_1 \ v_2 \ v_3] = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \hookrightarrow P^{-1} = ? = \frac{\text{adj}(P)}{|P|}$$

$$\begin{matrix} v_1 & v_2 & v_3 \\ \lambda=2 & \lambda=2 & \lambda=4 \end{matrix}$$

Diagonal Matrix $B = P^{-1}AP$

$$B = P^{-1}AP = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

→ Diagonal Matrix
with diagonal
elements as
eigen value of
[A]

$$P = [v_1 \ v_2 \ v_3]$$

$$\text{If } P = [v_2 \ v_3 \ v_1]$$

$$; \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{3 \times 3}$$

E. values of [B] = E. values of [A]

↳ A: Given Matrix, B: Diagonal Matrix

$$B = P^{-1} A P$$

A and B are known to be Similar Matrices.

$$B^2 = B \cdot B = P^{-1} A P \cdot P^{-1} A P$$

$$= P^{-1} A \underbrace{P P^{-1}} A P = P^{-1} \underbrace{A A} P$$

$$B^2 = P^{-1} A^2 P$$

$$B = P^{-1} A P$$

$$B^2 = P^{-1} A^2 P$$

⋮

$$B^n = P^{-1} A^n P$$

↳ $B = P^{-1} A P$

$$P \cdot B = P P^{-1} A P = A P \quad \left\{ \text{Pre Multiplication of } P \right\}$$

$$P B = A P$$

Post multiplication of P^{-1}

$$P B P^{-1} = A P P^{-1}$$

↳ $A = P B P^{-1}$

$$B = P^{-1} A P$$

$$A = P B P^{-1}$$

$$A^2 = P B^2 P^{-1}$$

⋮

$$A^n = P B^n P^{-1}$$

Q.

Let A be a square matrix such that λ is an eigenvalue of A with corresponding eigenvector $x \neq 0$, i.e.,

$$Ax = \lambda x.$$

Let P be an invertible matrix, and define

$$B = P^{-1}AP. \quad \curvearrowright \quad A = PBP^{-1}$$

Find the eigenvalues and corresponding eigenvectors of the matrix B in terms of λ , x , and P .

↳

$$Ax = \lambda x$$

$$PBP^{-1}x = \lambda x \quad \{A = PBP^{-1}\}$$

pre. Multiplication with P^{-1}

$$BP^{-1}x = \lambda P^{-1}x$$

$$B[P^{-1}x] = \lambda[P^{-1}x]$$

$$A \rightarrow \lambda, x$$

$$P^{-1}AP \rightarrow \lambda, P^{-1}x$$

Q.

Let A be a square matrix such that λ is an eigenvalue of A with corresponding eigenvector $x \neq 0$, i.e.,

$$Ax = \lambda x.$$

Let P be an invertible matrix, and define

$$B = PAP^{-1}. \quad \hookrightarrow \quad P^{-1}BP = A$$

Find the eigenvalues and corresponding eigenvectors of the matrix B in terms of λ , x , and P .

$$Ax = \lambda x$$

$$P^{-1}BPx = \lambda x$$

Pre Multiplication of P

$$BPx = \lambda Px$$

$$B[Px] = \lambda [Px]$$

$$A \rightarrow \lambda, x$$

$$PAP^{-1} \rightarrow \lambda, Px$$

Q. Let A be a square matrix such that λ is an eigenvalue of A with corresponding eigenvector $x \neq 0$, i.e.,

$$Ax = \lambda x.$$

Let P be an invertible matrix, and define

$$B = P A^n P^{-1}. \quad \text{where } A^n = P^{-1} B P$$

Find the **eigenvalues and corresponding eigenvectors** of the matrix B in terms of λ , x , and P .

$$Ax = \lambda x$$

$$A^n x = \lambda^n x$$

$$P^{-1} B P x = \lambda^n x$$

$$B P x = \lambda^n P x$$

$$B [Px] = \lambda^n [Px]$$

$$x \rightarrow \lambda, x$$

$$A^n \rightarrow \lambda^n, x$$

$$P A^n P^{-1} \rightarrow \lambda^n, Px$$

Q.

Let A be a square matrix such that λ is an eigenvalue of A with corresponding eigenvector $x \neq 0$, i.e.,

$$Ax = \lambda x.$$

Let P be an invertible matrix, and define

$$\leadsto A^n = P B P^{-1}$$

$$B = P^{-1} A^n P, \quad n \in \mathbb{N}.$$

Find the eigenvalues and corresponding eigenvectors of the matrix B in terms of λ , x , P , and n .

$$Ax = \lambda x$$

$$A^n x = \lambda^n x$$

$$P B P^{-1} x = \lambda^n x$$

$$B P^{-1} x = \lambda^n P^{-1} x$$

$$B [P^{-1} x] = \lambda^n [P^{-1} x]$$

$$A \rightarrow \lambda, x$$

$$P^{-1} A^n P \rightarrow \lambda^n, P^{-1} x$$

Q.

Let A be a non-zero column vector of order 2027×1 such that

$$\|A\| = 2.$$

Define the matrix

$$B = AA^T.$$

Which of the following statements is/are true?

- ☒ (A) The matrix B is diagonalizable
- ☒ (B) $\text{rank}(B) = 1$
- ☒ (C) The only non-zero eigenvalue of B is 4
- ☒ (D) The only non-zero eigenvalue of B is 2

$B = [AA^T]_{2027 \times 2027}$
 $\downarrow$ Eigen values
 $0, 0, \dots, 0, \dots$ $\rightarrow$ zero / non-zero
2027 Eigen values

$$\eta(B) = 2026$$

$\downarrow$
 B will at least
 have 2026 zero E. values.

$$A = \begin{bmatrix} \\ \\ \\ \end{bmatrix}_{2027 \times 1} \quad \text{rank}(A) = 1$$

$$A^T = \begin{bmatrix} & & & \end{bmatrix}_{1 \times 2027}$$

$$AA^T = \begin{bmatrix} \sim & \sim & \sim & \sim \\ 0 & 0 & 0 & 0 \\ 0 & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{2027 \times 2027}$$

$$\rho(A) = \rho(AA^T) = 1$$

$$\rho(AA^T) = 1$$

$$\eta(AA^T) = 2027 - 1 = 2026$$

$$\text{Tr}(B) = \text{Tr}(AA^T) = \|A\|^2 = \underbrace{\lambda_1 + \lambda_2 + \dots + \lambda_{2026}}_0 + \lambda_{2027}$$

$$\|A\|^2 = \lambda_{2027}$$

$$\lambda_{2027} = 4$$

$$\begin{array}{c}
 [B]_{2027 \times 2027} \\
 \swarrow \quad \searrow \\
 \underbrace{0, 0, \dots, 0}_{\downarrow} \quad \quad \quad 4 \downarrow \\
 2026 \text{ L.I.D. E. vectors} \quad + \quad 1 \text{ I.D. E.V.}
 \end{array}$$

$$\eta(B) = 2026 = \text{G.M.} [\lambda = 0]$$

$$2027 \text{ L.I.D. E.V.} \Rightarrow \text{Diagonalizable}$$

Q.

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{\lambda_1, \lambda_2, \lambda_3} \xrightarrow{E \cdot V} ?$$

Let $P = [v_1 \ v_2 \ v_3]$ be an invertible matrix whose columns are linearly independent eigenvectors of A . Define

$$B = P^{-1}AP.$$

$$A = B = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Find the sum of absolute values of all the elements of the matrix B .

$\Downarrow$
magnitude

$$\text{ans.} \rightarrow |\lambda_1| + |\lambda_2| + |\lambda_3| = ?$$

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{Tr}(A)$$

$$= 2 + 3 - 5$$

$$= \boxed{0} \text{ wrong}$$

$$\lambda^2 - 5\lambda + 5 = 0$$

$$\lambda = \frac{5 \pm \sqrt{5}}{2}$$

$$\lambda \rightarrow -5, \frac{5 + \sqrt{5}}{2}, \frac{5 - \sqrt{5}}{2}$$

$$5 + \frac{5}{2} + \frac{\sqrt{5}}{2} + \frac{5}{2} - \frac{\sqrt{5}}{2} = \boxed{10} \text{ ans.}$$

Hermitian

Q. Prove that; for a Real Symmetric Matrix eigen values are real.

↳ Real symmetric Matrix:- $A^T = A$; $A^\theta = (A^*)^T = A^T = A$
 $A^\theta = A$

$$A \rightarrow \lambda, x$$

$$Ax = \lambda x \text{ --- (1)}$$

$$(Ax)^\theta = (\lambda x)^\theta$$

$$x^\theta A^\theta = \lambda^* x^\theta$$

$$\rightarrow \underline{x^\theta A} = \lambda^* x^\theta \text{ --- (2)}$$

from eqn (1); pre multiplication of x^θ

$$\rightarrow \underline{x^\theta A x} = \lambda x^\theta x \quad \rightarrow \lambda^* x^\theta x = \lambda x^\theta x$$

from eqn (2)

$$\boxed{\lambda^* = \lambda}$$

$$\rightarrow \boxed{\lambda \text{ : REAL}}$$

$$A^T = A$$

Q. Prove that, For a real Symmetric Matrix, Eigen vectors corresponding to different eigen values are orthogonal to each-other.

$$A \rightarrow \lambda_1, \lambda_2 : v_1, v_2$$

$$\text{To prove: } v_1^T v_2 = v_2^T v_1 = 0$$

$$Av_1 = \lambda_1 v_1 \text{ --- (1)}$$

pre m. of v_2^T

$$v_2^T Av_1 = \lambda_1 v_2^T v_1$$

from eqn (1)

$$\lambda_2 v_2^T v_1 = \lambda_1 v_2^T v_1$$

$$(\lambda_2 - \lambda_1) v_2^T v_1 = 0$$

$$\lambda_1 \neq \lambda_2$$

$$Av_2 = \lambda_2 v_2 \text{ --- (2)}$$

Taking Transpose

$$(Av_2)^T = (\lambda_2 v_2)^T$$

$$v_2^T A^T = \lambda_2 v_2^T$$

$$\underline{v_2^T A = \lambda_2 v_2^T} \text{ --- (3)}$$

$$v_2^T v_1 = 0$$



$$v_1 \perp v_2$$

↪ Real Symmetric Matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\hookrightarrow A^T = A$$

- Eigenvalues:

$$\boxed{1, 1, 4}$$

↪ Eigen values are Real

- Eigenvectors:

$$\underline{\lambda = 1} \Rightarrow v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda = 4 \Rightarrow v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

- Orthogonality:

$$\boxed{v_1, v_2 \perp v_3}$$

$$v_1 \perp v_2 \quad \times \times \quad \underline{\underline{\text{NO}}}$$

$$\begin{array}{l} v_1 \perp v_3 \\ v_2 \perp v_3 \end{array} \quad \left. \vphantom{\begin{array}{l} v_1 \perp v_3 \\ v_2 \perp v_3 \end{array}} \right\} \rightarrow \underline{\underline{\text{YES}}} \quad \checkmark$$

Q. Are vectors $\overset{\curvearrowright \text{REAL}}{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}$ and $\overset{\curvearrowright \text{REAL}}{\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}}$ orthogonal?
over Real plane

↳ $1(-1) + 2(1) + 1(-1) = 0 \rightarrow$ orthogonal ✓

↓
Q. Are vectors $\overset{\curvearrowright \text{complex}}{\begin{bmatrix} i \\ -2 \\ -i \end{bmatrix}}$ and $\overset{\curvearrowright \text{complex}}{\begin{bmatrix} -1 \\ i \\ 1 \end{bmatrix}}$ orthogonal?
↳ over complex plane

↳ $\begin{aligned} (-i)(-1) + (-2)(i) + (i)(1) &= i - 2i + i = 0 \\ i(-1) + (-2)(-i) + (-i)(1) &= -i + 2i - i = 0 \end{aligned} \rightarrow$ orthogonal

↳ Remember:

* skew symmetric Matrix of odd order : $|A| = 0$ ↳ $[A]_{3 \times 3} \rightarrow |A| = 0$
" " " of even order : $|A| \neq 0$ or zero $[A]_{4 \times 4} \rightarrow |A| \neq 0$ or 0

Q. Prove that, for a real skew symmetric Matrix, Eigen values are either purely Imaginary or zero.

↳ $\underbrace{A^T = -A}_{\text{skew-symmetric}} ; A^H = (A^T)^* = (-A)^* = -A^T = -(-A) = A ; \underbrace{A^H = -A}_{\text{skew-Hermitian}}$

$A \rightarrow \lambda, x$

$Ax = \lambda x \text{ --- (1)}$

$(Ax)^H = (\lambda x)^H$

$x^H A^H = \lambda^* x^H$

$-x^H A = \lambda^* x^H \text{ --- (2)}$

$x^H A = -\lambda^* x^H$

pre multiplication of x^H

$x^H A x = \lambda x^H x$

$-\lambda^* x^H x = \lambda x^H x$

$-\lambda^* = \lambda$

$\lambda^* = -\lambda$

λ : purely Imag. or zero

$\lambda = 2j \quad \underline{\lambda^*} = -2j = -\underline{\lambda}$

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -2 \\ 0 & 2 & 0 \end{bmatrix}$$

Real
Skew Symm.

$$\hookrightarrow A^T = -A$$

$$0, i\sqrt{5}, -i\sqrt{5}$$

$\hookrightarrow$ purely Imag.
or zero

$$\lambda = 0 \Rightarrow v_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$\hookrightarrow$

Different

$$\lambda = i\sqrt{5} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -i\sqrt{5} \\ -2 \end{bmatrix}$$

$\hookrightarrow v_1 \perp v_2 \perp v_3$

$$\lambda = -i\sqrt{5} \Rightarrow v_3 = \begin{bmatrix} 1 \\ i\sqrt{5} \\ -2 \end{bmatrix}$$

$\hookrightarrow$

$$B = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$$

$\hookrightarrow$ Real skew
Symm.

$$3i, -3i$$

$\hookrightarrow$ purely Imag

$$\lambda = 3i \Rightarrow v_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\lambda = -3i \Rightarrow v_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$\hookrightarrow v_1 \perp v_2$

$$A^T = A$$

Symm.

$$A^H = A$$

$\leftrightarrow$ Hermitian

Skew Symm. $\leftrightarrow$ skew Hermitian

$$A^T = -A$$

$$A^H = -A$$

↪ Hermitian

$$H = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

$$\hookrightarrow H^{\theta} = H$$

$$\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} = H$$

$$[0, 2]$$

: Real Eigen value

$$\lambda = 2 \Rightarrow v_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\lambda = 0 \Rightarrow v_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

orthogonal
over complex
plane

$$1(1) + i(i) = 0$$

$$S^{\theta} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = -S$$

$$S = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}_{2 \times 2}$$

$$\hookrightarrow S^{\theta} = -S$$



Skew Hermitian Matrix

$$[i, -i]$$

: Imag. Eigen values
or zero

$$\lambda = i \Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = -i \Rightarrow v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

orthogonal =

↳ Properties:-

1. Eigenvalues (Real case)

For a real symmetric matrix and a Hermitian matrix:

All eigenvalues are real

2. Eigenvalues (Skew case)

For a real skew-symmetric matrix and a skew-Hermitian matrix:

Eigenvalues are either purely imaginary or zero

3. Orthogonality of Eigenvectors

For a real symmetric matrix, real skew-symmetric matrix, Hermitian matrix, and skew-Hermitian matrix:

Eigenvectors corresponding to distinct eigenvalues are orthogonal ✓

- For a real symmetric matrix: orthogonality is in real space ←
- For real skew-symmetric, Hermitian, and skew-Hermitian matrices: orthogonality is in complex space ←

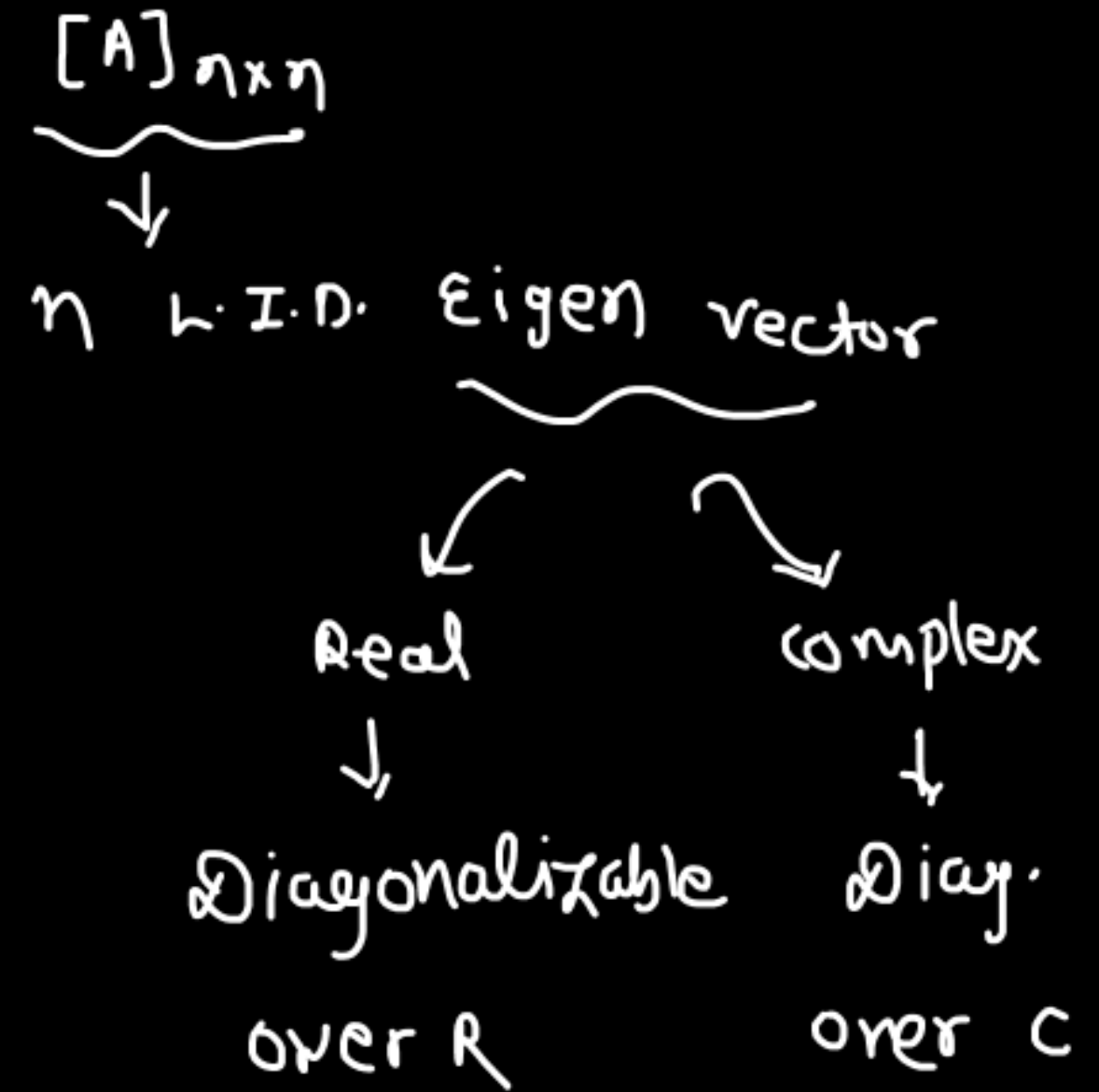
4. Diagonalization

- A real symmetric matrix:

Always diagonalizable over $\mathbb{R}$

- A Hermitian, real skew-symmetric, and skew-Hermitian matrix:

Always diagonalizable over $\mathbb{C}$



Unitary Matrix $AA^H = I = A^H A$

Q. Prove that Eigen values of orthogonal Matrix lie on unity circle i.e. magnitude of all possible eigen value is 1.

↳ $[A]$: Orthogonal Matrix

↓
 λ, x

$$||Ax||^2 = (Ax)^T (Ax)$$

$$= x^T \underbrace{A^T A} x$$

$$= x^T x$$

$$||Ax||^2 = ||x||^2 \quad \hookrightarrow ||Ax|| = ||x||$$

$$\underbrace{AA^T = I = A^T A}$$

$$Ax = \lambda x$$

$$||Ax|| = ||\lambda x||$$

$$||x|| = ||\lambda x||$$

$$||x|| = |\lambda| \cdot ||x||$$

$$\boxed{|\lambda| = 1}$$

$$||x||^2 = x^T x$$

$$\begin{matrix} & x & x & x \\ x & & & \\ & & AA^T = I & \\ & \downarrow & & \downarrow \\ & & & \end{matrix}$$

$$x \quad \lambda \cdot \lambda^* = 1 \quad x \quad x$$

$$x \quad x \quad \underbrace{\lambda^2 = 1} \quad x \quad x$$

↓
A and A^T have diff. eigen vectors

1. Eigenvalues

- For an orthogonal matrix and a unitary matrix:

$$|\lambda| = 1$$



2. Orthogonality of Eigenvectors

For an orthogonal matrix and a unitary matrix:



Eigenvectors corresponding to distinct eigenvalues are orthogonal



3. Diagonalization



- For an orthogonal matrix and a unitary matrix:

Always diagonalizable over $\mathbb{C}$

↳ Three special Matrices:-

1. Involutory:- $A^2 = I$
 $\lambda^2 = 1 \rightsquigarrow \boxed{\lambda = \pm 1}$

2. Idempotent:- $A^2 = A$
 $\lambda^2 = \lambda \rightsquigarrow \lambda^2 - \lambda = 0$
 $\lambda(\lambda - 1) = 0 \rightsquigarrow \boxed{\lambda = 0, 1}$

3. Nilpotent:- $A^k = 0 \rightsquigarrow \lambda^k = 0 \rightsquigarrow \boxed{\lambda = 0}$

$A^2 = 0$: $[A]$ is a Nilpotent Matrix of Index 2.

$A^3 = 0$: $[A]$ " " " " " 3.

$A^k = 0$: $[A]$ " " " " " k .

Eg:- ① Involutory:- $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightsquigarrow A^2 = I \rightsquigarrow \boxed{\lambda = 1, 1}$
 $|A| = 1$

$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \rightsquigarrow A^2 = I$
 $\rightsquigarrow \boxed{\lambda = -1, -1}$
 $\rightsquigarrow |A| = 1$

$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightsquigarrow A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $|A| = -1$
 $\rightsquigarrow \boxed{\lambda = 1, -1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

② Idempotent:-

$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow A^2 = A \quad |A| = 0$
 $\rightsquigarrow \boxed{\lambda = 0, 0}$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^2 = A \quad |A| = 1$
 $\boxed{\lambda = 1, 1}$

$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \rightsquigarrow \boxed{\lambda = 0, 1}$

$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
 $A^2 = A \quad |A| = 0$

③ Nilpotent:-

👉 Take

(1)

$$\hookrightarrow A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Nilpotent Matrix
of Index 2

◆ Check A^2

$$\hookrightarrow A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = [0]$$

(2)

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Nilpotent Mat. of
Index 3

$$B^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0$$

$$B^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

Q. Which of the following statements are true?

- ~~(A)~~ For an involutory matrix, $\lambda = 1$ will surely be one of the eigenvalues.
- ~~(B)~~ For an involutory matrix, $\lambda = -1$ will surely be one of the eigenvalues.
- ~~(C)~~ For an idempotent matrix, $\lambda = 0$ will surely be one of the eigenvalues.
- ~~(D)~~ For an idempotent matrix, $\lambda = 1$ will surely be one of the eigenvalues.
- ~~(E)~~ For a nilpotent matrix of index k , all eigenvalues are zero.
- ~~(F)~~ Nilpotent matrices are always non-invertible.
- ~~(G)~~ An involutory matrix is always invertible.
- ~~(H)~~ An idempotent matrix may or may not be invertible.
- ~~(I)~~ If A and B are involutory matrices, then $A + B$ is involutory.
- ~~(J)~~ If $AB + BA + I = 0$, then $A + B$ is involutory. Given that A and B are I_n
- ~~(K)~~ For a nilpotent matrix, the nullity is always ≥ 1 . $|A| = 0$
- ~~(L)~~ For an involutory matrix, the nullity is always 0.
- ~~(M)~~ For an idempotent matrix, the nullity is always > 0 .
- ~~(N)~~ For an idempotent matrix, the nullity can be any value from 0 to n .

$$I_n \rightarrow A^2 = I$$

$$|A| = \lambda_1 \lambda_2 \quad \lambda^2 = 1$$

$$\Rightarrow \lambda = \frac{1}{\lambda}, -\frac{1}{\lambda}$$

$$\boxed{\pm 1}$$

$$I_n \rightarrow A^2 = A$$

$$\lambda^2 = \lambda$$

$$\boxed{|A| = \frac{1}{\lambda}, 0} \quad \lambda = 0, 1$$

$$nil \rightarrow A^k = 0$$

$$|A| = \lambda_1 \lambda_2 \dots \lambda_n = 0 \quad \lambda^k = 0 \Rightarrow \underline{\lambda = 0}$$

$$\hookrightarrow (I) A^2 = I, B^2 = I; (A+B) = ? \quad (A+B)^2 = A^2 + B^2 + AB + BA$$

If $A+B$ is involutory then $I = A^2 + B^2 + AB + BA$

$$(A+B)^2 = I$$

$$I = I + I + AB + BA$$

$$\hookrightarrow \boxed{AB + BA + I = 0} \quad \checkmark$$

(h) Involutory $|A| \neq 0$; 0 can never be an eigen value

$$\eta(A) = G.M. (\lambda = 0)$$

$$A.M. (\lambda = 0) = 0$$

$$G.M. (\lambda = 0) = 0$$

$$\boxed{\eta(A) = 0}$$

$$\boxed{\eta(A) = 0}$$

IMPORTANT MATRICES – PROPERTIES SUMMARY

Matrix Type	Defining Property	Eigenvalues	Orthogonality of Eigenvectors	Diagonalization
Symmetric	$A = A^T$	Real	Orthogonal in real space	Always diagonalizable over $\mathbb{R}$
Hermitian	$A = A^\theta$	Real	Orthogonal in complex space	Always diagonalizable over $\mathbb{C}$
Skew-Symmetric	$A^T = -A$	Purely imaginary or 0	Orthogonal in complex space	Always diagonalizable over $\mathbb{C}$
Skew-Hermitian	$A^\theta = -A$	Purely imaginary or 0	Orthogonal in complex space	Always diagonalizable over $\mathbb{C}$
Orthogonal	$A^T A = I$	Magnitude of eigenvalues = 1	Orthogonal in complex space	Always diagonalizable over $\mathbb{C}$
Unitary	$A^\theta A = I$	Magnitude of eigenvalues = 1	Orthogonal in complex space	Always diagonalizable over $\mathbb{C}$
Involutory	$A^2 = I$	± 1	Not guaranteed	Always diagonalizable
Idempotent	$A^2 = A$	0, 1	Not guaranteed	Always diagonalizable
Nilpotent	$A^k = 0$ (for some $k \geq 1$)	0 only	Not guaranteed	Not diagonalizable (except zero matrix)

Question:

Let M be a real symmetric 3×3 matrix with eigenvalues $0, 2, \alpha$, where the algebraic multiplicity of eigenvalue 0 is 1 .

The corresponding eigenvectors are:

$$\lambda=0 \leftarrow u = \begin{bmatrix} 4 \\ \beta \\ \gamma \end{bmatrix}, \quad \lambda=2 \leftarrow v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \quad \lambda=2 \leftarrow w = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Find the value of $\alpha + \beta - \gamma$.

$$A.M. [\lambda=0] = 1$$

$$\alpha \neq 0$$

Real Symm. Matrix $\rightarrow$ E. vectors $\perp$, corresponding to different E. values

$$\vec{v} \cdot \vec{w} = (-1)(1) + 2(1) + 0(1) = 1 \rightarrow \underbrace{v \perp w \times \times}$$

$$u \perp v \Rightarrow 4(-1) + 2\beta + \gamma(0) = 0 \Rightarrow \boxed{\beta = 2}$$

$\hookrightarrow$ same eigen values

$$\boxed{\alpha = 2}$$

$$u \perp w \Rightarrow 4(1) + \beta + \gamma = 0 \Rightarrow \boxed{\gamma = -6}$$

$$\boxed{\alpha + \beta - \gamma = 10} \text{ Ans.}$$

Q.

Question:

Let A be a 5×5 matrix with eigenvalues

$$\hookrightarrow 2i, -2i, 6i, -6i, 0$$

Which of the following types of matrices can A possibly be?

- ✓ (A) Skew-symmetric matrix $\rightarrow$ Imag. or zero
- ✓ (B) Skew-Hermitian matrix $\rightarrow$ Imag. or zero
- ✗ (C) Orthogonal matrix $\rightarrow |\lambda| = 1$
- ✗ (D) Real symmetric matrix $\rightarrow$ REAL

Q.

Which of the following can be an eigenvalue of an orthogonal matrix?

$$\hookrightarrow |\lambda| = 1$$

✓ (A) i , (B) $\frac{2+i}{\sqrt{5}}$, (C) $\frac{3-4i}{5}$, ✗ (D) $\frac{7+3i}{6}$

$$\downarrow$$

$$= \frac{\sqrt{5}}{5} \neq 1$$

$$= \frac{\sqrt{25}}{5} = 1$$

$$\neq 1$$

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Q.

Question:

Let α and β be two non-zero real numbers, and let v_1, v_2 be non-zero real vectors in $\mathbb{R}^3$ such that

$$v_1^T v_2 = 0, \quad \underbrace{v_1^T v_1 = 1}, \quad \underbrace{v_2^T v_2 = 1}.$$

Define the 3×3 matrix

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T.$$

The eigenvalues of A are:

☒ (A) $0, \alpha, \beta, 0, \alpha + \beta, \alpha - \beta, 0, \frac{\alpha + \beta}{2}, \sqrt{\alpha\beta}, 0, 0, \sqrt{\alpha^2 + \beta^2}$

$$v_1^T v_2 = 0 \rightarrow v_1 \perp v_2$$

$$v_2^T v_1 = 0$$

$$Ax = \lambda x$$

$$v_1^T v_1 = \|v_1\|^2$$

$$\text{Tr}(v_1 v_1^T) = \|v_1\|^2$$

↳ Method - 1 :-

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T$$

$$A \cdot v_1 = \alpha v_1 v_1^T \overset{1}{v_1} + \beta v_2 v_2^T \overset{0}{v_1}$$

$$\boxed{A \cdot v_1 = \alpha v_1} \rightarrow \text{Eigen value} = \alpha, \text{ Eigen vector} = v_1$$

$$A \cdot v_2 = \alpha v_1 v_1^T \overset{0}{\cancel{v_2}} + \beta v_2 v_2^T \overset{1}{\cancel{v_2}}$$

$$\boxed{A \cdot v_2 = \beta v_2} \hookrightarrow \text{Eigen value} = \beta, \text{ Eigen vector} = v_2$$

$$\hookrightarrow A = \alpha v_1 v_1^T + \beta v_2 v_2^T$$

$$\text{Tr}(A) = \alpha \text{Tr}(v_1 v_1^T) + \beta \text{Tr}(v_2 v_2^T)$$

$$\text{Tr}(A) = \alpha \|v_1\|^2 + \beta \|v_2\|^2$$

$$\boxed{\text{Tr}(A) = \alpha + \beta}$$

$$A \rightarrow \lambda_1, \alpha, \beta$$

$$\text{Tr}(A) = \alpha + \beta = \lambda_1 + \alpha + \beta$$

$$\boxed{\lambda_1 = 0}$$

$$\text{Tr}[\alpha A + \beta B]$$

$$= \alpha \text{Tr}(A) + \beta \text{Tr}(B)$$

$$\text{Tr}(v_1 v_1^T) = \|v_1\|^2$$

$$\boxed{A \rightarrow 0, \alpha, \beta}$$

M-II

$$v_1^T v_2 = 0 \Rightarrow v_1 \perp v_2$$

$$v_1^T v_2 = v_2^T v_1 = 1 \Rightarrow \|v_1\| = \|v_2\| = 1$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow v_1 \perp v_2$$
$$\|v_1\| = \|v_2\| = 1$$

$$v_1 v_1^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \beginbegin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = v_1 v_1^T$$

$$v_2 v_2^T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = v_2 v_2^T$$

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\nearrow$ E. values
 $= \underbrace{\alpha, \beta, 0}$

NOTES: TELEGRAM GROUP

↳ Linear Algebra Eng. Maths ↓